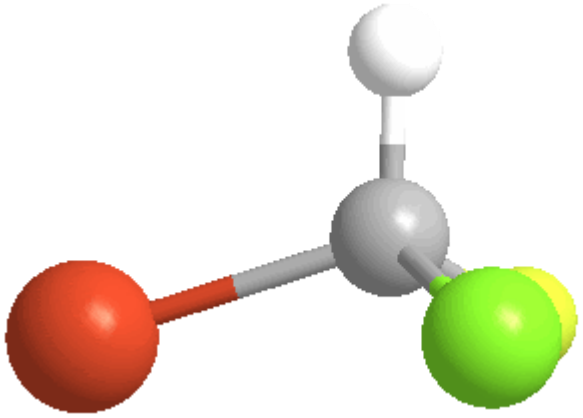
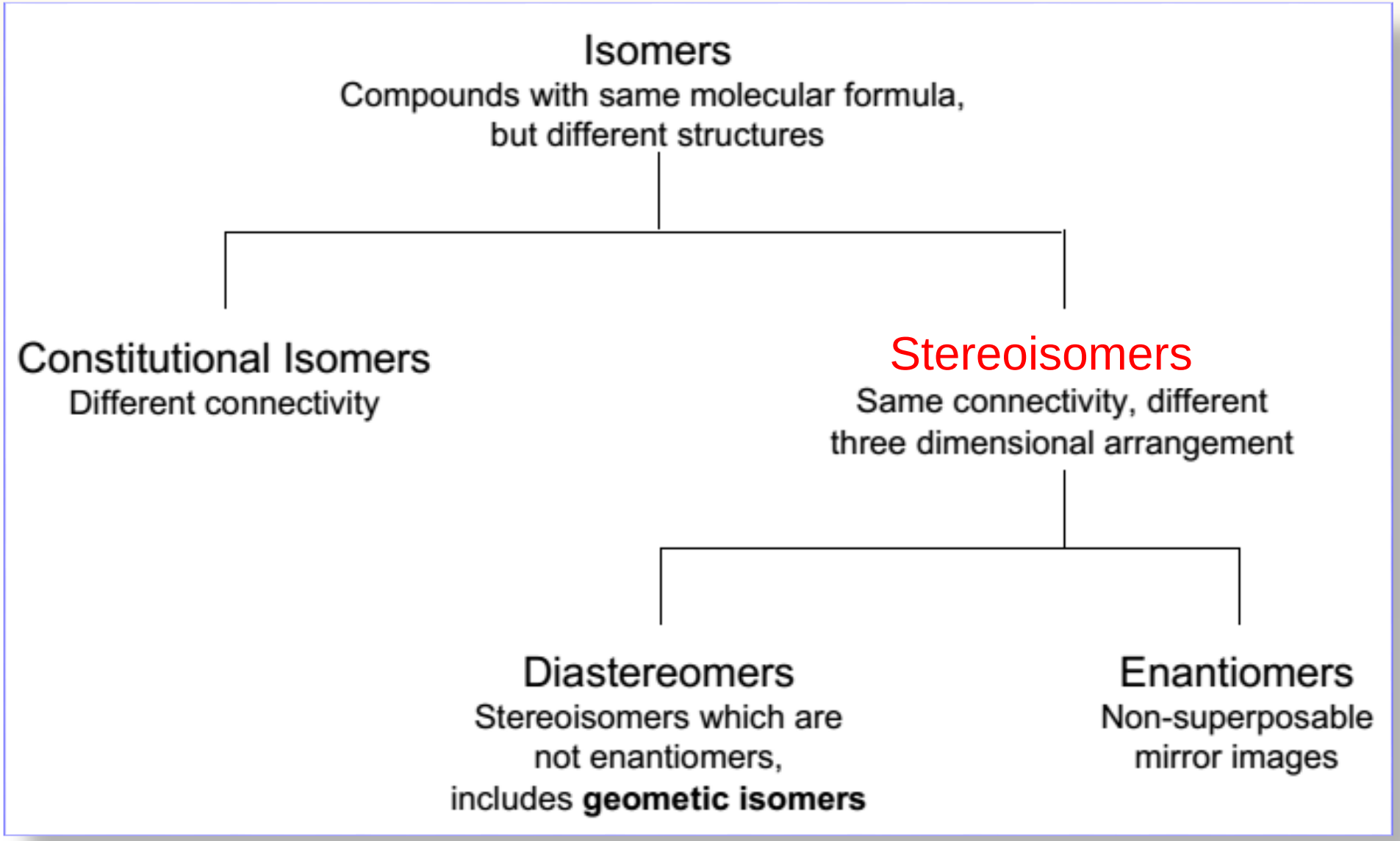
Advanced Stereochemistry

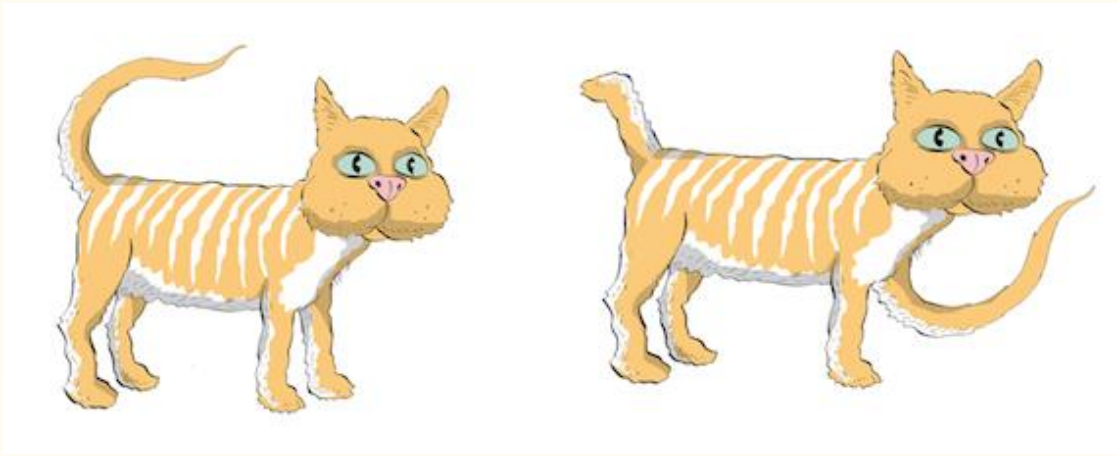


What You Will Learn In This course

1. Prochirality
2. Configuration and conformation of cyclic compounds
3. Conformational analysis of biopolymers
4. Chirality in molecules devoid of chiral centers
5. ORD curve, CD, octant rule, α-haloketone rule
6. Fused rings & bridged ring systems

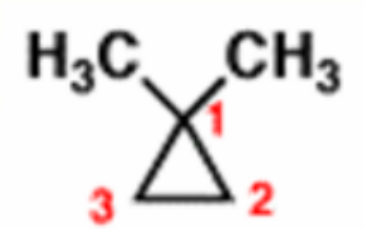
Isomers and Stereoisomers





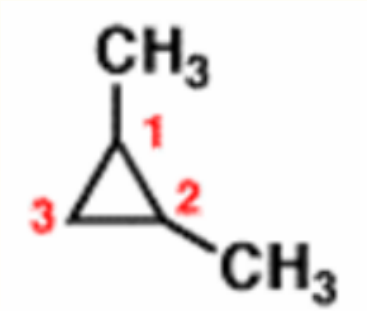
Constitutional Isocats!

Same components, different connectivity



1,1-Dimethylcyclopropane

C_5H_{10}

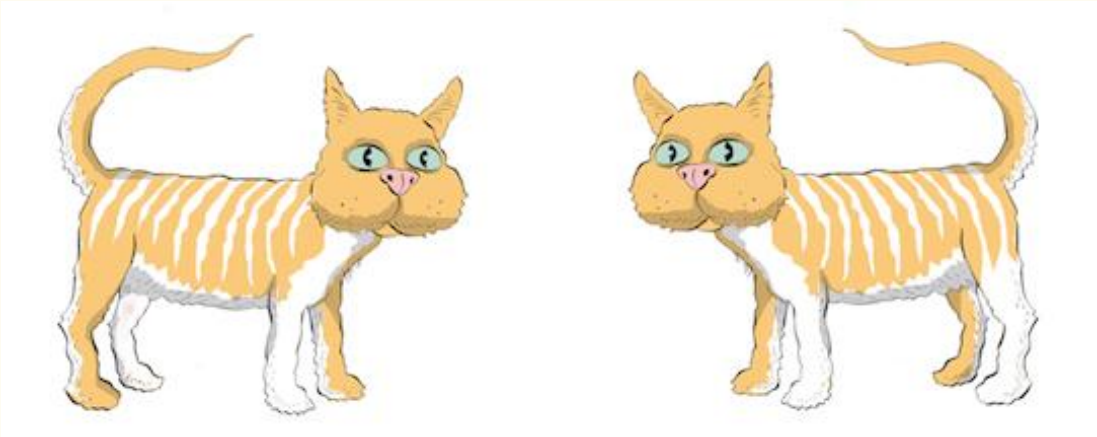


1,2-Dimethylcyclopropane

C_5H_{10}

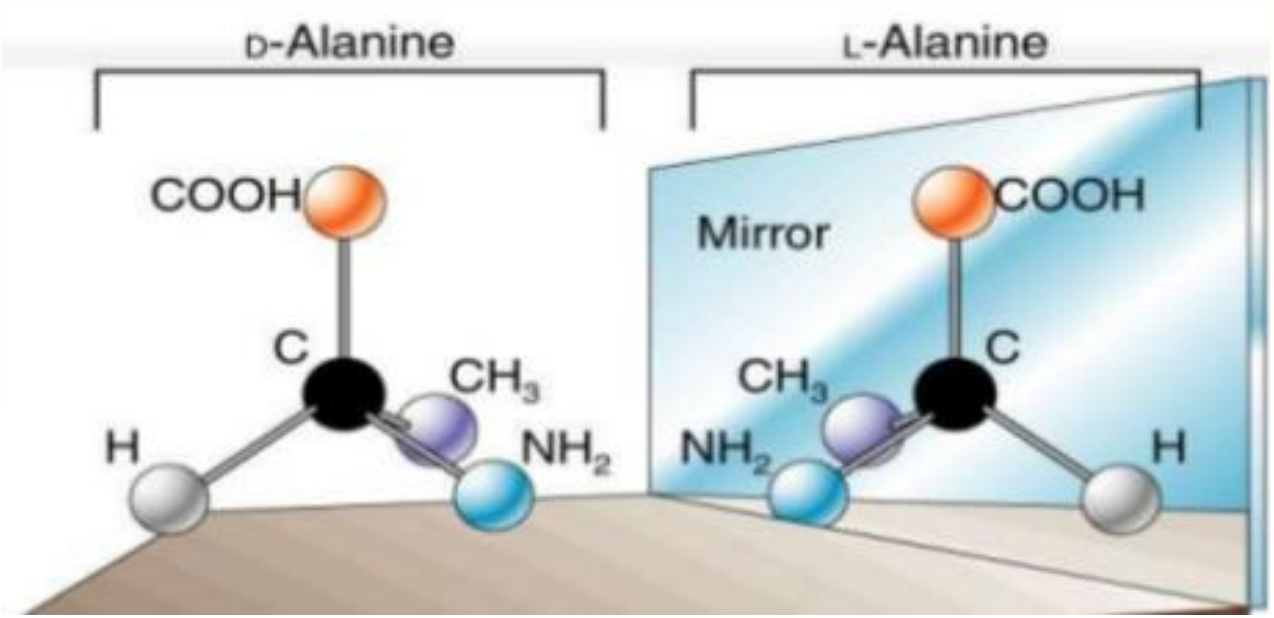
Constitutional isomers

Same molecular formula, but different connectivity



Stereoisocats

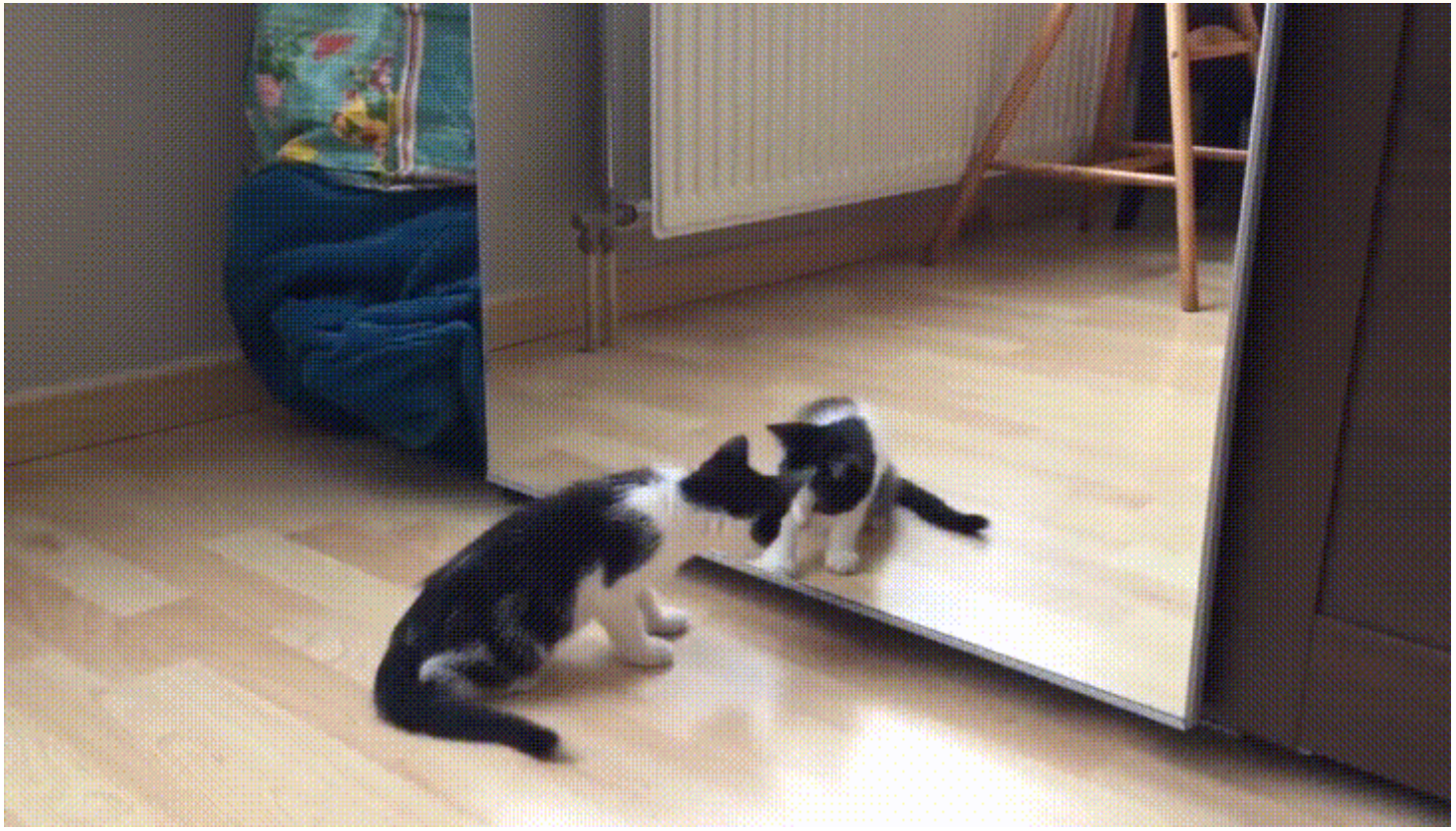
Same connectivity, but different arrangement in space



Stereoisomers

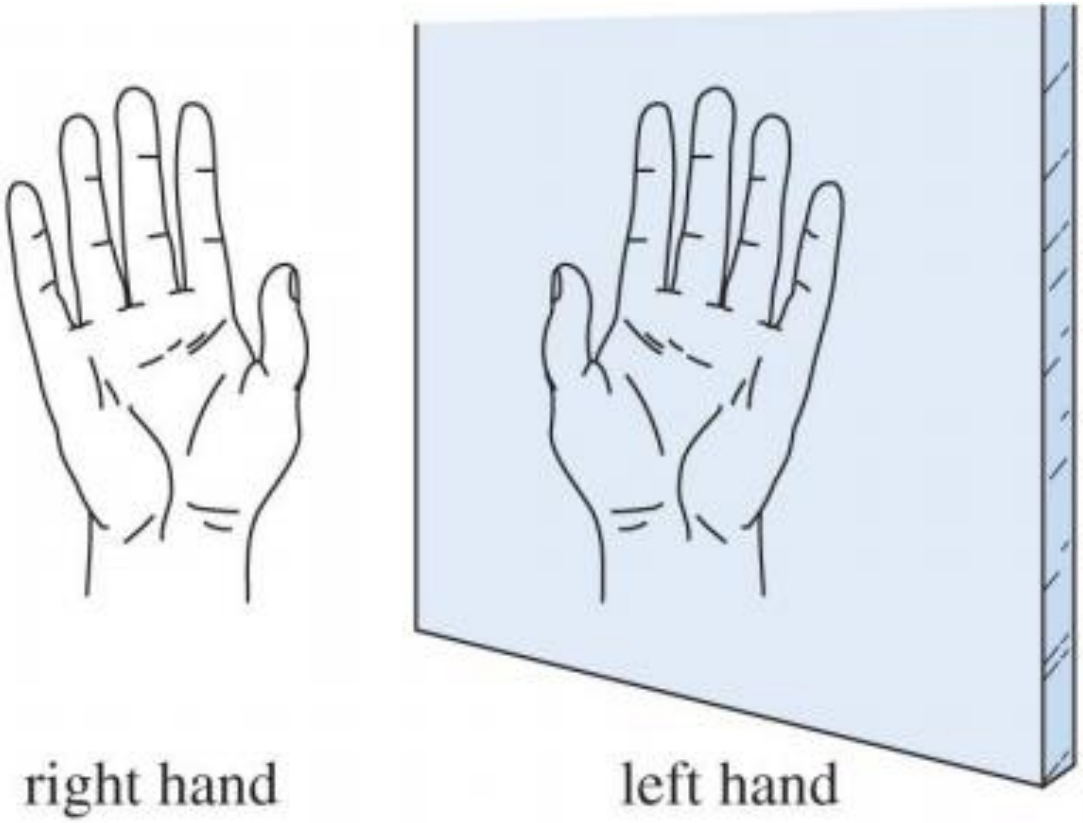
So, tell me again
What are Enantiomers?

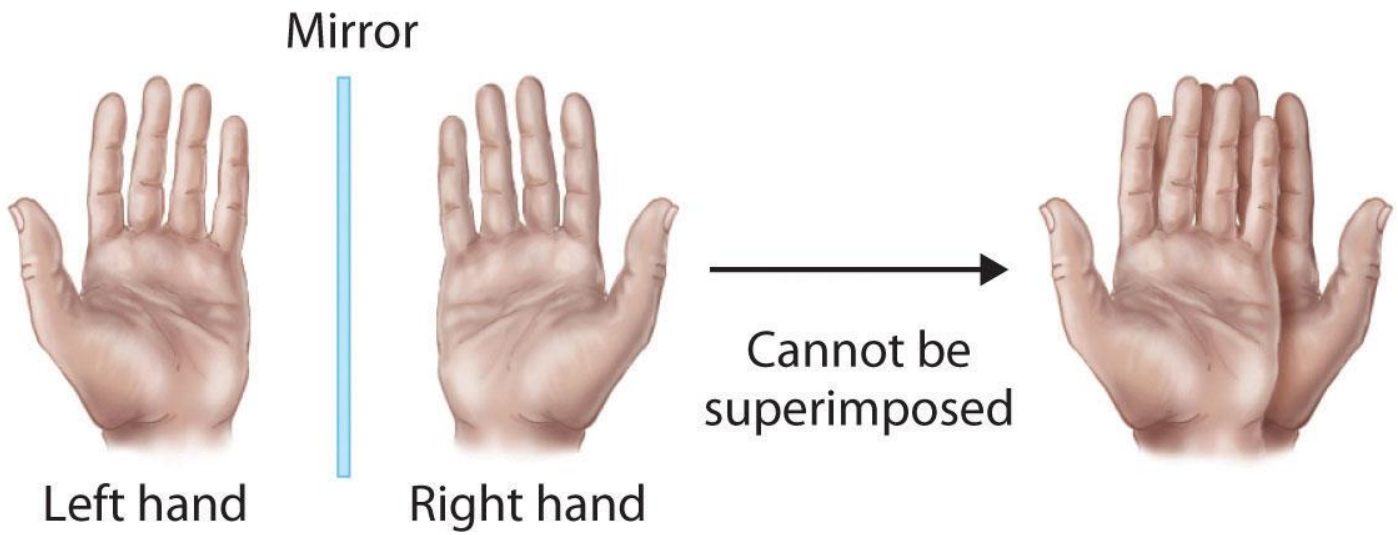
- **Enantiomers** are chiral molecules that are non-superimposable mirror images of one another



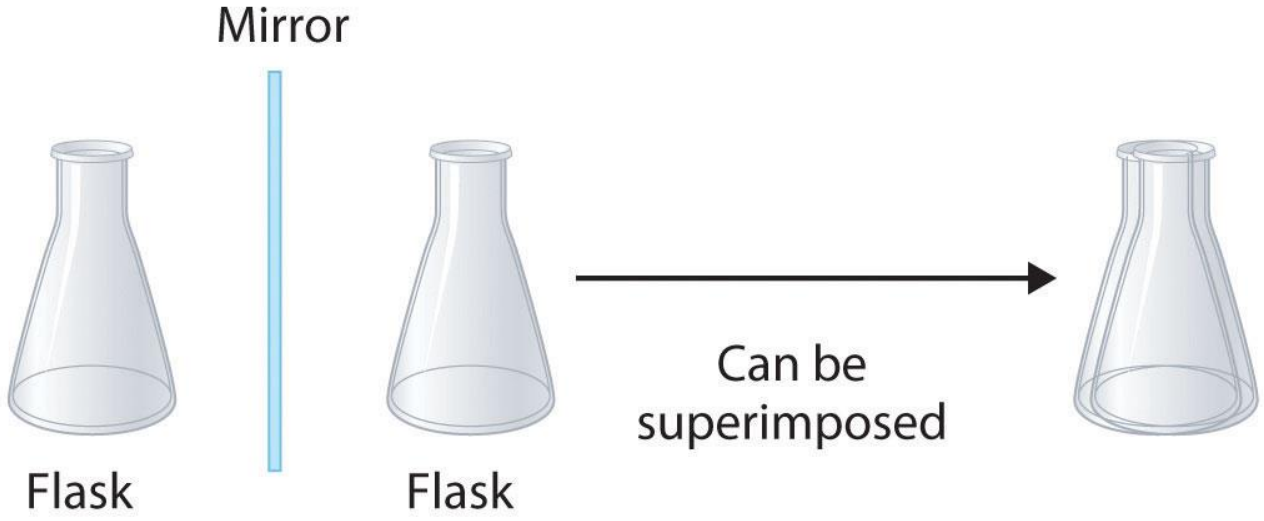
Chirality (Handedness)

- An object is "chiral" if and only if it is not super imposable on its mirror image





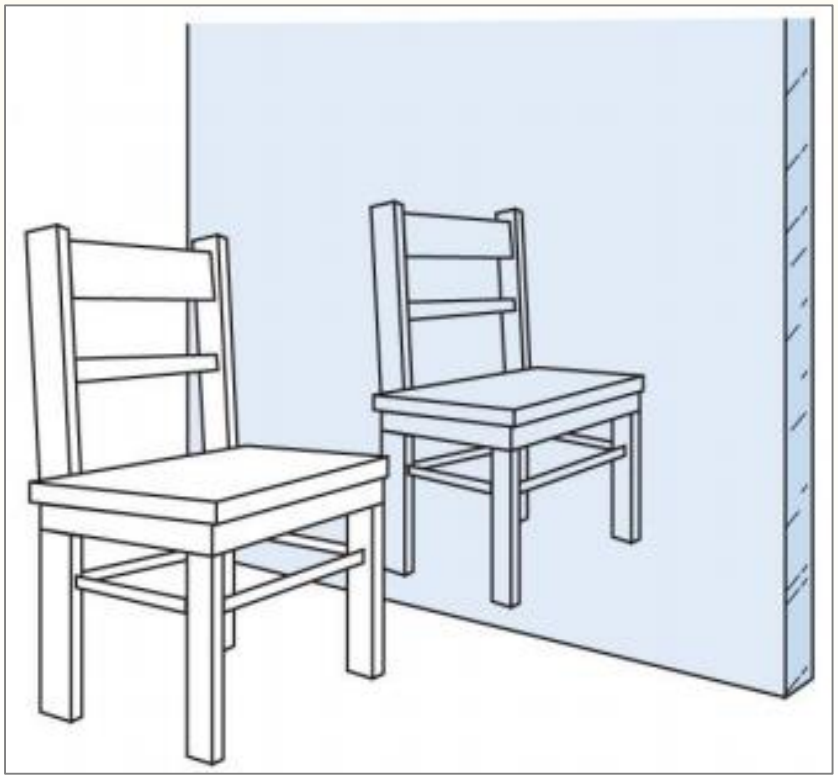
(a) Chiral objects



(b) Achiral objects

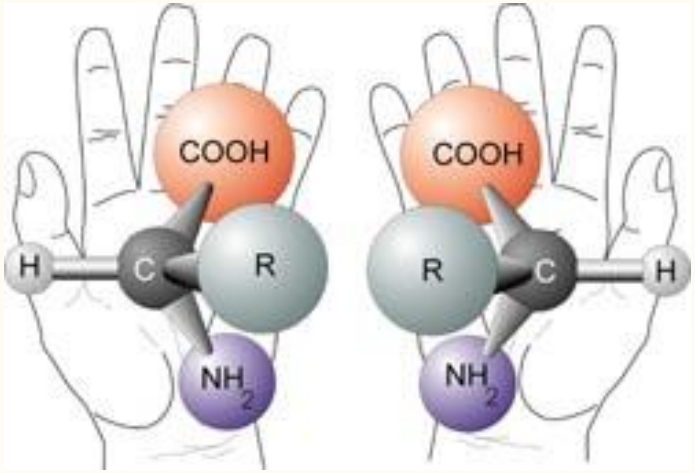
Chiral or Achiral?

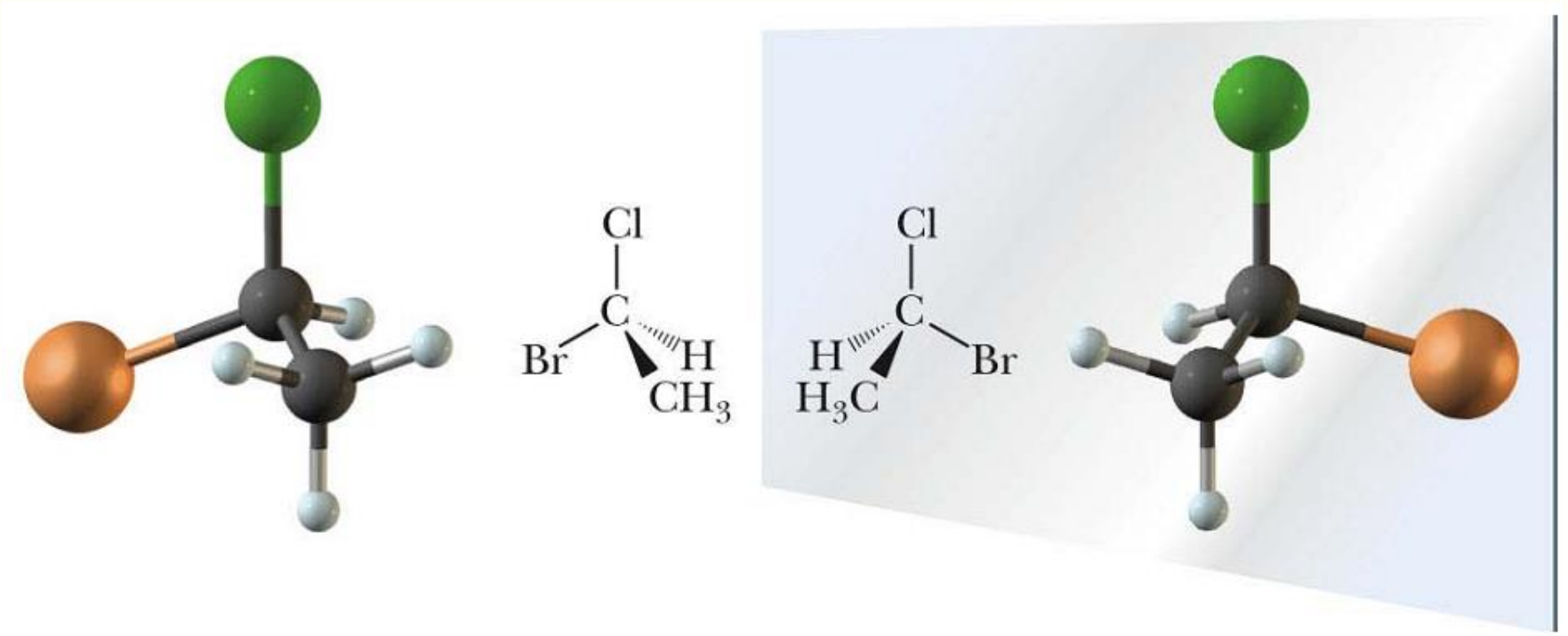


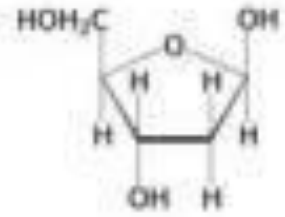


Chiral Molecules

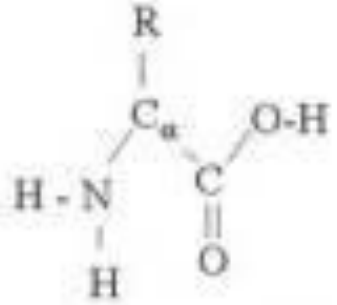
Molecules that are non-superimposable on their mirror image.



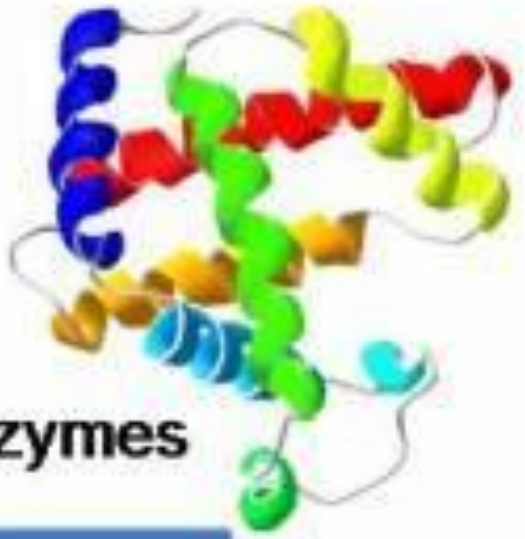




sugars



amino-acids

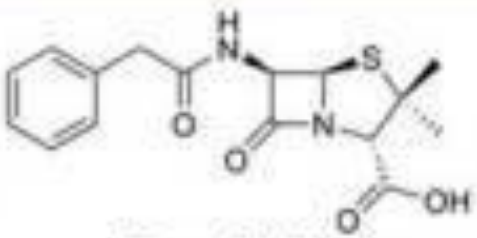


enzymes

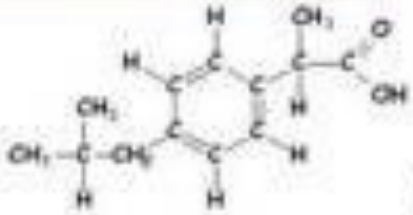
CHIRAL MOLECULES



DNA



Penicillin



Ibuprofen



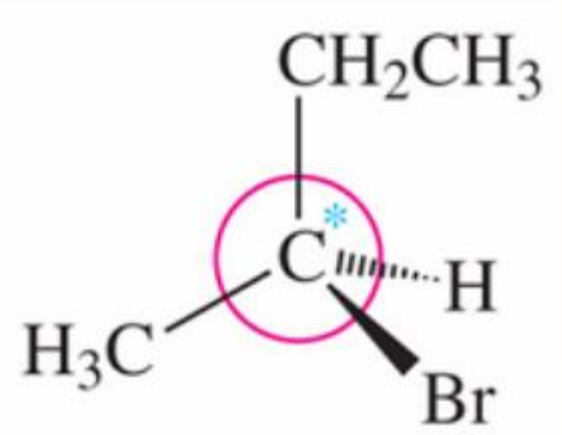
carbon nanotubes

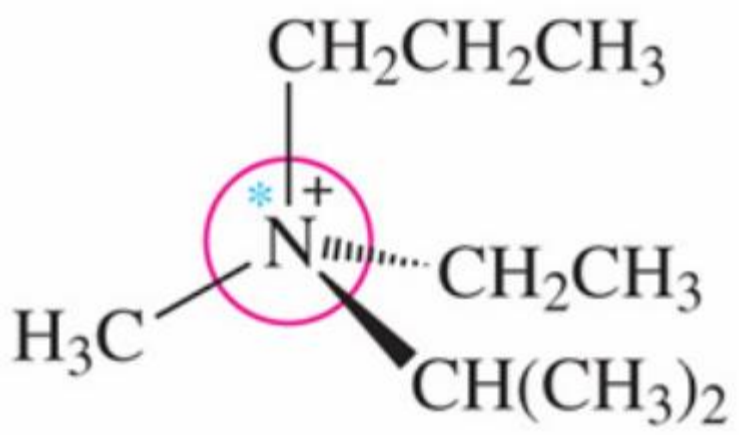
https://www.rowland.harvard.edu/rjf/fischer/background.php

How Can We Tell If A Molecule Is Chiral?

1. Look for a tetrahedral atom with 4 different substituents.

A tetrahedral atom that is bonded to 4 different substituents constitutes a stereocenter (chiral center or stereogenic center).

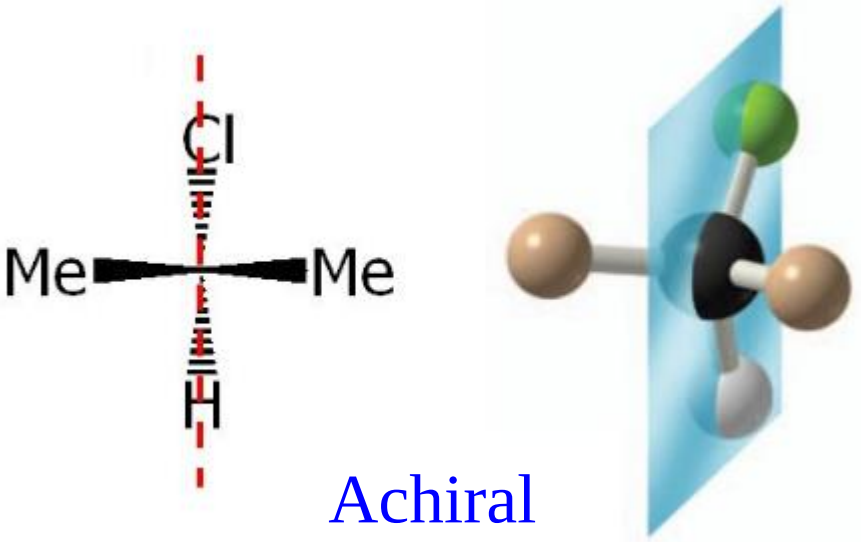




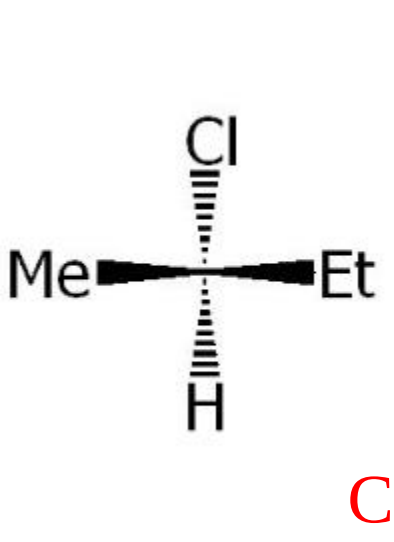
How Can We Tell If A Molecule Is Chiral?

2. Look for a plane of symmetry

- A plane of symmetry divides an object or molecule into two equal halves that are mirror images of each other.
- An object that has a plane of symmetry is superimposable on its mirror image and is achiral.
- A chiral molecule DOES NOT have a plane of symmetry.

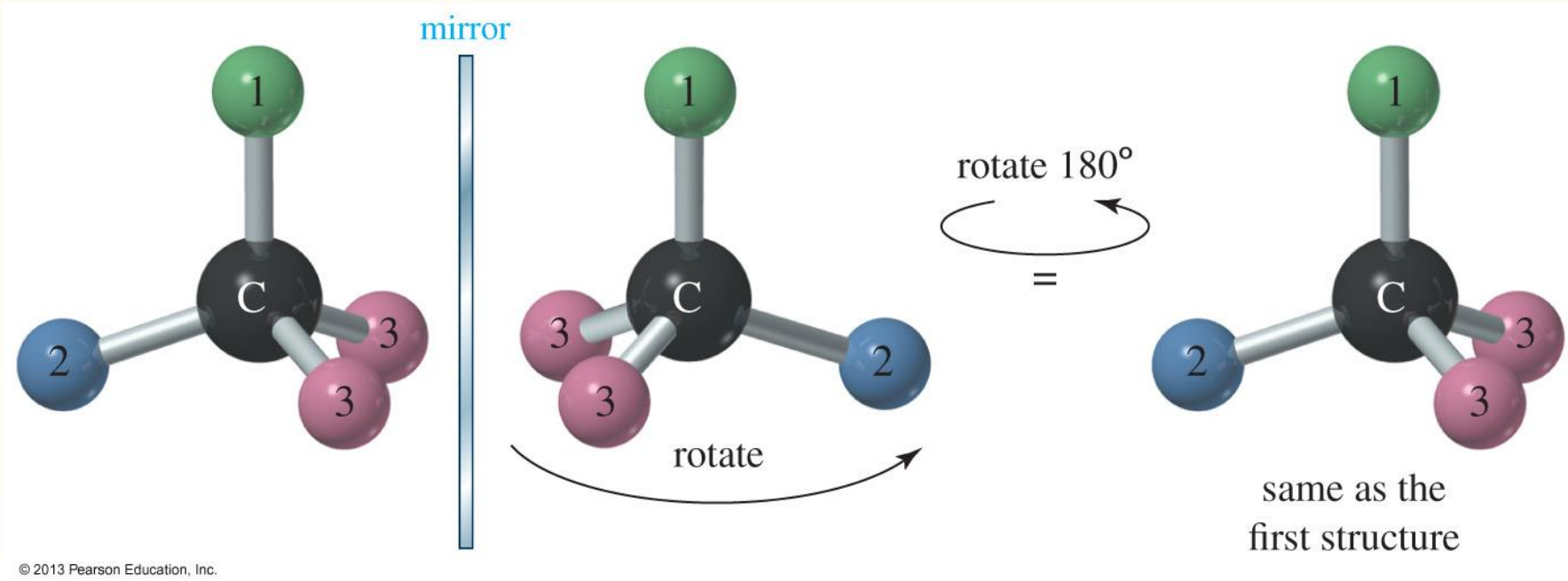


Achiral



Chiral

Achiral Compounds



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Take this mirror image and try to superimpose it on the one to the left matching all the atoms. Everything will match.

When the images can be superposed,
the compound is ***achiral***.



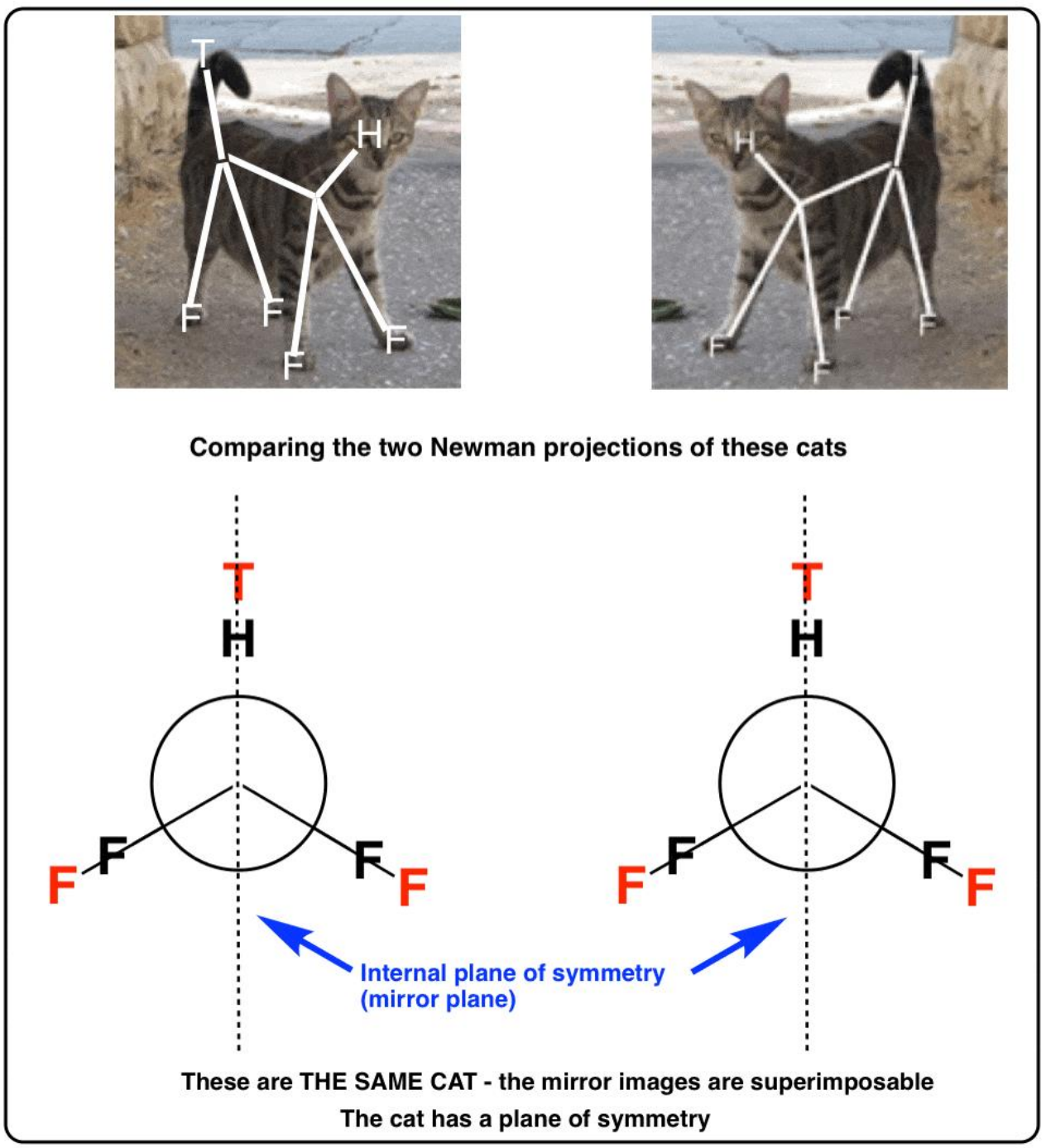


The Same Cat?

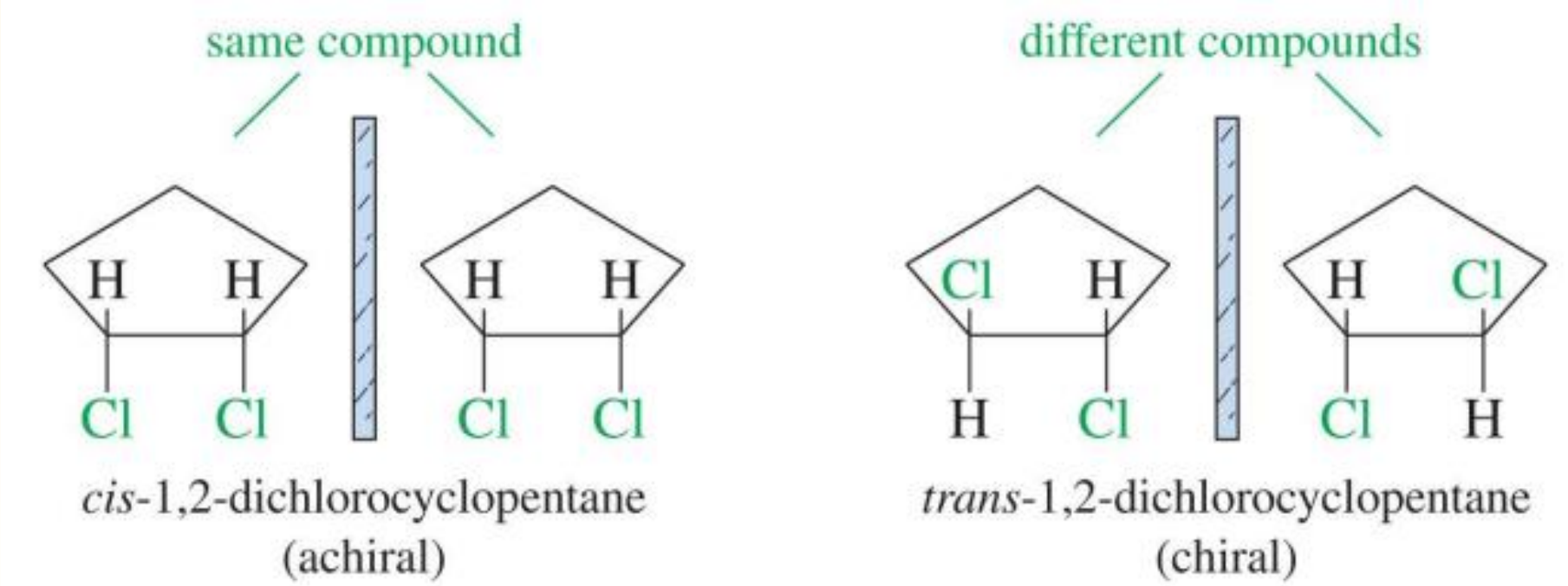
Looking at the Newman projections it's clear that they are completely identical.

Furthermore, if you look closely, you'll see that Sawhorse cat has an **internal mirror plane**, looking down the spine.

This means that Sawhorse Cat is superimposable on his mirror image.



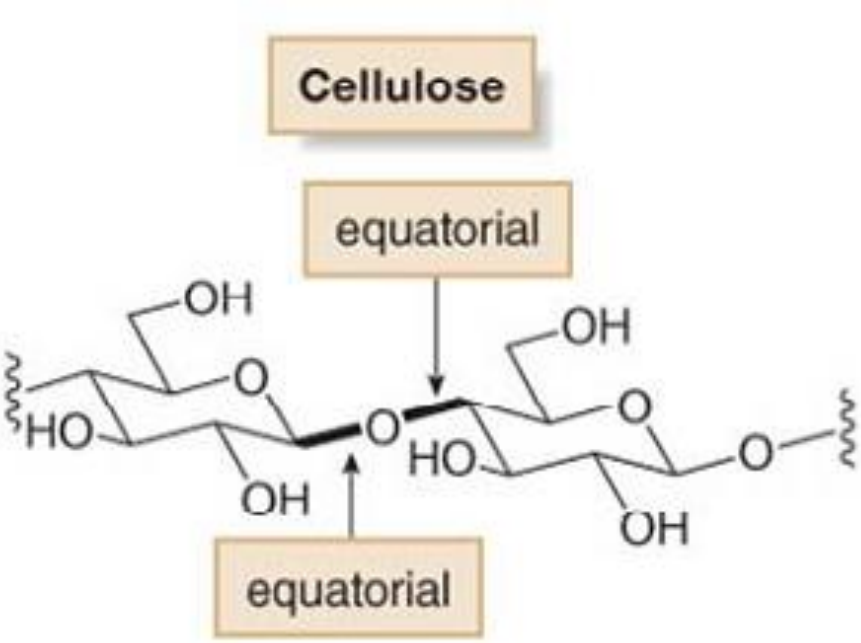
Chiral and Achiral Compounds

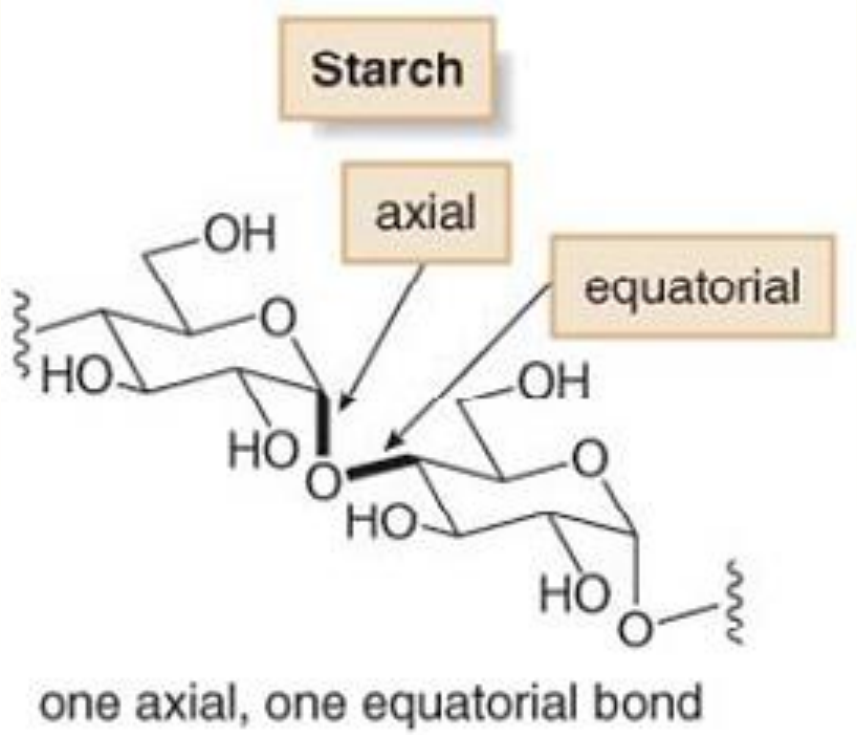


- The *cis* isomer has a superimposable mirror image – Achiral.
- The *trans* isomer is non-superimposable on its mirror image – Chiral.

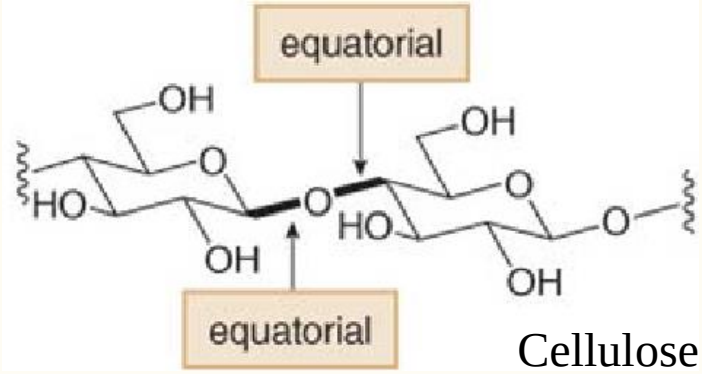
Why Stereochemistry Is So Important?

- Cellulose and starch are both composed of the same repeating unit, glucose, joined by an oxygen atom.
- The only difference is one ring carbon.
- In cellulose, it is equatorial, and in starch, it is axial

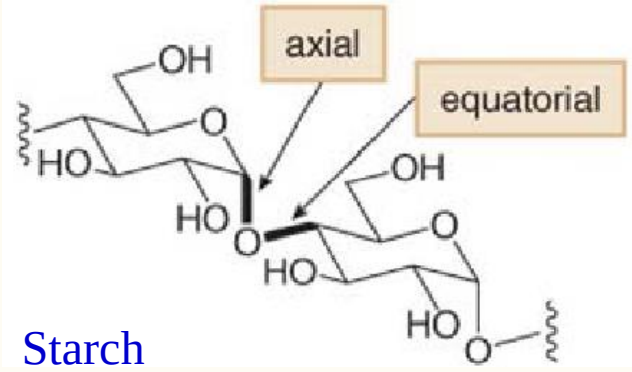




Why Stereochemistry Is So Important?

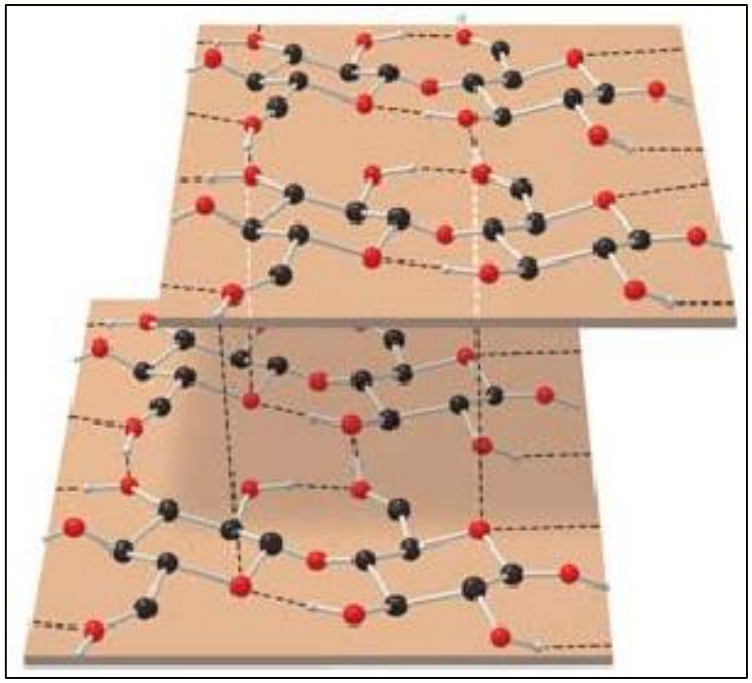


Cellulose

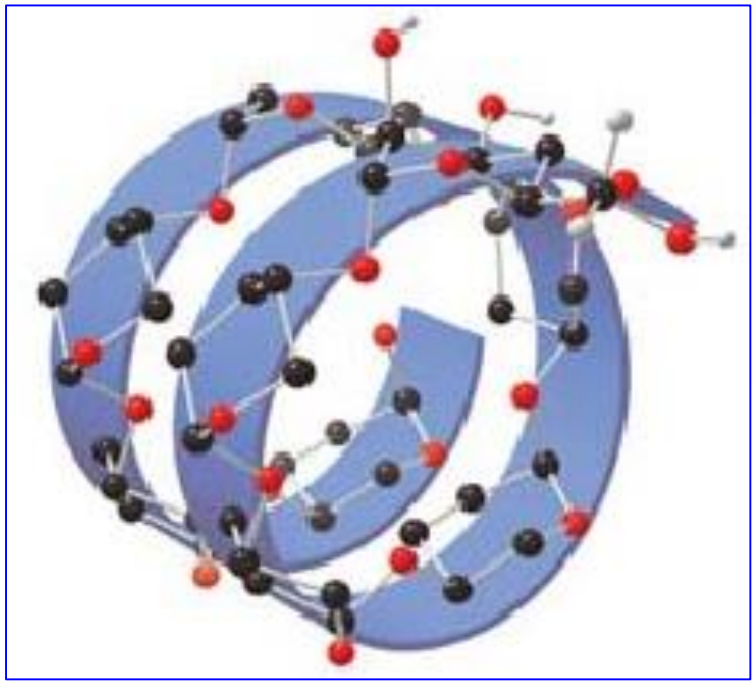


Starch

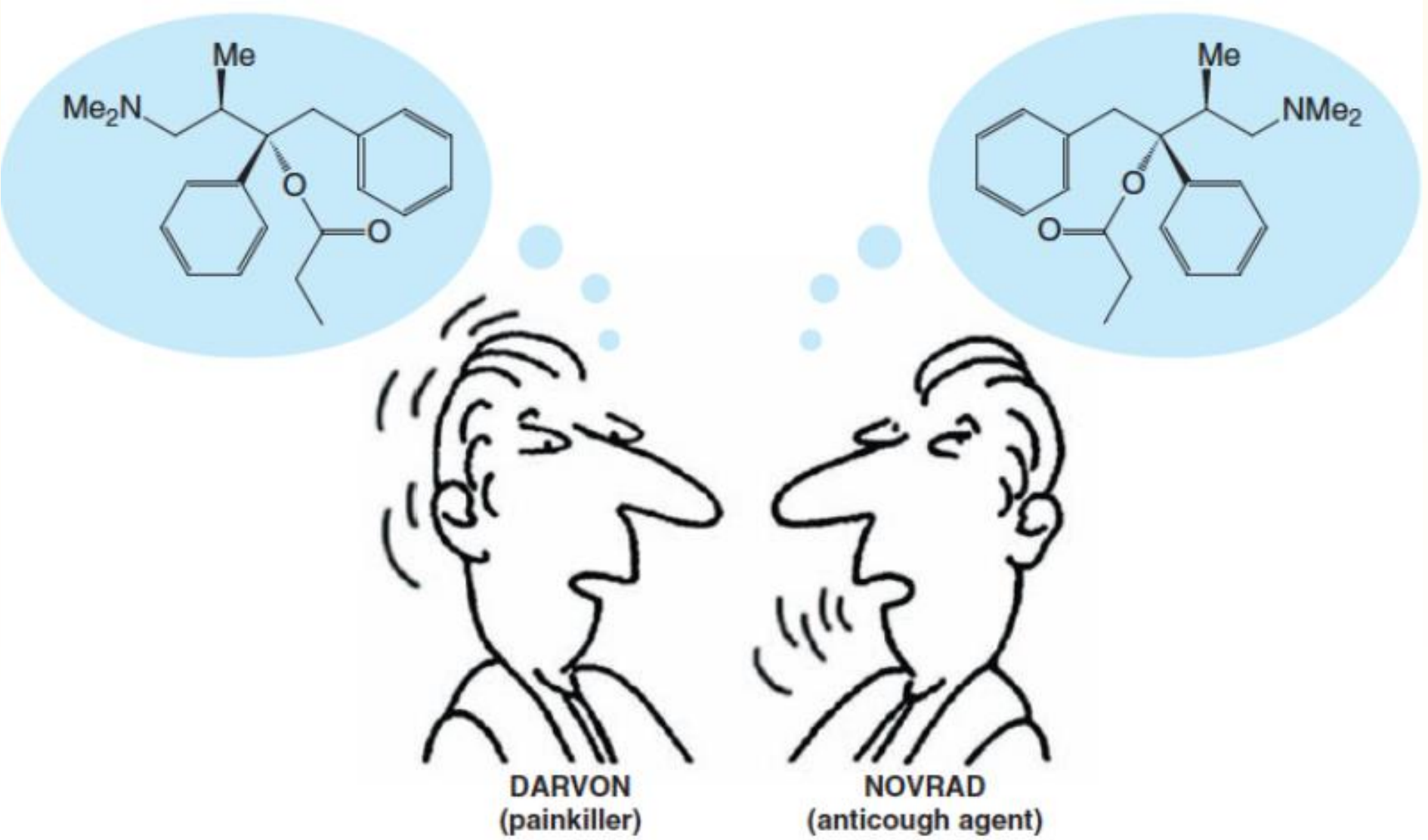
An extensive 3-D network



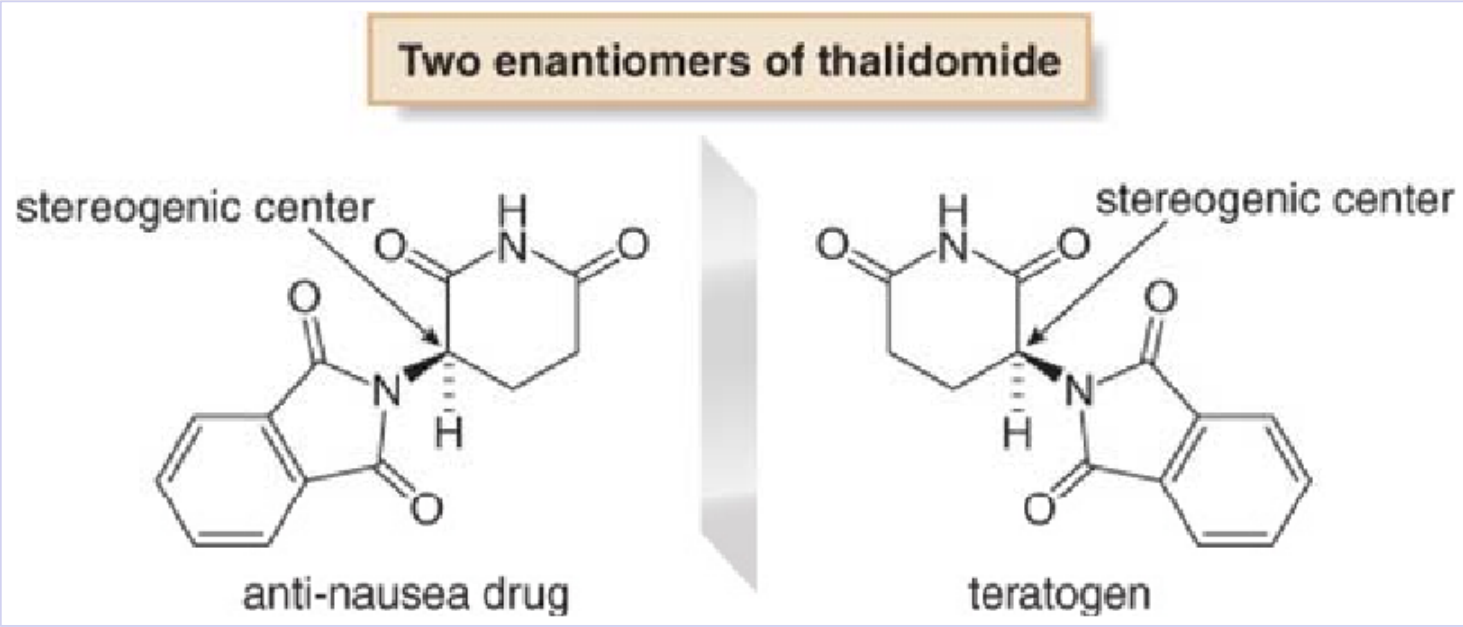
A helix structure



Why Stereochemistry Is So Important?



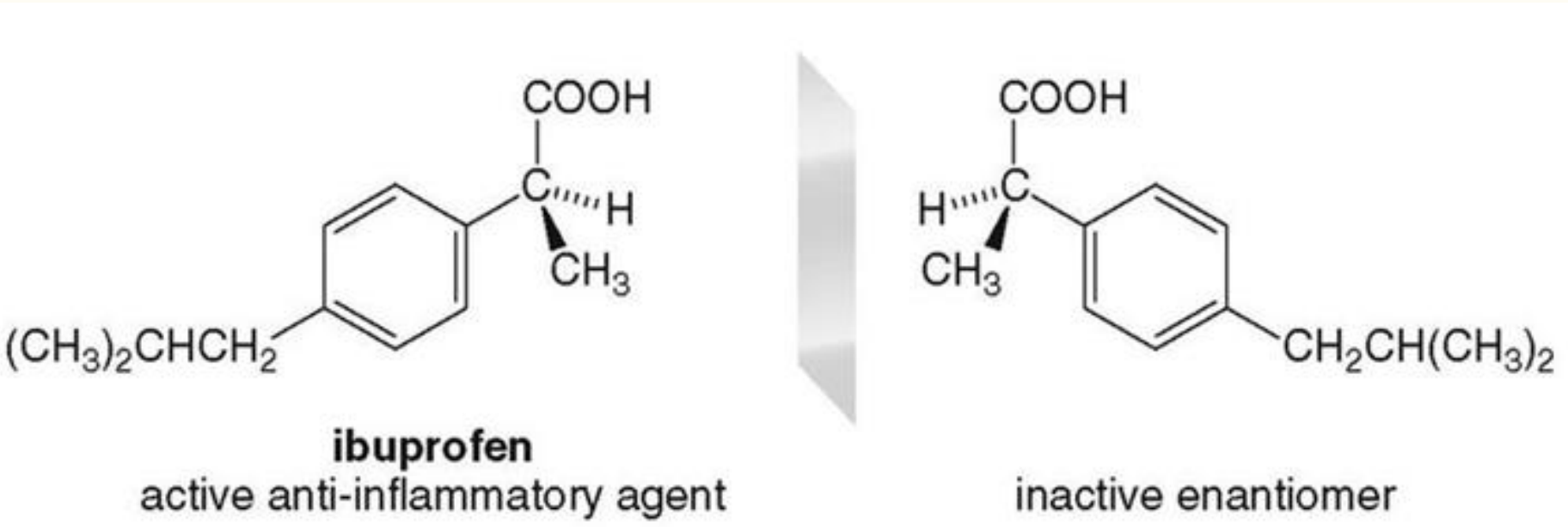
Why Stereochemistry Is So Important?



- **Thalidomide** was released in 1956 as a mild sedative used to treat nausea in pregnant women. (Withdrawn from the market in 1961 once it was discovered thalidomide was a human teratogen.)
- As little as one dose could cause a significant birth defect.
- Over 10,000 infants were born with birth defects to women who ingested thalidomide during pregnancy.

Why Stereochemistry Is So Important?

Ibuprofen is an active anti-inflammatory agent whose enantiomer is inactive, and it is sold as a racemic mixture.



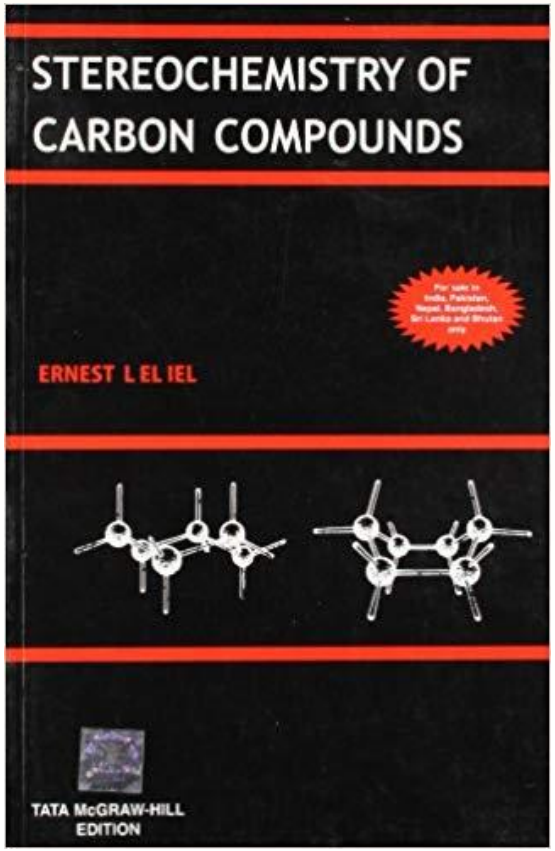
ibuprofen
active anti-inflammatory agent

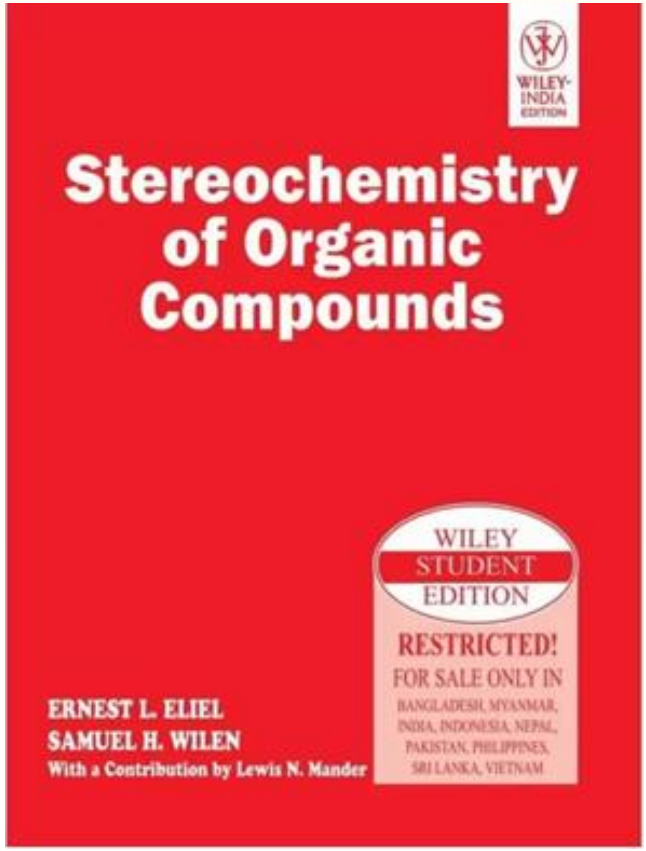
inactive enantiomer

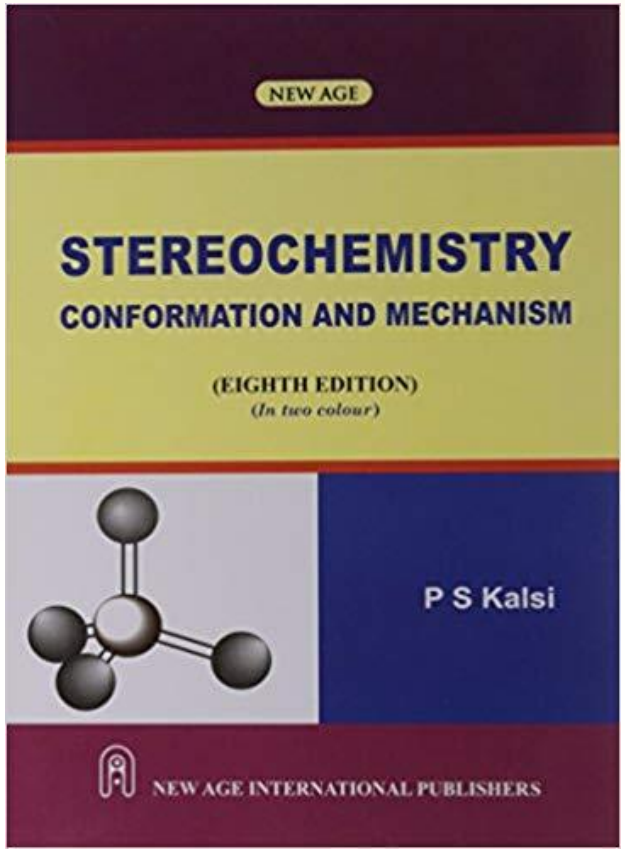
Smith. General Organic & Biological Chemistry 2nd Ed.

Suggested Readings

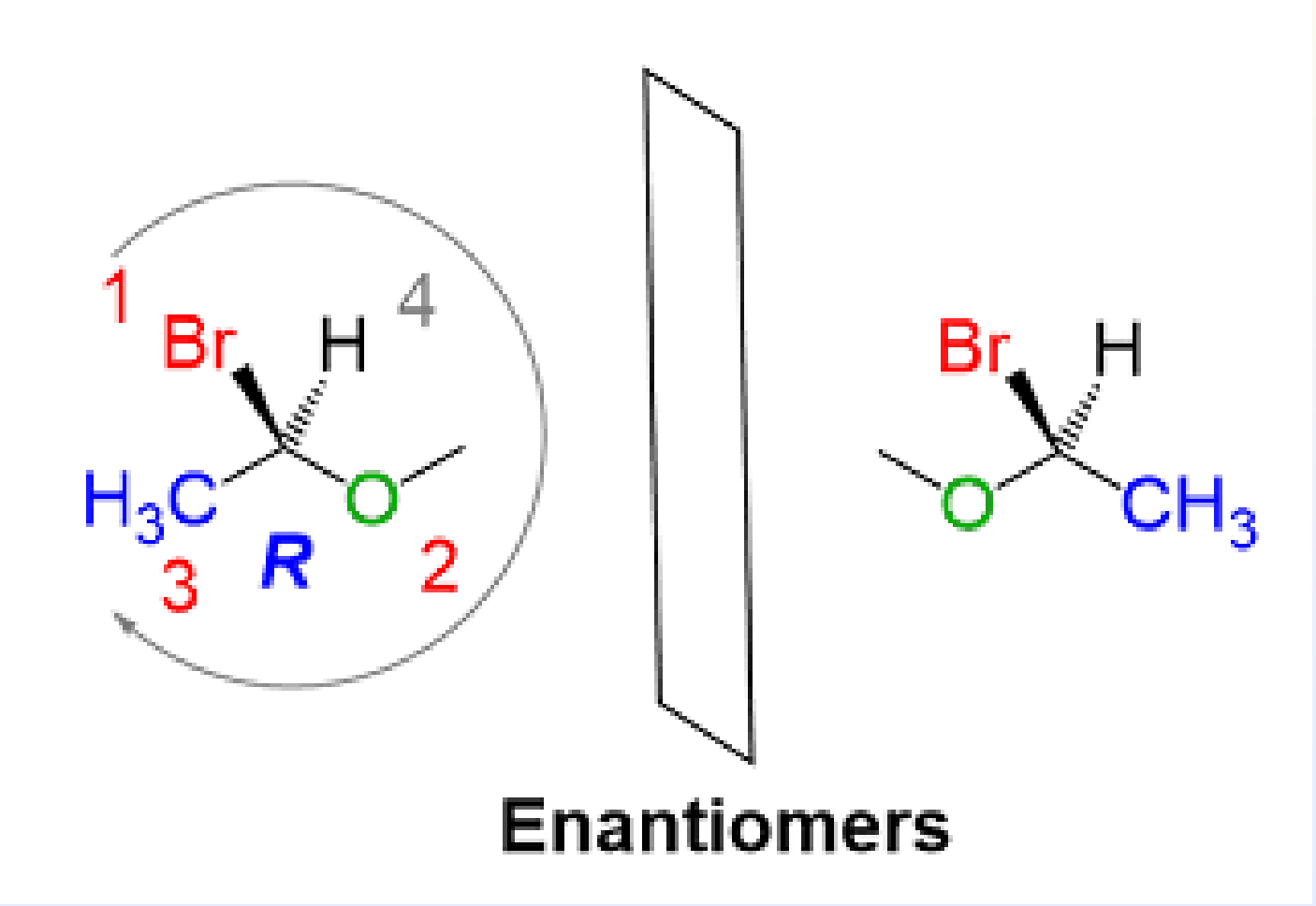
1. Stereochemistry of carbon compounds by E. L. Eliel.
2. Stereochemistry, conformation and Mechanism, P. S. Kalsi.
3. Organic Chemistry, I. L. Finar, Vol. 2, ELBS, Longman, Pearson.





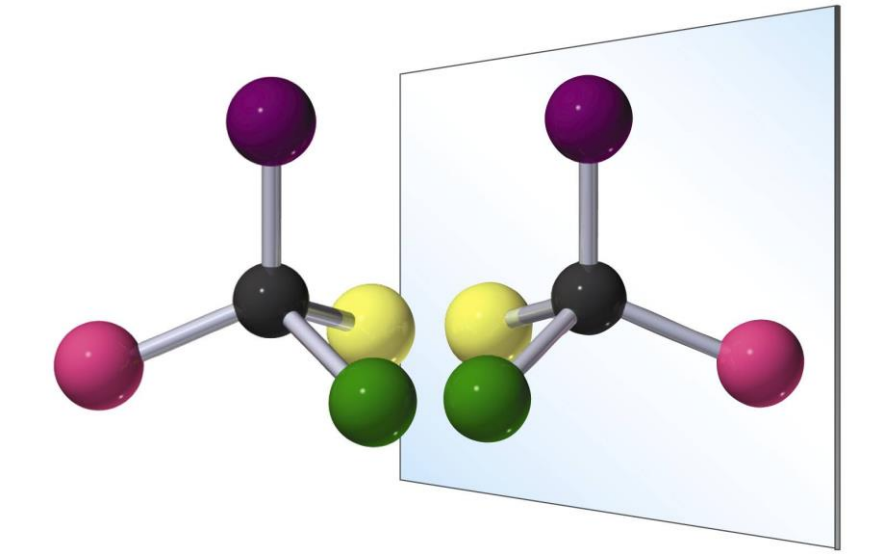


Advanced Stereochemistry



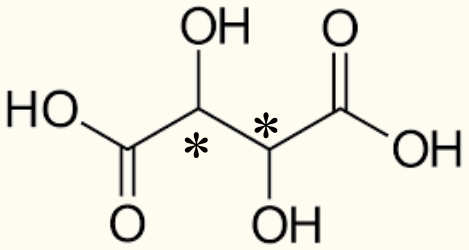
Properties Of Enantiomers

1. Enantiomers are chiral.
2. Pure sample of a single enantiomer is usually optically active.
3. Enantiomers have identical physical and chemical properties in an achiral environment.
4. Different biological properties.

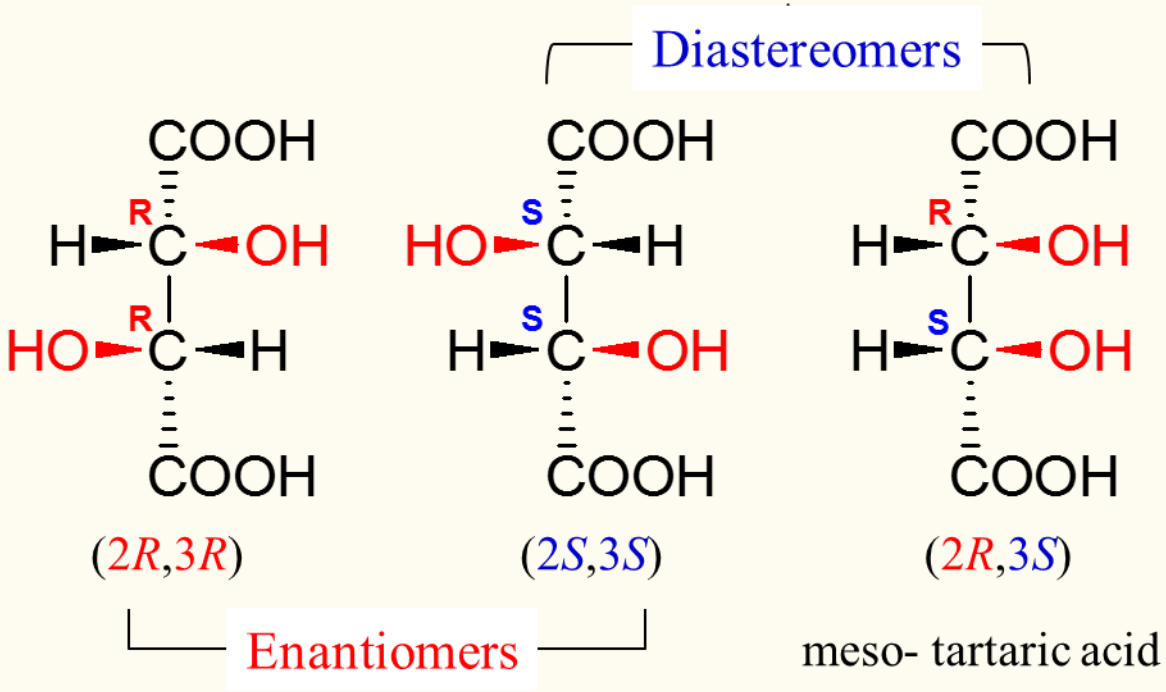


nonsuperimposable
mirror images

Properties Of Enantiomers



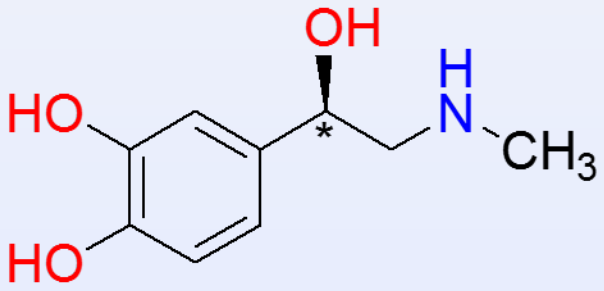
Tartaric acid



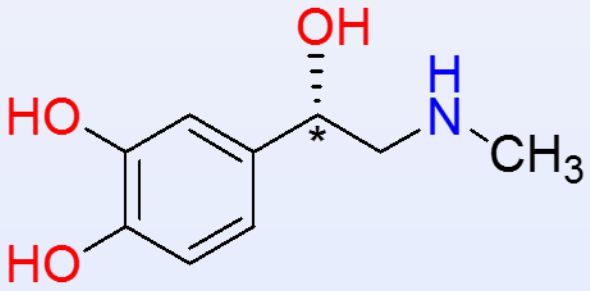
Specific rotation	+ 12.7	– 12.7	0
Melting point (°C)	171 - 174	171 - 174	146 - 148
Density at 20 °C (g/cm^3)	1.7598	1.7598	1.660
Solubility in water at 20 °C (g / 100 mL)	139	139	125
pK_1 (25 °C)	2.98	2.98	3.23
pK_1 (25 °C)	4.34	4.34	4.82

Biological Activity Of Enantiomers

- Enantiomers can have biological activities which are altogether different.
- One enantiomer of a biologically active molecule may cause a particular biological response, whereas the other enantiomer is usually completely inactive or less active.
- The hormone ***adrenaline*** mediates vasoconstriction (inducing increased blood pressure and heart rate), but the (*R*)-enantiomer is more active than the (*S*).



(*R*)- (–)-adrenaline



(*S*)- (+)-adrenaline

https://personalpages.manchester.ac.uk/staff/T.Wallace/20412tw2/chem20412_stereo_Lect4.htm

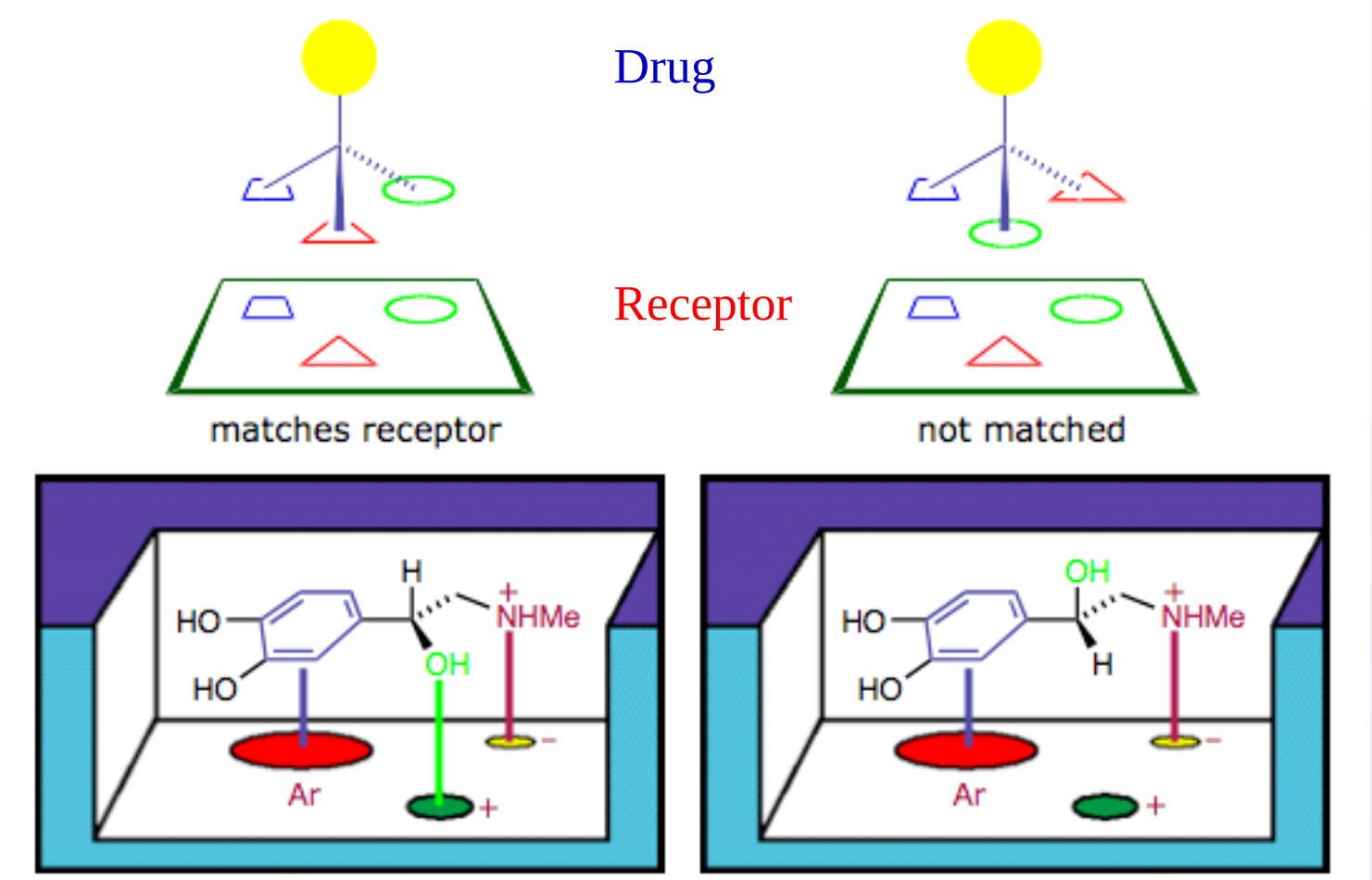
Biological Activity Of Enantiomers

Adrenaline or epinephrine is a hormone released into the body of someone feeling extreme emotions, which causes the person to have more energy. Adrenaline triggers the body's fight-or-flight response.



Biological Activity Of Enantiomers

Three Point Model for Drug Receptor Interaction



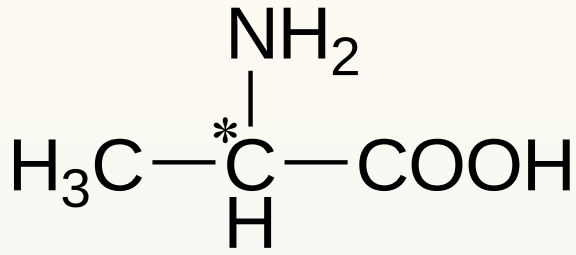
Representation of ***(R)*-adrenaline** binding to a receptor

The ***(S)*-adrenaline** has a lower binding energy

https://personalpages.manchester.ac.uk/staff/T.Wallace/20412tw2/chem20412_stereo_Lect4.htm

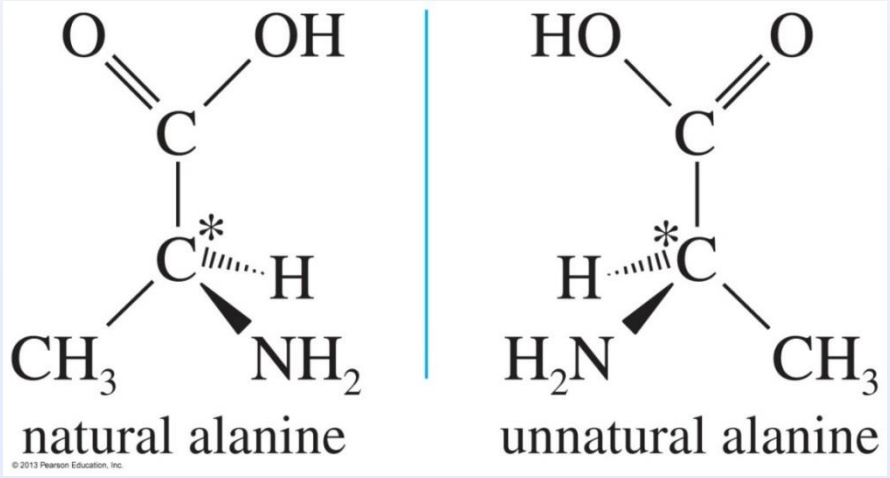
(*R*) and (*S*) Configuration

- Alanine has an asymmetric center, so it exists in two enantiomeric forms.
- Both enantiomers of alanine receive the same name in the IUPAC system: 2-aminopropanoic acid.
- Only one enantiomer is biologically active.
- A way to distinguish between them is to use stereochemical modifiers (R) and (S).



Alanine

2-aminopropanoic acid. (IUPAC name)



natural alanine

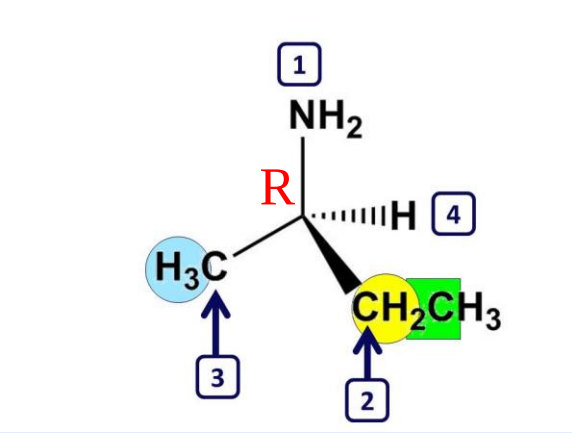
unnatural alanine

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Cahn–Ingold–Prelog Priority System

- Enantiomers have different spatial arrangements of the four groups attached to the asymmetric carbon.
- The two possible spatial arrangements are called **configurations**.
- Each asymmetric carbon atom is assigned a letter (*R*) or (*S*) based on its three-dimensional configuration.
- **Cahn–Ingold–Prelog** convention is the most widely accepted system for naming the configurations of chirality centers.

An **absolute configuration** refers to the spatial arrangement of atoms or groups in a chemical compound about an asymmetric atom and its stereochemical description *e.g.* *R* or *S*.



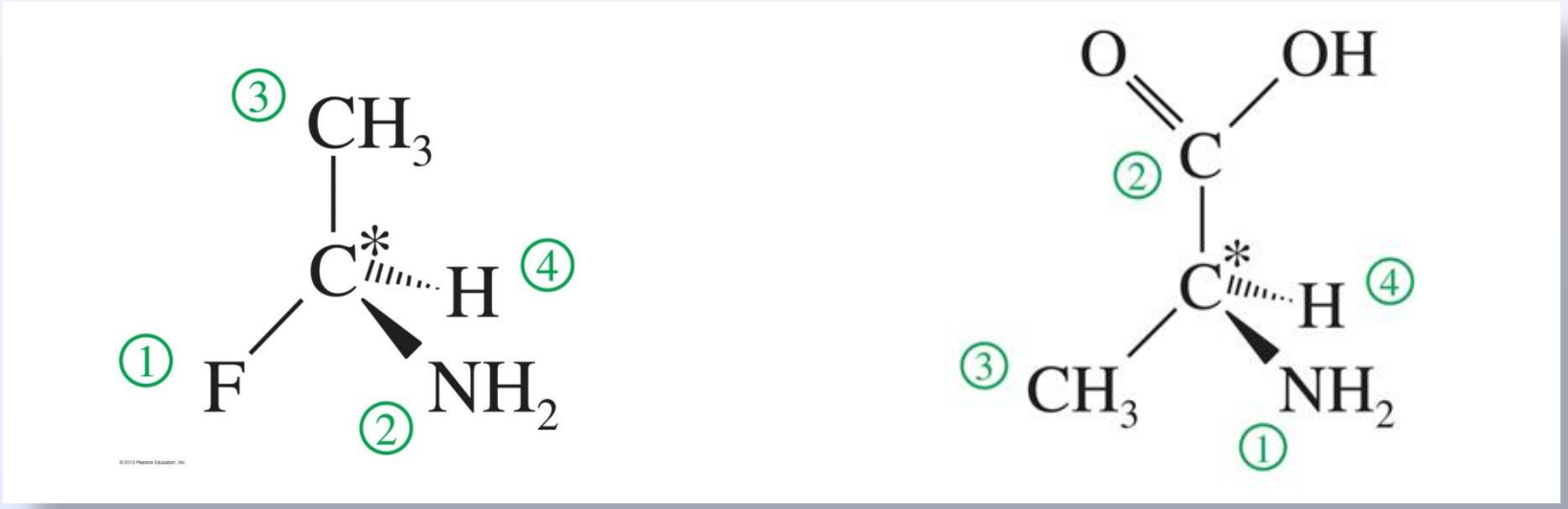
Assigning R & S Stereocenters

1. **Assign priority to the four atoms attached to the stereocenters**

   a) Each atom is assigned a priority based on atomic number. Heavier isotopes get higher priority.

(1) —H, (6) —CH_3, (7) —NH_2, (8) —OH, (16) —SH, (17) —Cl, (35) —Br, (53) —I

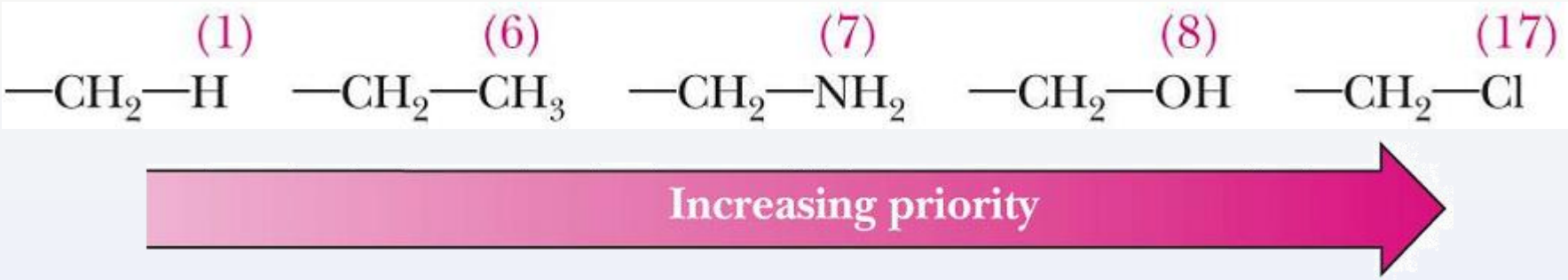
Increasing priority

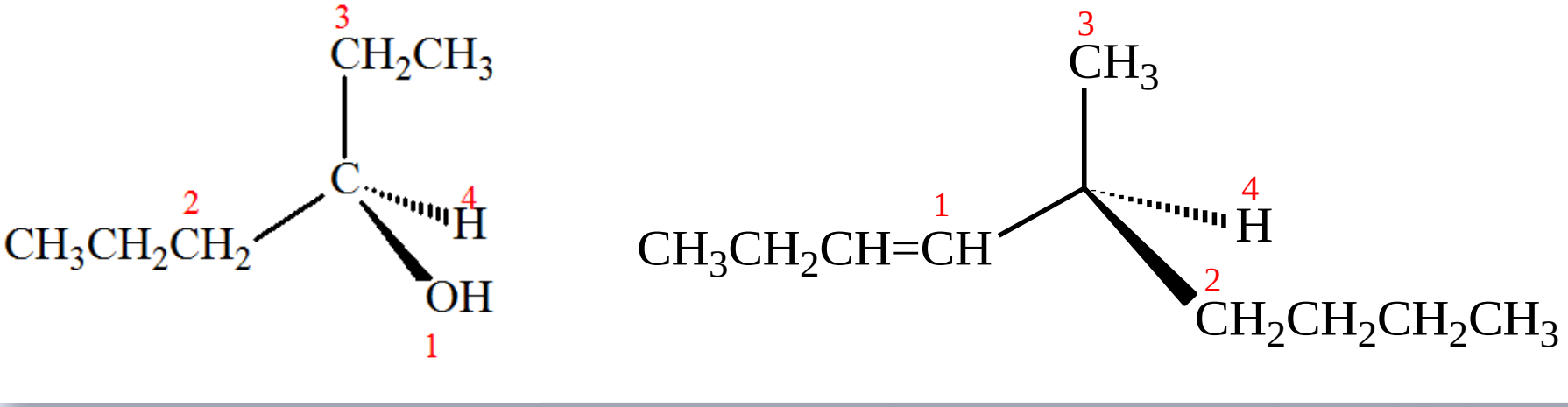


Assigning R & S Stereocenters

1. Assign priority to the four atoms attached to the stereocenters

b) If priority cannot be assigned based solely on atomic number, then look at the next atom attached. Priority is assigned by the first point of difference.

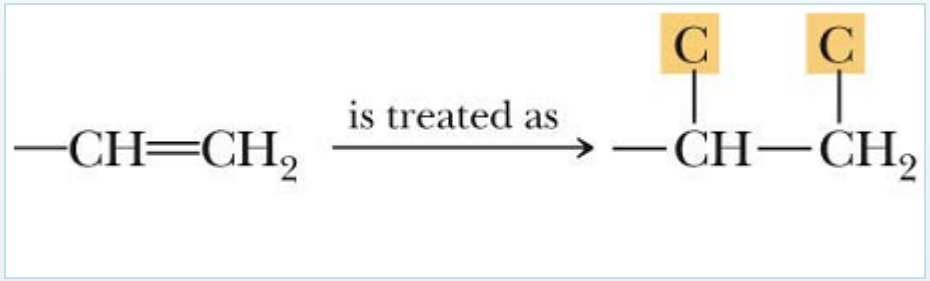


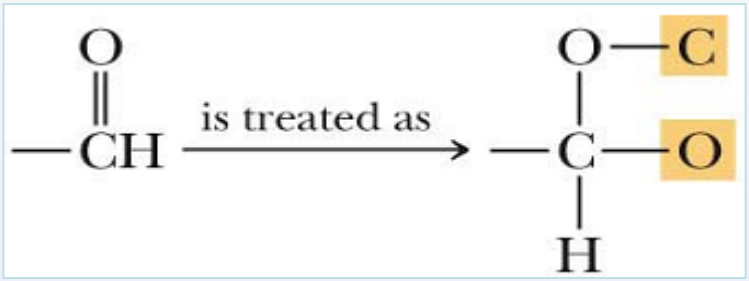


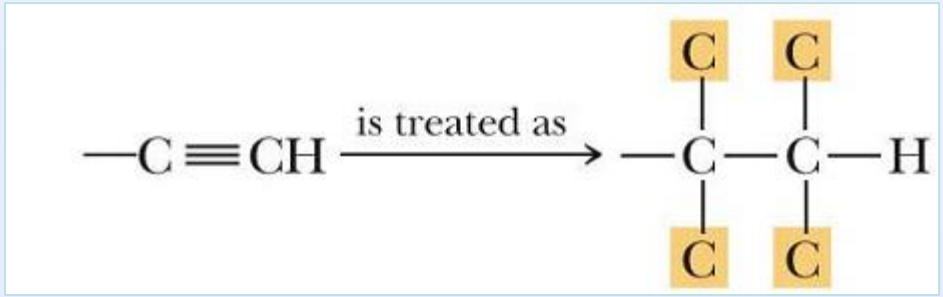
Assigning R & S Stereocenters

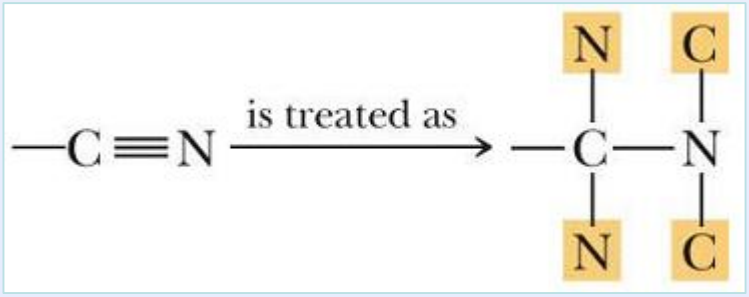
1. Assign priority to the four atoms attached to the stereocenters

c) Atoms participating in multiple bonds are considered to be bonded to the equivalent number of similar atoms by a single bond.



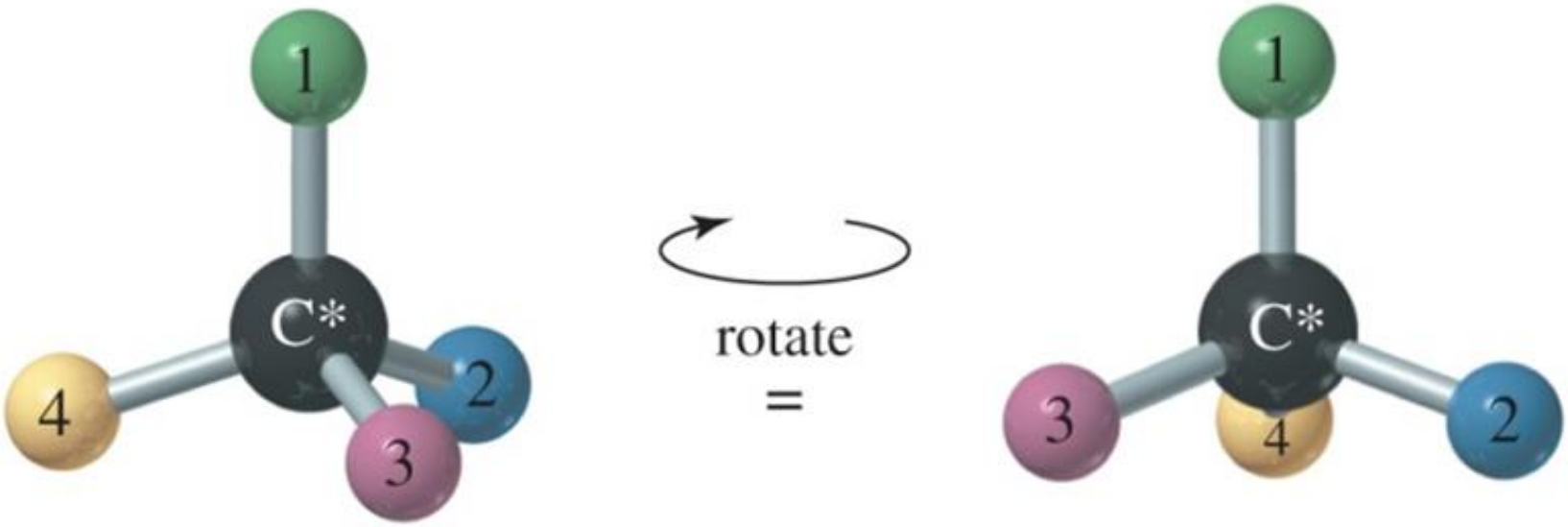






Assigning R & S Stereocenters

2. **Orient the molecule in space so that the lowest priority is directed away from you.**

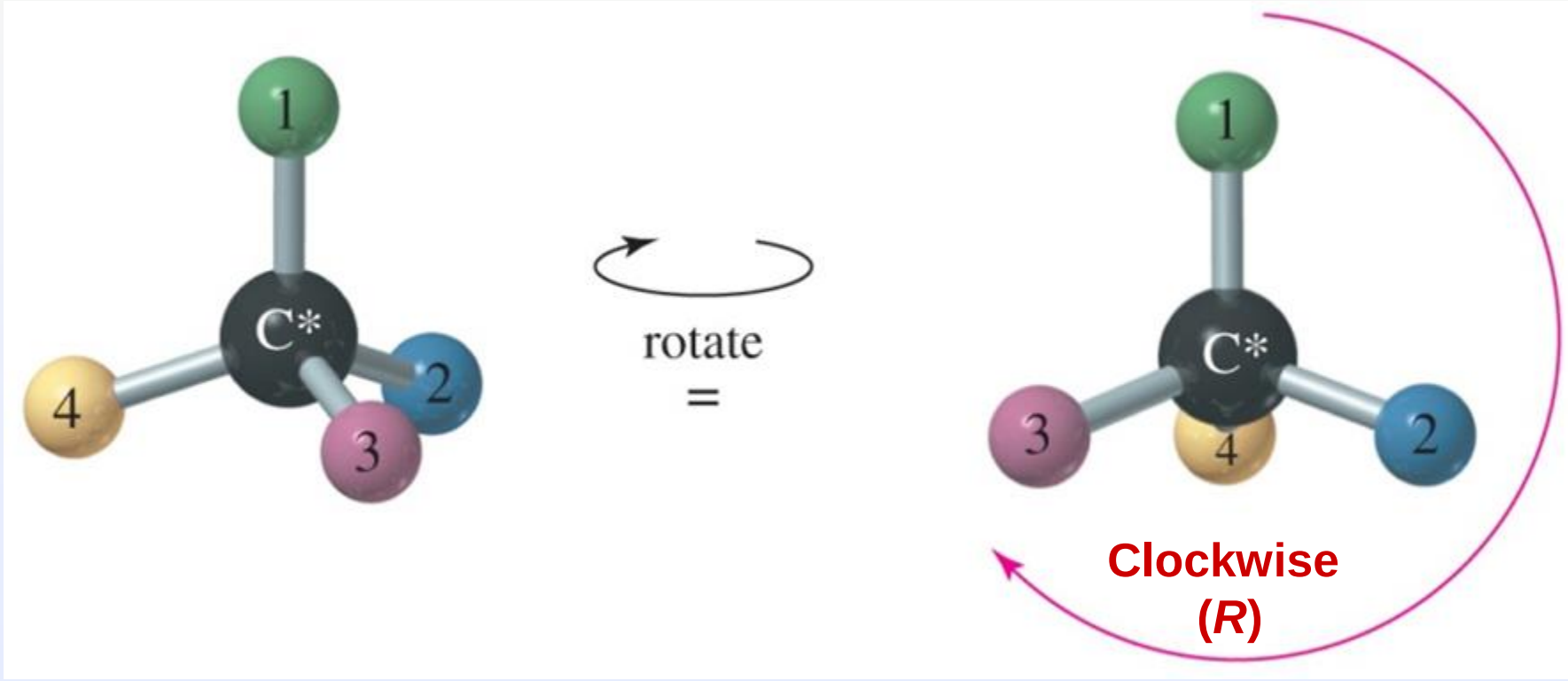


When rotating to put the lowest priority group in the back, keep one group in place and rotate the other three.

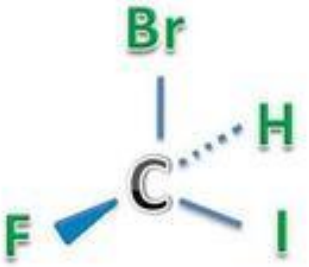
Assigning R & S Stereocenters

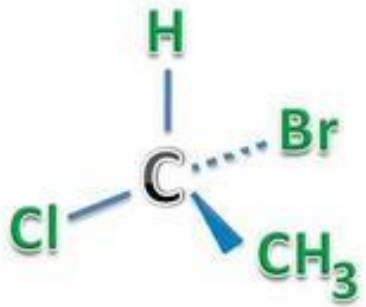
3. "Read" the groups projected toward you in order from highest to lowest.

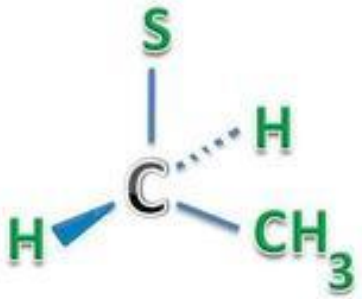
Groups in a clockwise fashion → **R** (R stands for Latin, *rectus*)
Groups in a counterclockwise fashion → **S** (S stands for Latin *sinister*)

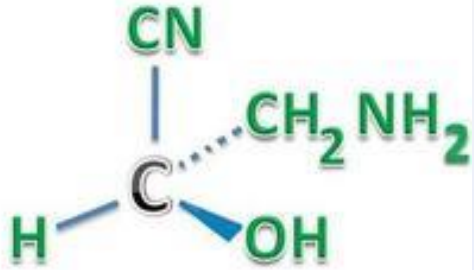


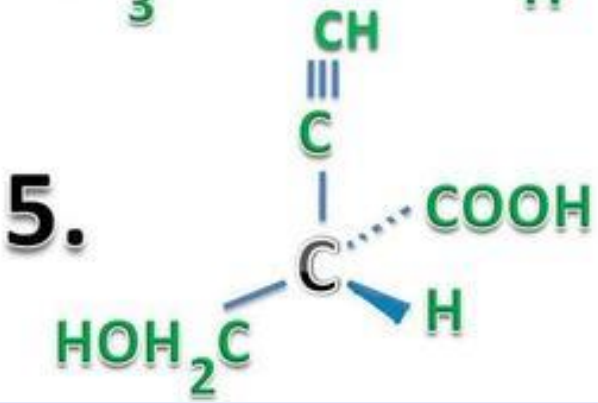
Assign (R) Or (S) Configuration To The Following Compounds

1. 

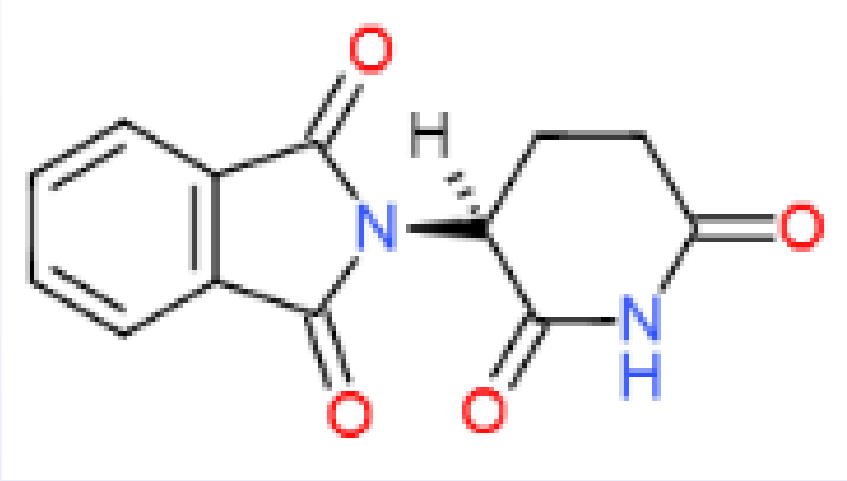
2. 

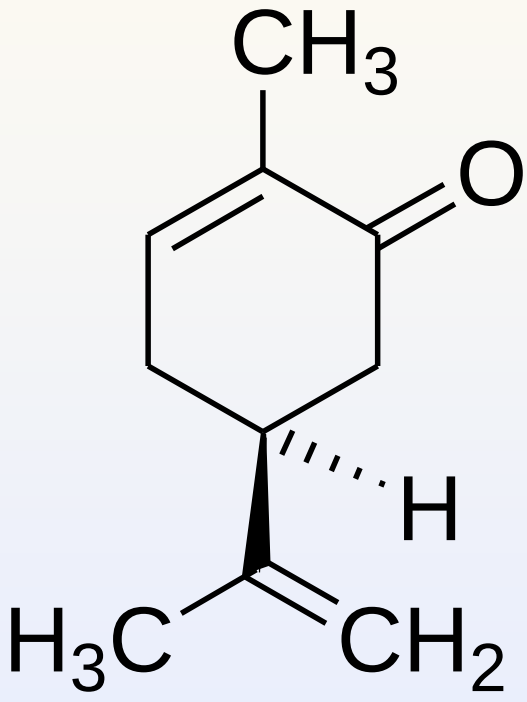
3. 

4. 

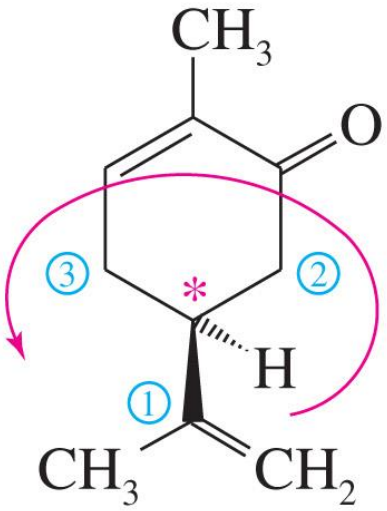
5. 

Configuration In Cyclic Compounds





Configuration In Cyclic Compounds



(*S*)-carvone

Group 1: C* — C(C)(C) — CH_2, with CH_3 below

Group 2: C* — CH_2 — C(O)(C) — O, with C below

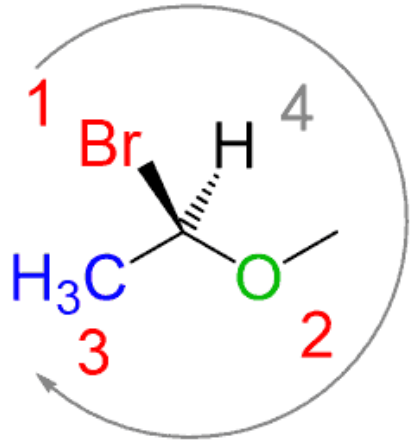
Group 3: C* — CH_2 — C(C) — C(C) — C=O, with H and CH_3 below

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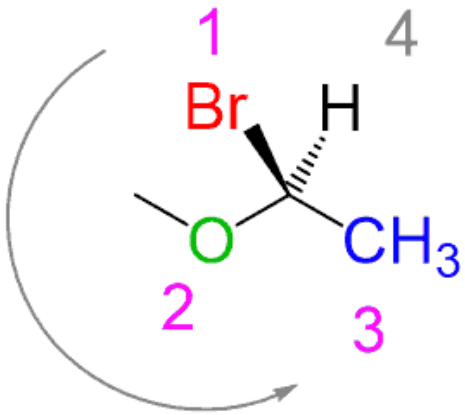
Advanced Stereochemistry

(*R*) and (*S*) Configuration

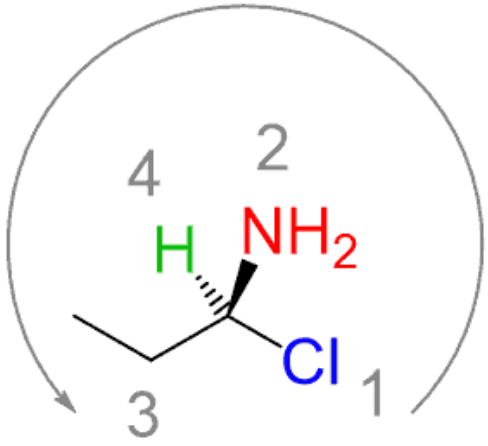
The lowest priority (in this case the H) must be pointing back:



R



S



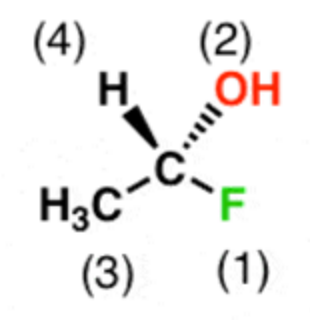
S

There are, however, different ways to determine the configuration if the lowest priority is not pointing back.

https://www.chemistrysteps.com/how-to-determine-r-and-s-configuration/

Determining R/S Configuration When the no. 4 Substituent is in the Front (i.e. on a "Wedge")

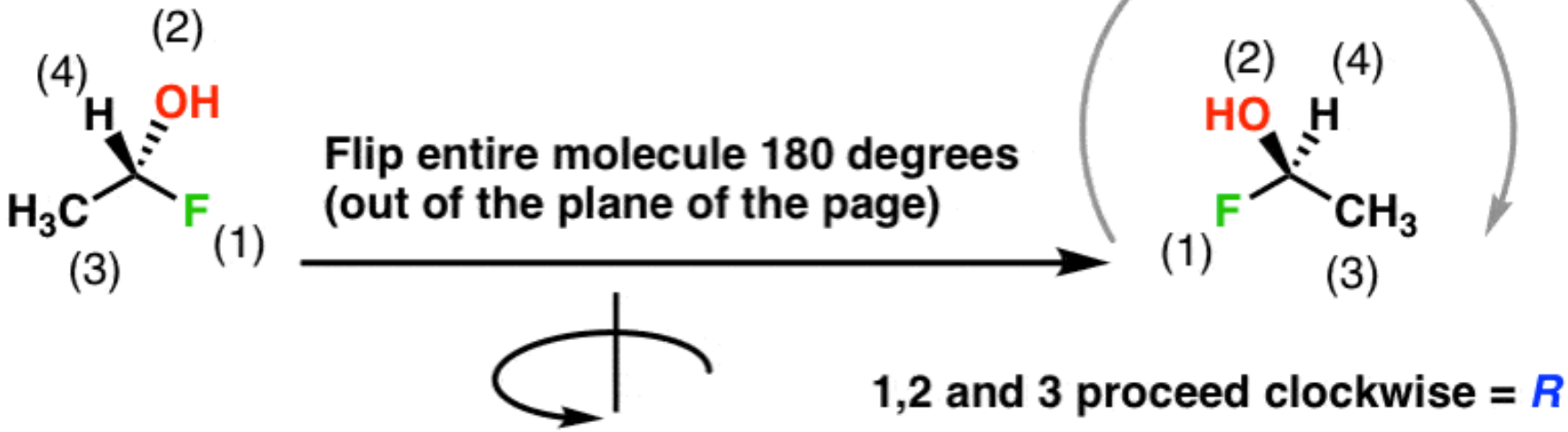
1-Fluoroethanol



How do we determine (R)/(S) in this case? There are many ways to do it.

Determining (R)/(S) When The #4 Group Is On A Wedge

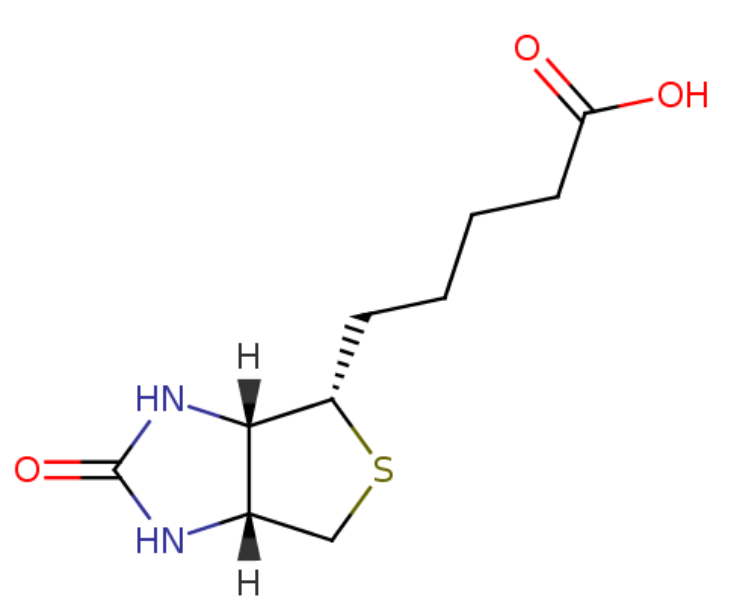
1st Rule: Rotation



https://www.masterorganicchemistry.com/2016/10/20/introduction-to-assigning-r-and-s-the-cahn-ingold-prelog-rules/

Determining R/S Configuration When the no. 4 Substituent is in the Front (i.e. on a "Wedge")

- However, it is not easy to rotate the molecule in your head so that the #4 priority is on a dash.
- Look at biotin with all these hydrogens pointing forward.
- Not the best option to redraw this molecule changing all the hydrogens and keeping the rest of the molecule as it should be.



- Thankfully, it is technically unnecessary to perform such a mental rotation!

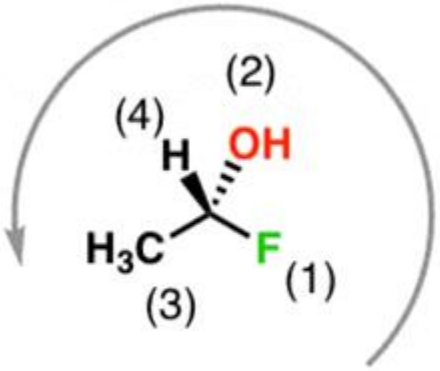
Determining R/S Configuration When the no. 4 Substituent is in the Front (i.e. on a "Wedge")

- When the #4 priority is on a wedge, we can apply the second rule.

Determining (R)/(S) When The #4 Group Is On A Wedge

2nd Rule: Apply the "Opposite Rule"

We can get the same result if we just apply the opposite of the normal (R)/(S) rules.



1,2 and 3 proceed counter-clockwise. Looks like (S)

However, group #4 (H) is pointing out of the plane of the page

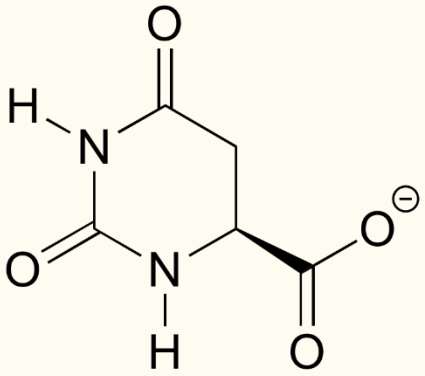
Therefore, flip the result

So in this case, flip (S) to give us the "real" configuration, R

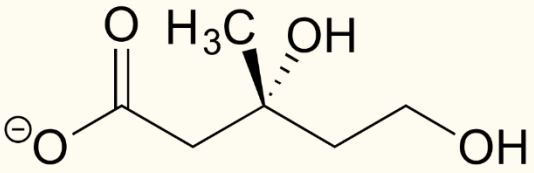
This is almost always easier than rotating the molecule

https://www.masterorganicchemistry.com/2016/10/20/introduction-to-assigning-r-and-s-the-cahn-ingold-prelog-rules/

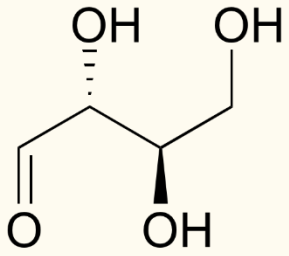
Determine the stereochemical configurations of the chiral centers in the biomolecules shown below:



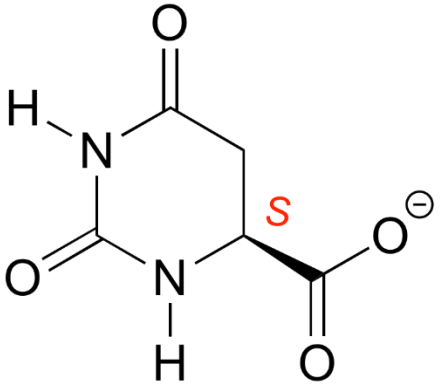
dihydroorotate



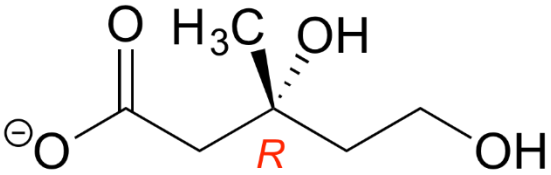
mevalonate



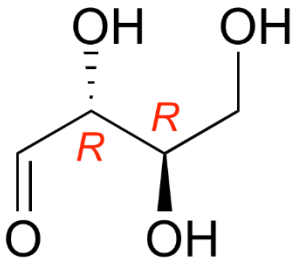
(D)-erythrose



dihydroorotate

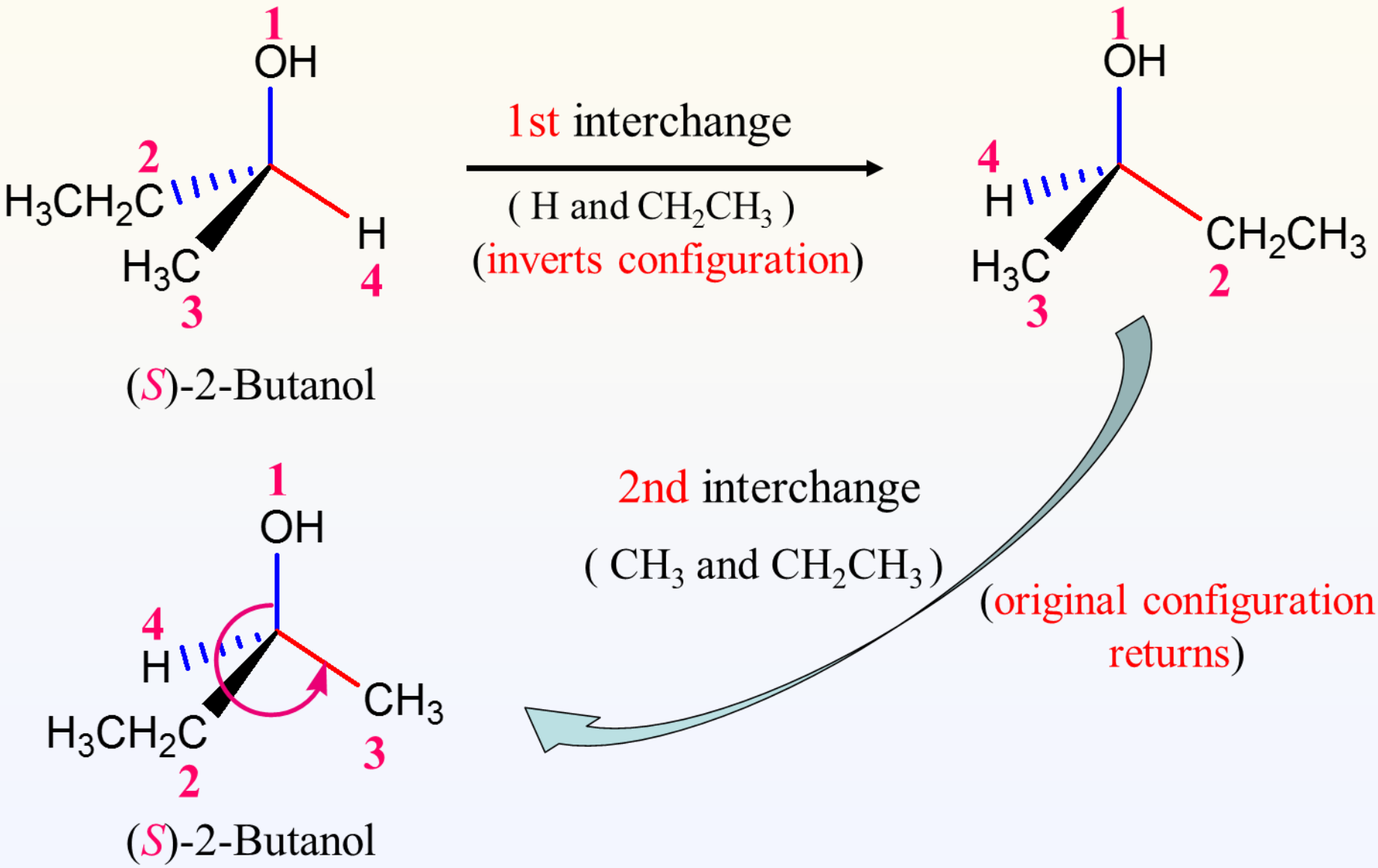


mevalonate

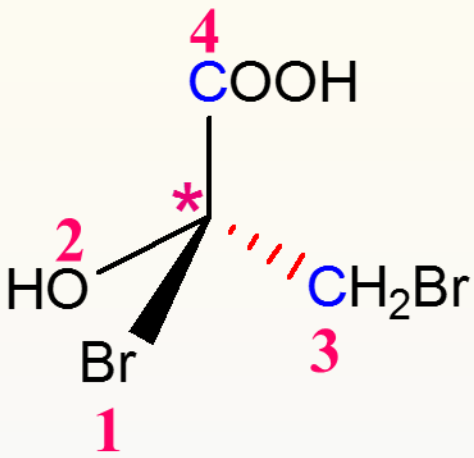


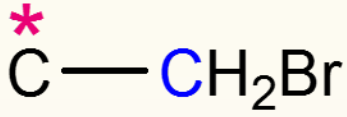
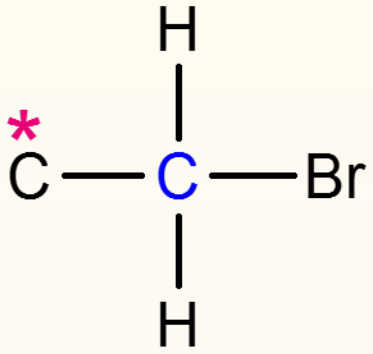
(D)-erythrose

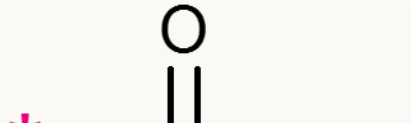
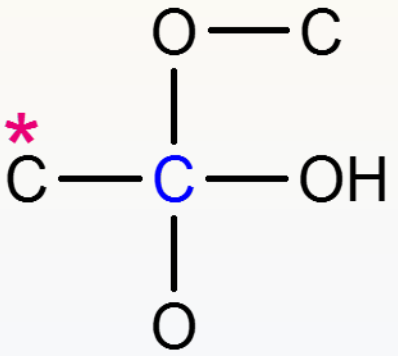
Determining R/S Configuration When the no. 4 Substituent is in the Plane



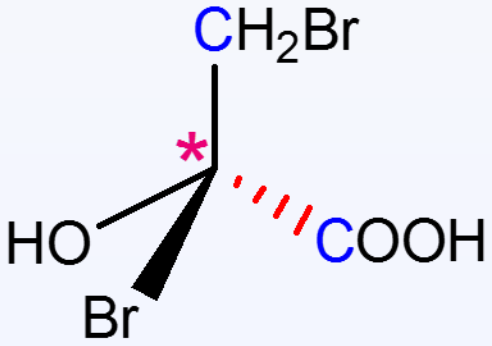
Practice Problem



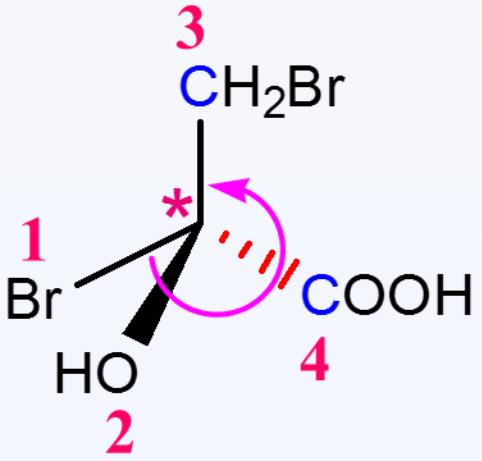
Group 3:  ≡ 

Group 4:  ≡ 

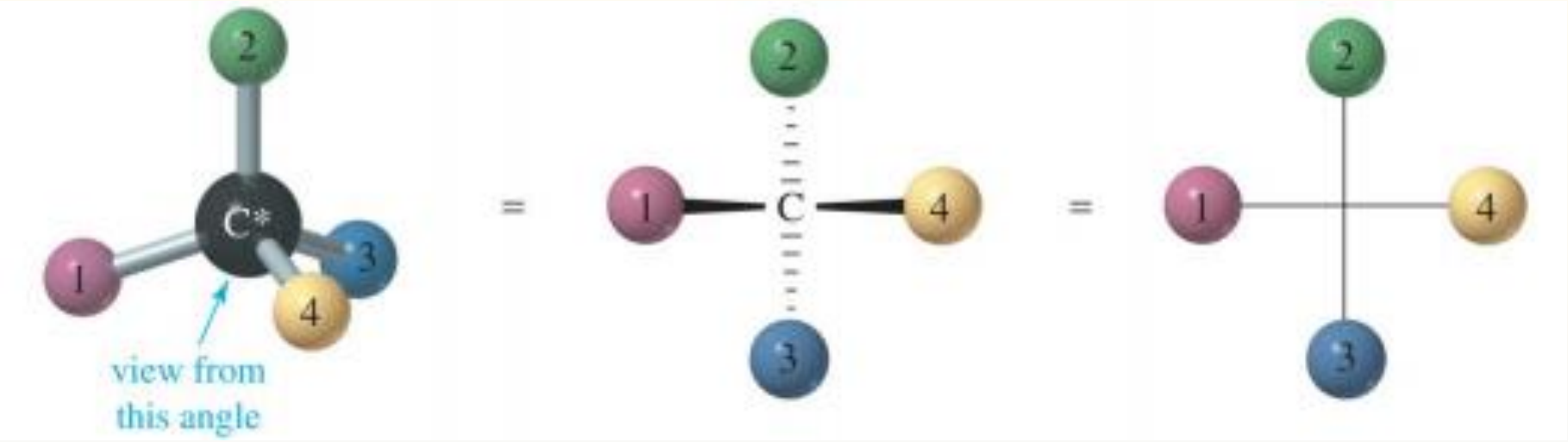
1st interchange

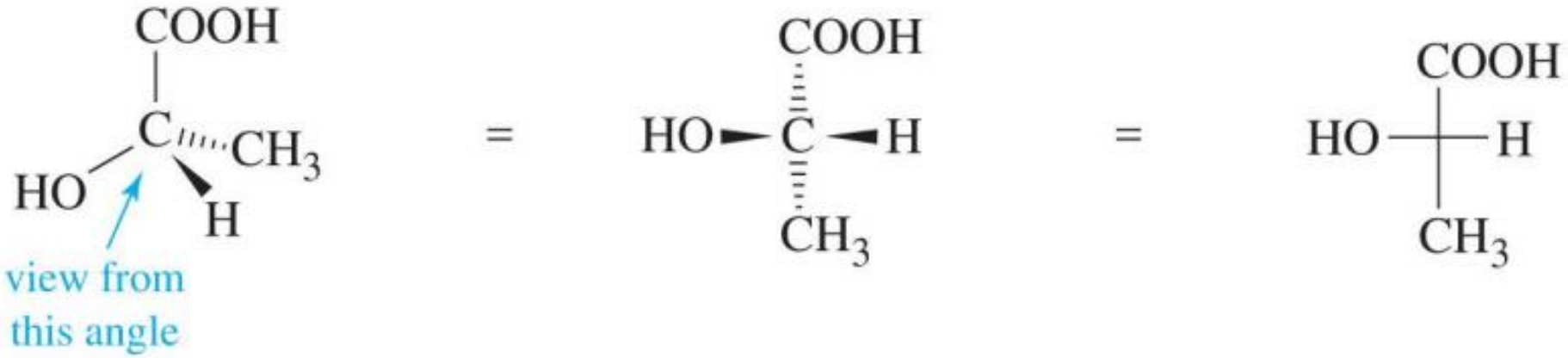


2nd interchange



Fischer Projections





(*S*)-lactic acid

(*S*)-lactic acid

Perspective drawing

Fischer projection

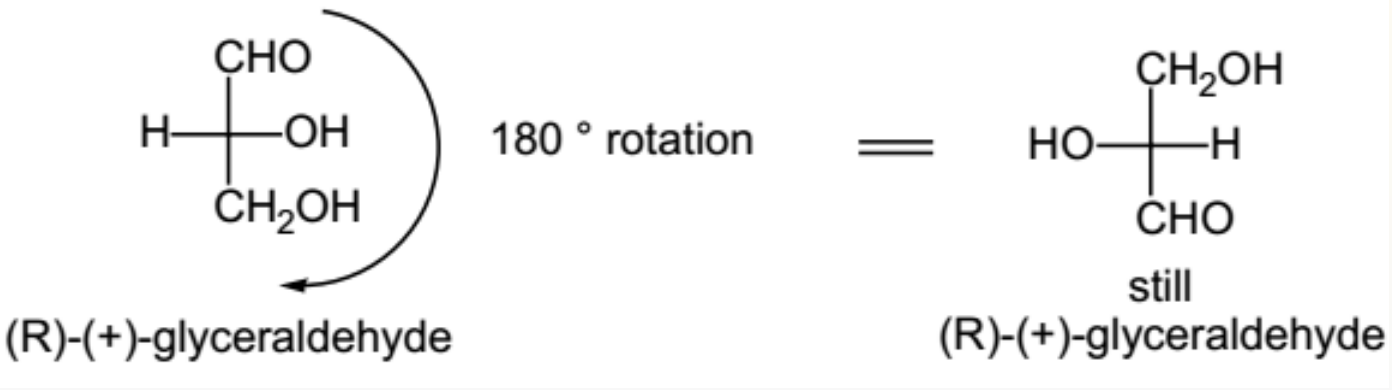
Fischer Projections



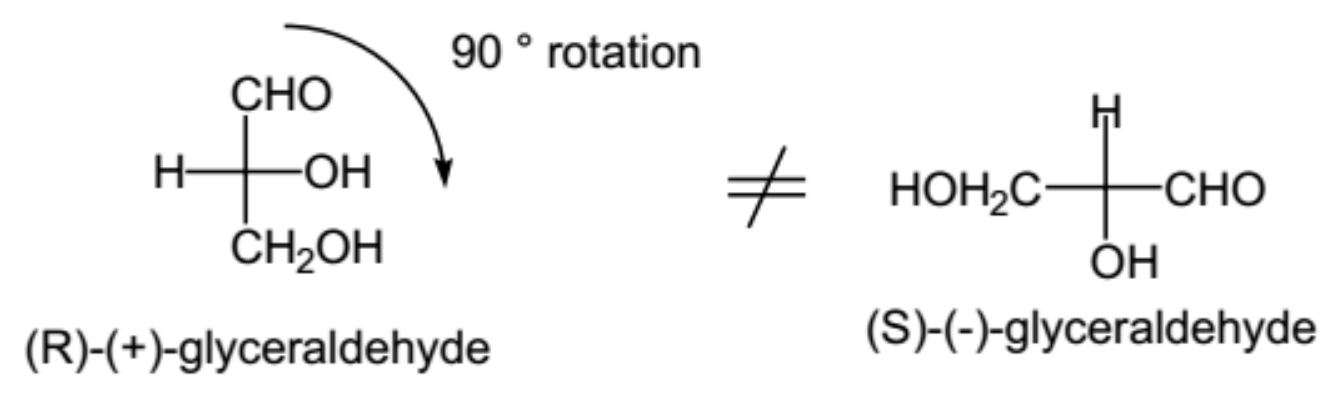
- Fischer projections are drawn with a cross, with the chiral atom at the center of the cross.
- The ***horizontal*** lines represent wedges (bonds) coming out of the plane of paper.
- The ***vertical*** lines represent dashed lines (bonds) going back behind of the plane the plane of the paper

Manipulation Of Fischer Projections

1. Fischer Projection can be rotated by 180° only!

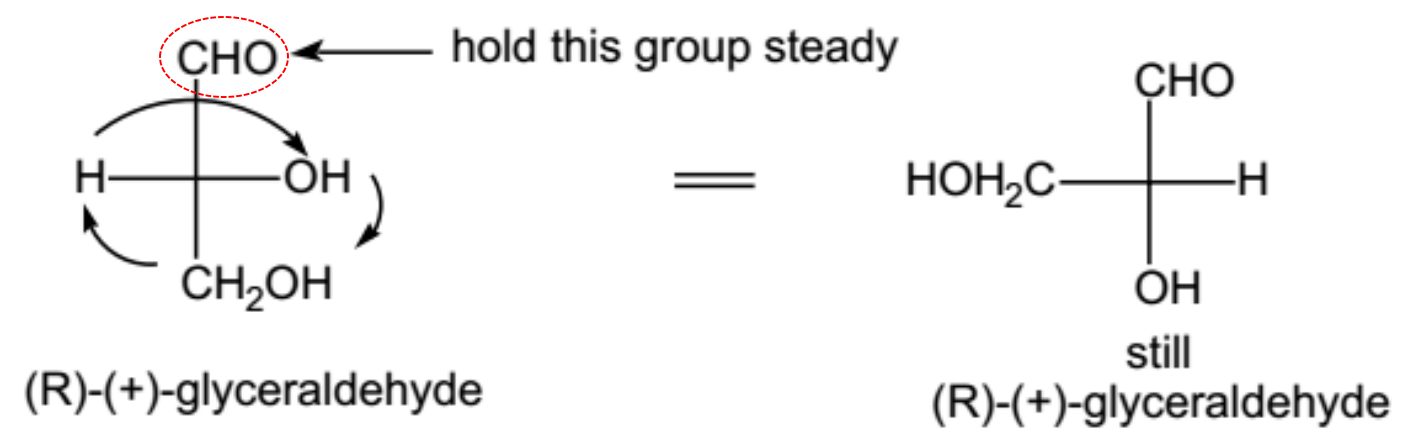


- Rotation by 90° invert the stereochemistry



Manipulation Of Fischer Projections

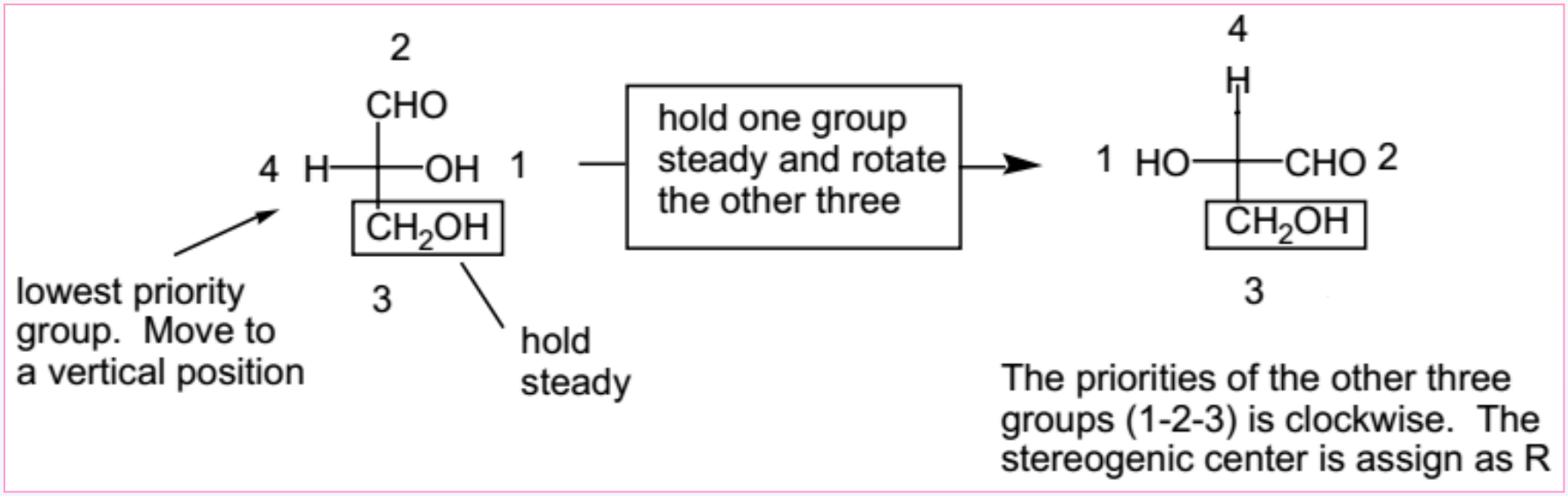
2. If one group of the Fischer projection is held steady, the other groups can be rotated either clockwise or counter clockwise.



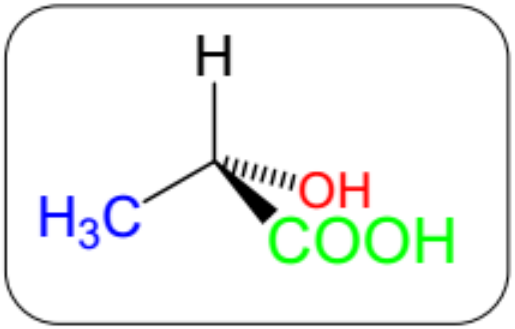
- Exchanging (one exchange!) any two groups around a Fischer projection generates the enantiomer of the original compound.
- Exchanging groups twice (two exchanges) regenerates the original stereochemistry.

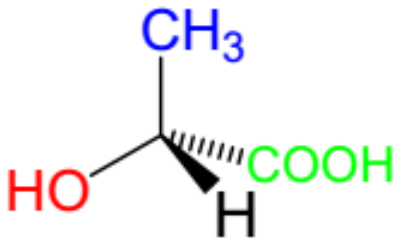
Assigning R And S-configurations To Fischer Projections

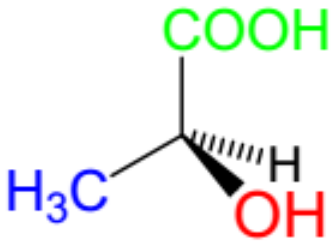
1. Assign priorities to 4 substituents according to the Cahn-Ingold-Prelog rule.
2. Perform the two allowed manipulations of the Fischer projection to place the lowest priority group on one of the vertical positions (either top or bottom).
3. If the priorities of the other three groups (1-2-3) proceed clockwise, the stereogenic center is assigned as R.
4. If the priorities of the other three groups (1-2-3) proceed counter-clockwise, the stereogenic center is assigned as S.



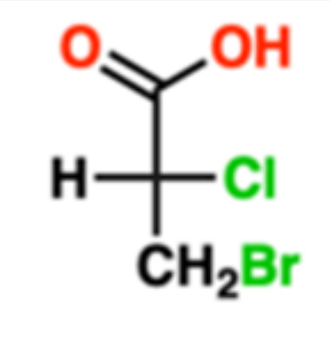
Q.1: Which of the followings are identical to first structure?

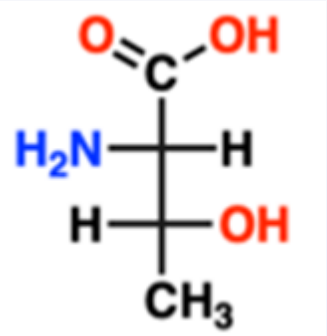


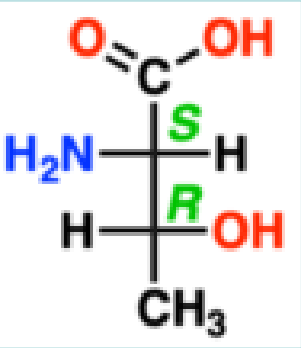


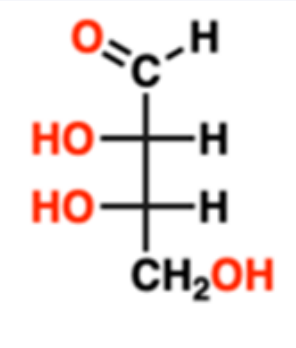


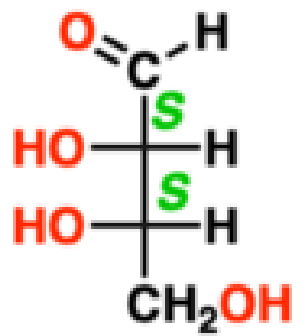
Q.2: What is the configuration of each stereocenter (chiral carbon atom) in each of the following compounds?

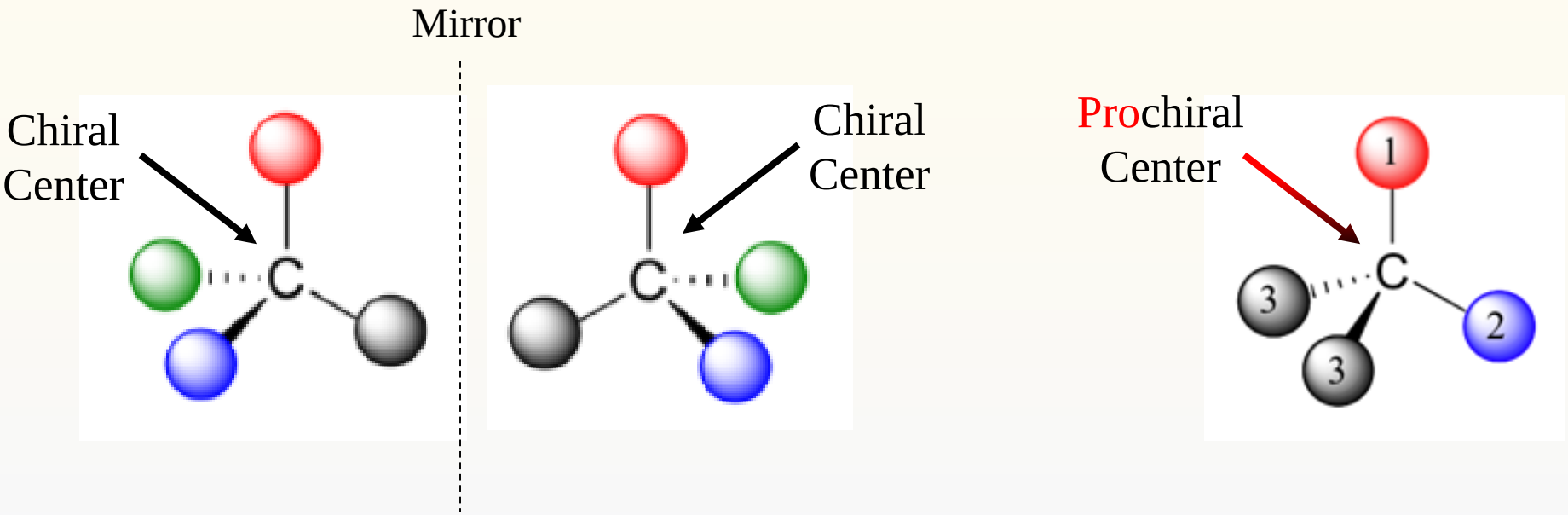










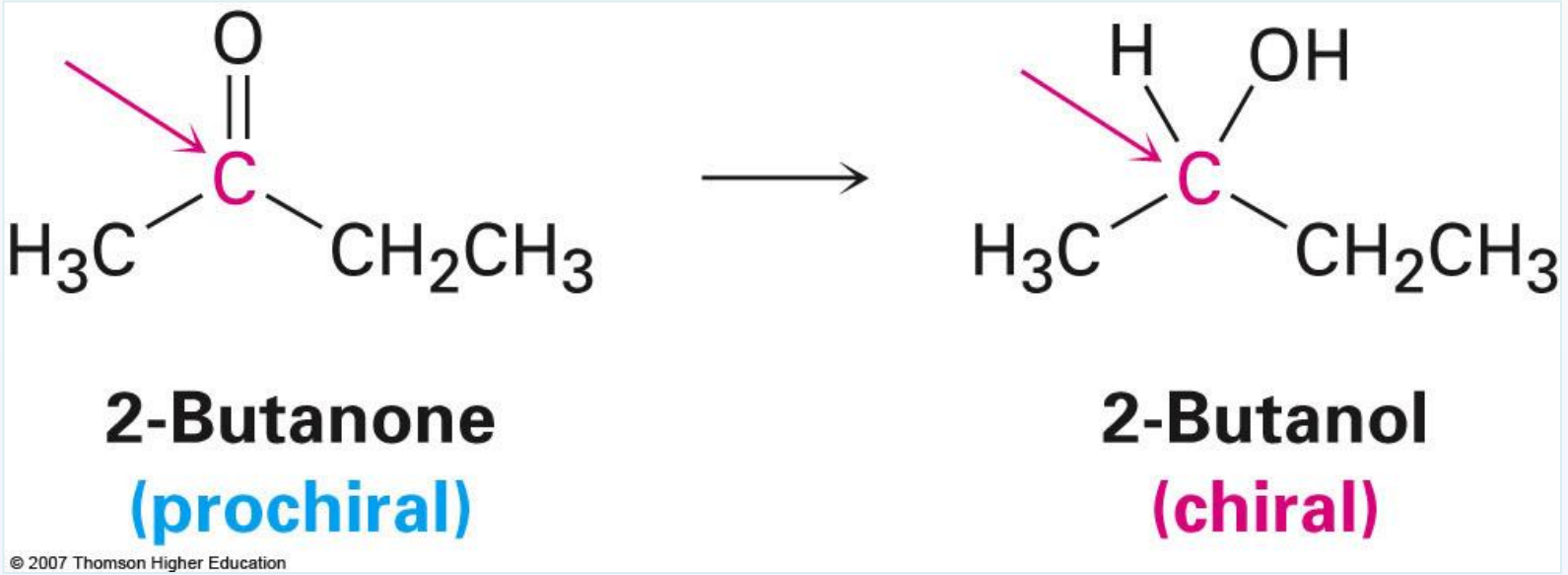


Prochirality

(Advanced Stereochemistry)

Prochirality

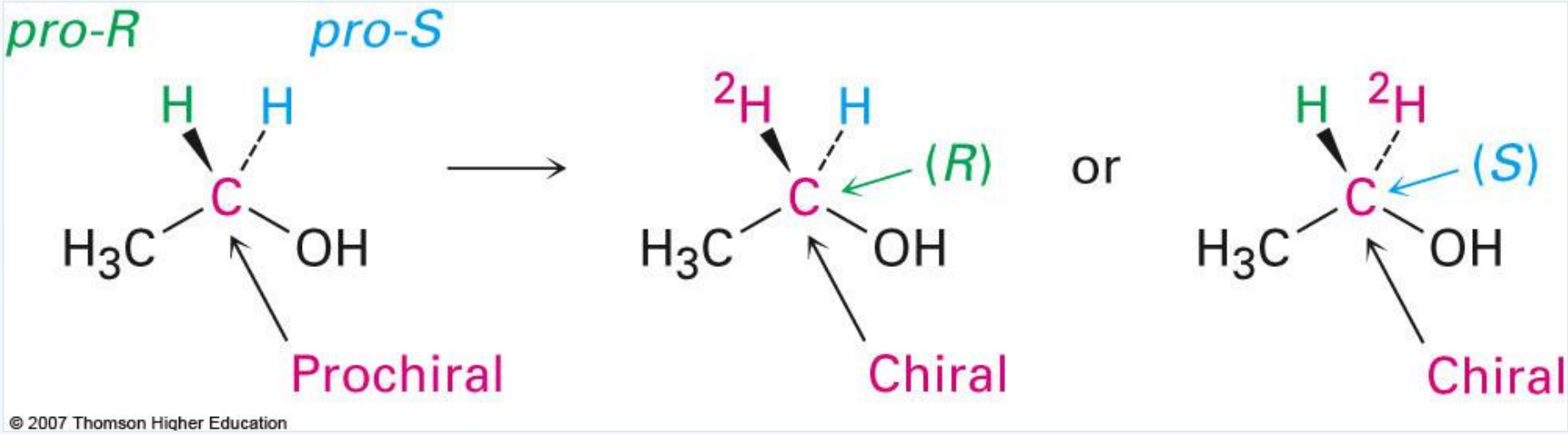
- Prochiral molecules are those that can be converted from a achiral molecule to a chiral molecule.
- Prochirality may be the result of an sp^2 carbon being converted to a chiral sp^3 carbon via a reaction or by substitution of a "prochiral" substituent (usually hydrogen) with another substituent that results in a chiral center.



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pro-R and *pro-S*

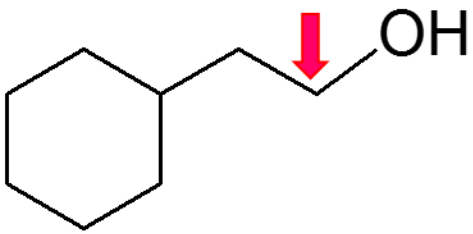
- An sp^3 carbon with two identical groups is a **prochiral center**
- If two identical substituents are attached to a sp^3-hybridized atom, the descriptors *pro*-R and *pro*-S are used to distinguish between the two.
- If the center becomes ***R*** the group is ***pro-R*** and ***pro-S*** if the center becomes ***S***.



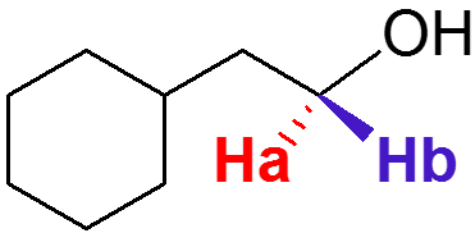
© 2007 Thomson Higher Education

pro-R and *pro-S*

Prochiral Center

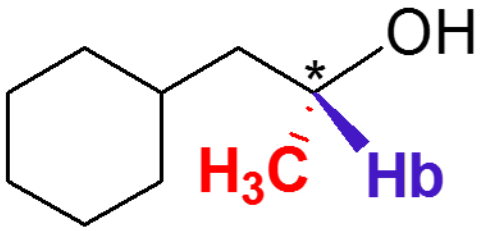


≡

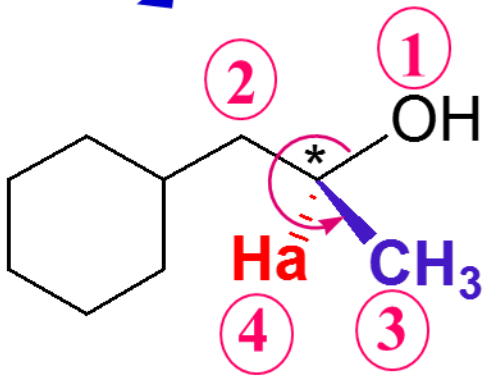


Substitute Ha with CH_3

Substitute Hb with CH_3



R-configuration

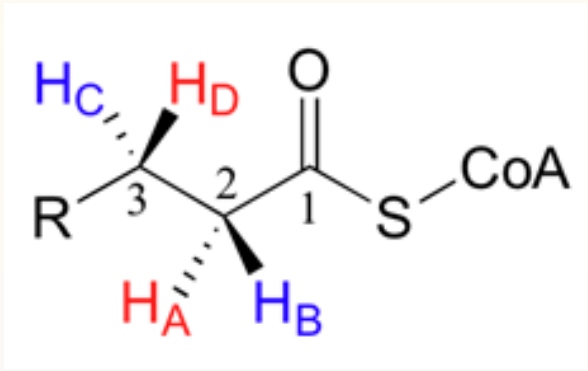


S-configuration

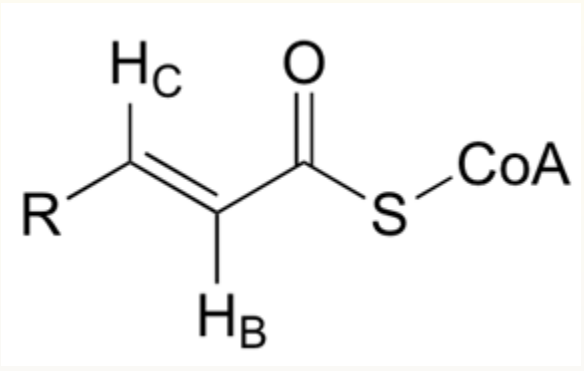
Ha is referred to as *pro-R*

Hb is referred to as *pro-S*

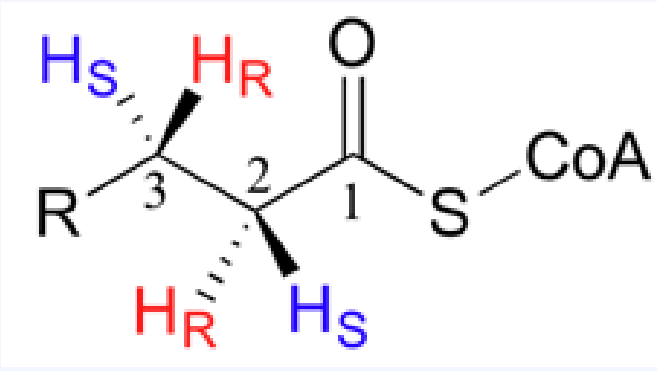
pro-R and *pro-S*





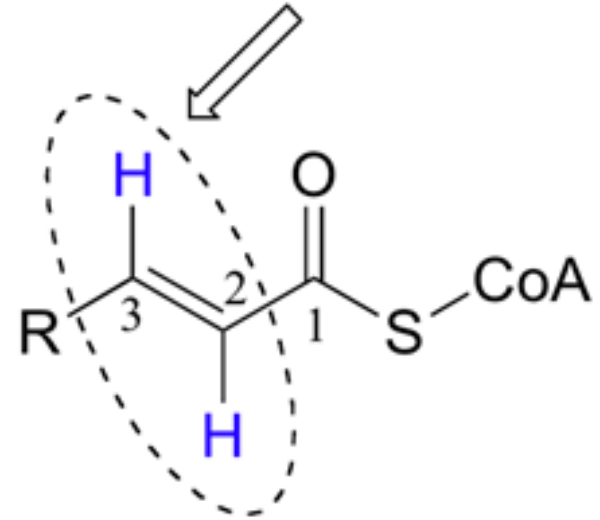






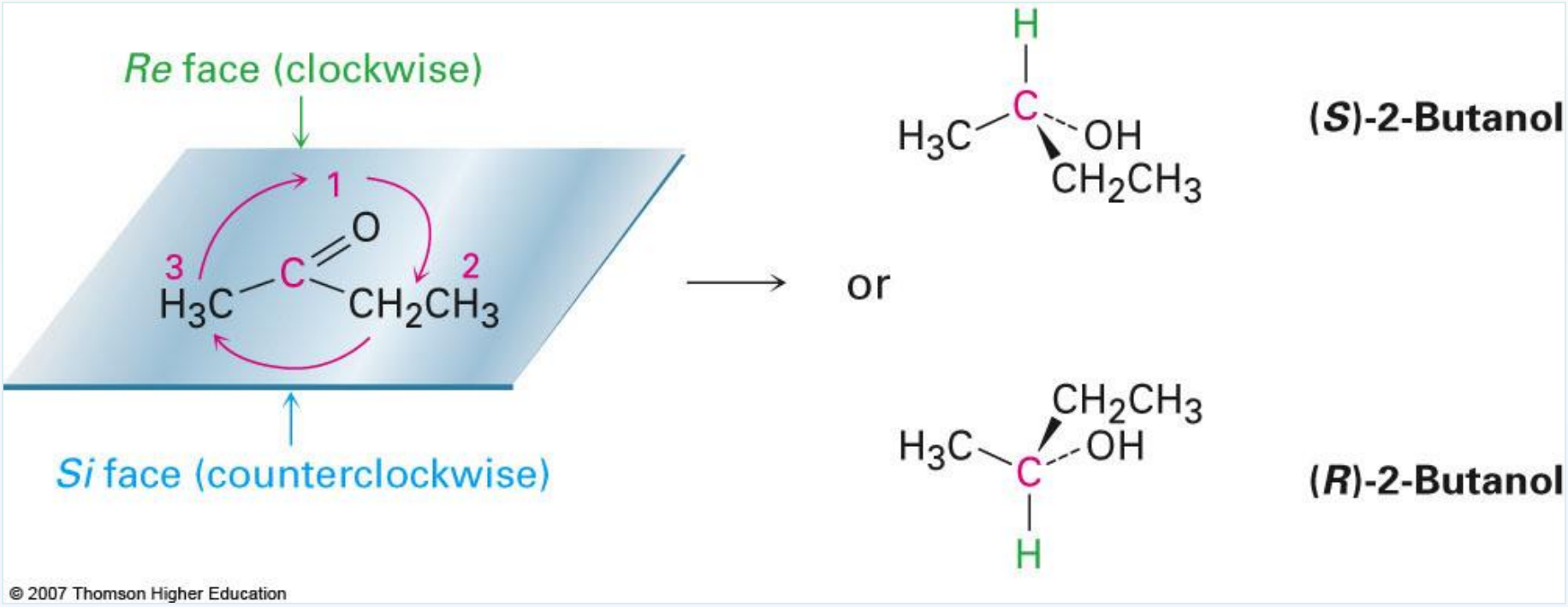
pro-R hydrogens lost

these hydrogens used to be H_S on starting molecule



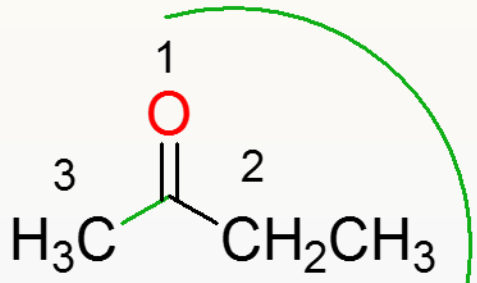
Re/*Si* face

- Planar faces that can become tetrahedral are different from the top or bottom
- A center at the planar face at a carbon atom is designated ***Re*** if the three groups in priority sequence are clockwise, and ***Si*** if they are counterclockwise



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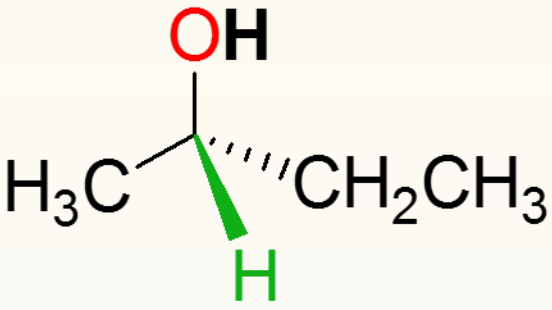
Re/*Si* face



2-Butanone

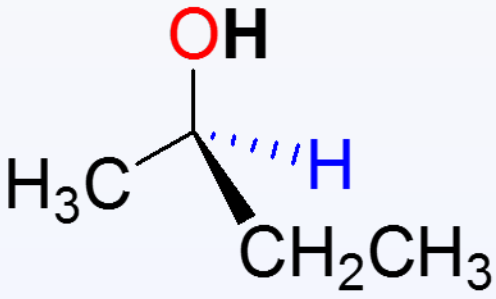
the *Re* face is closest to one's eye

H^- attack on the *Re*-face followed by H^+



(*S*)-2-Butanol

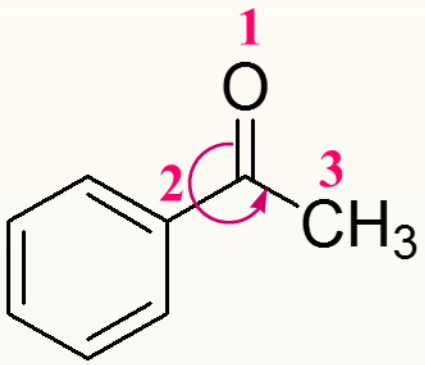
H^- attack on the *Si*-face followed by H^+



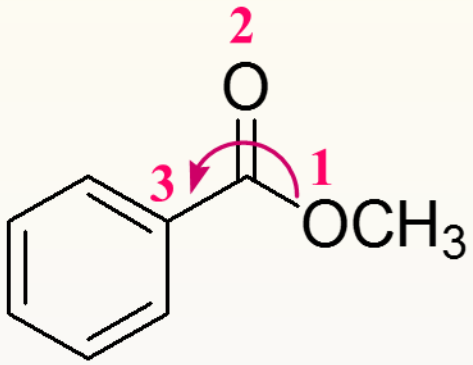
(*R*)-2-Butanol

Exercises

Q.1: Depict the top face in the following compounds as *Re* or *Si*.



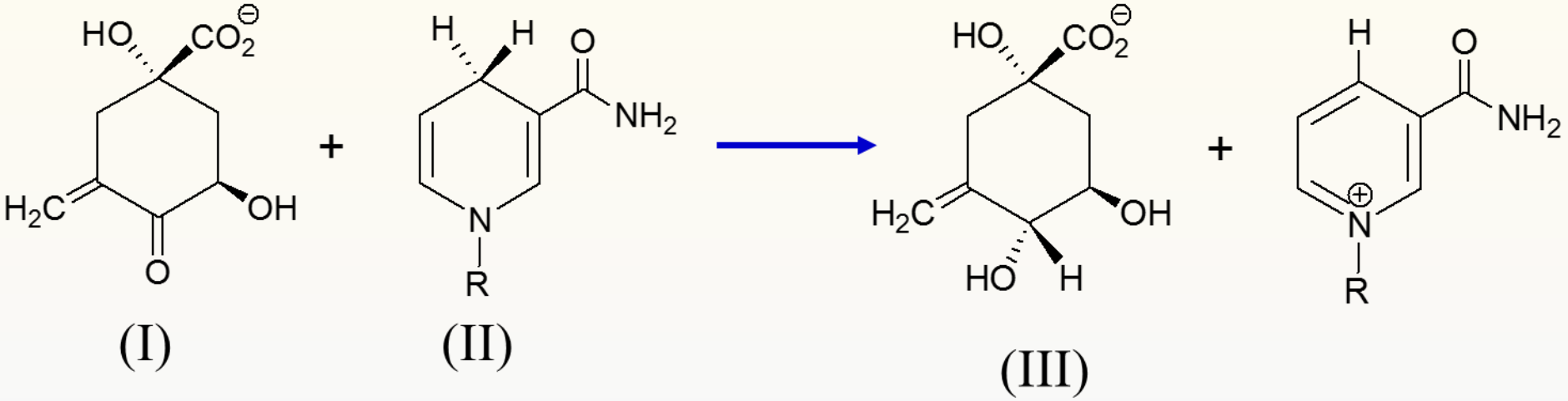
The top face is *Si* face



The top face is *Si* face



Exercises

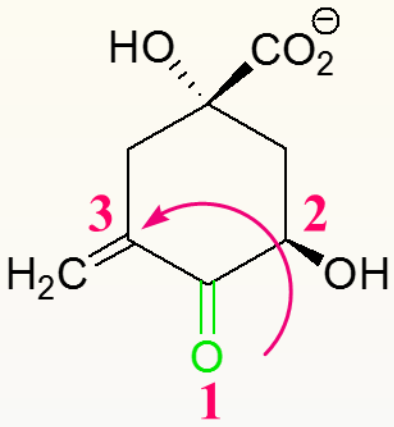


Q.2: Label the prochiral hydrogens in the compound (II) as H_R or H_S.

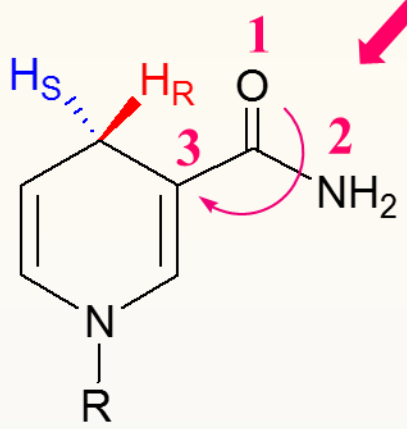
Q.3: For each of the carbonyl groups in (I) and (II), state whether we are looking at the *Re* or the *Si* face.

Exercises

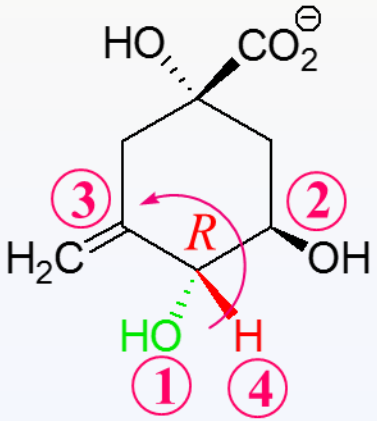
We are looking at *Re* face of the carbonyl group of amide



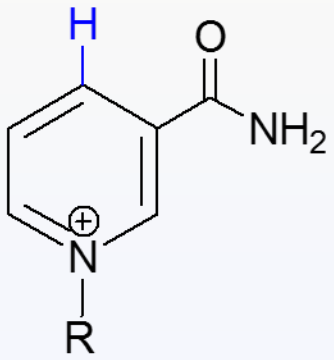
+



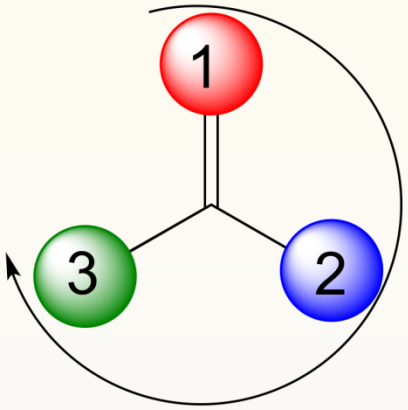
We are looking at *Si* face of the ketone



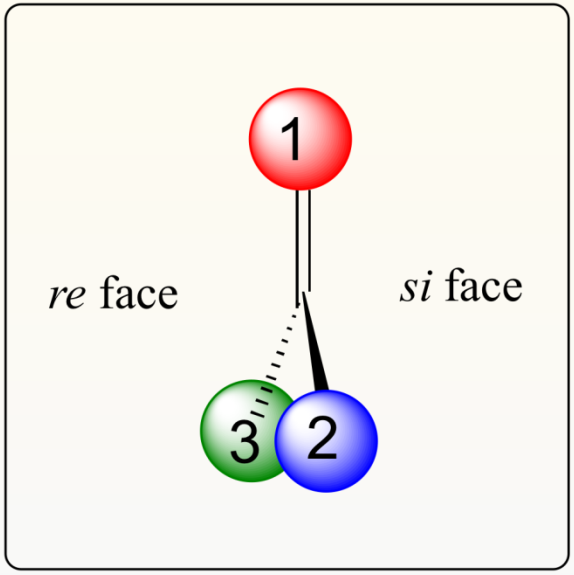
+

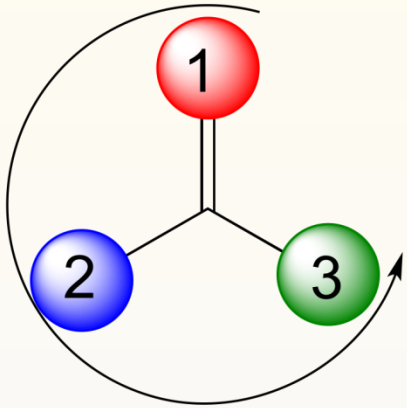


1, 2 and 3 proceed counterclockwise,
But the no. 4 priority group is in the front.
So the configuration is *R*



looking at the *re* face





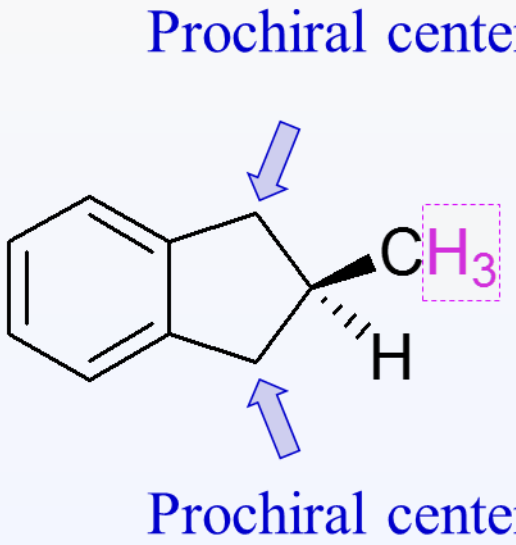
looking at the *si* face

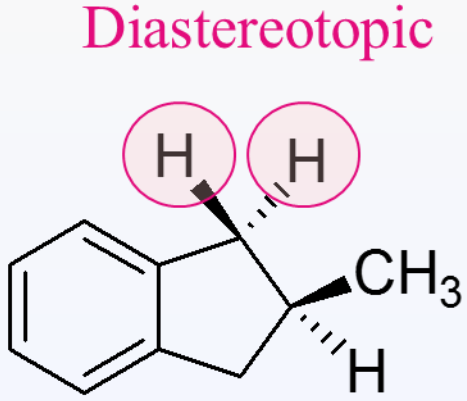
Prochirality

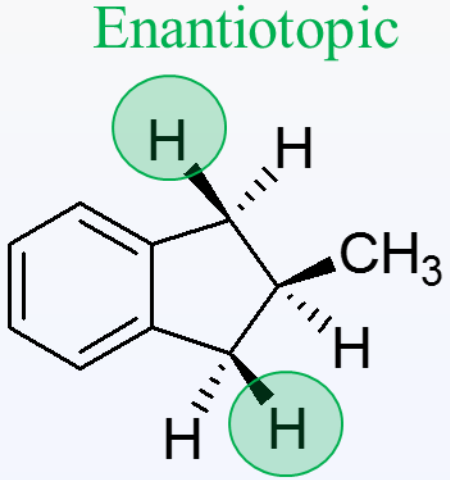
(Advanced Stereochemistry)

Topicity in Stereochemistry

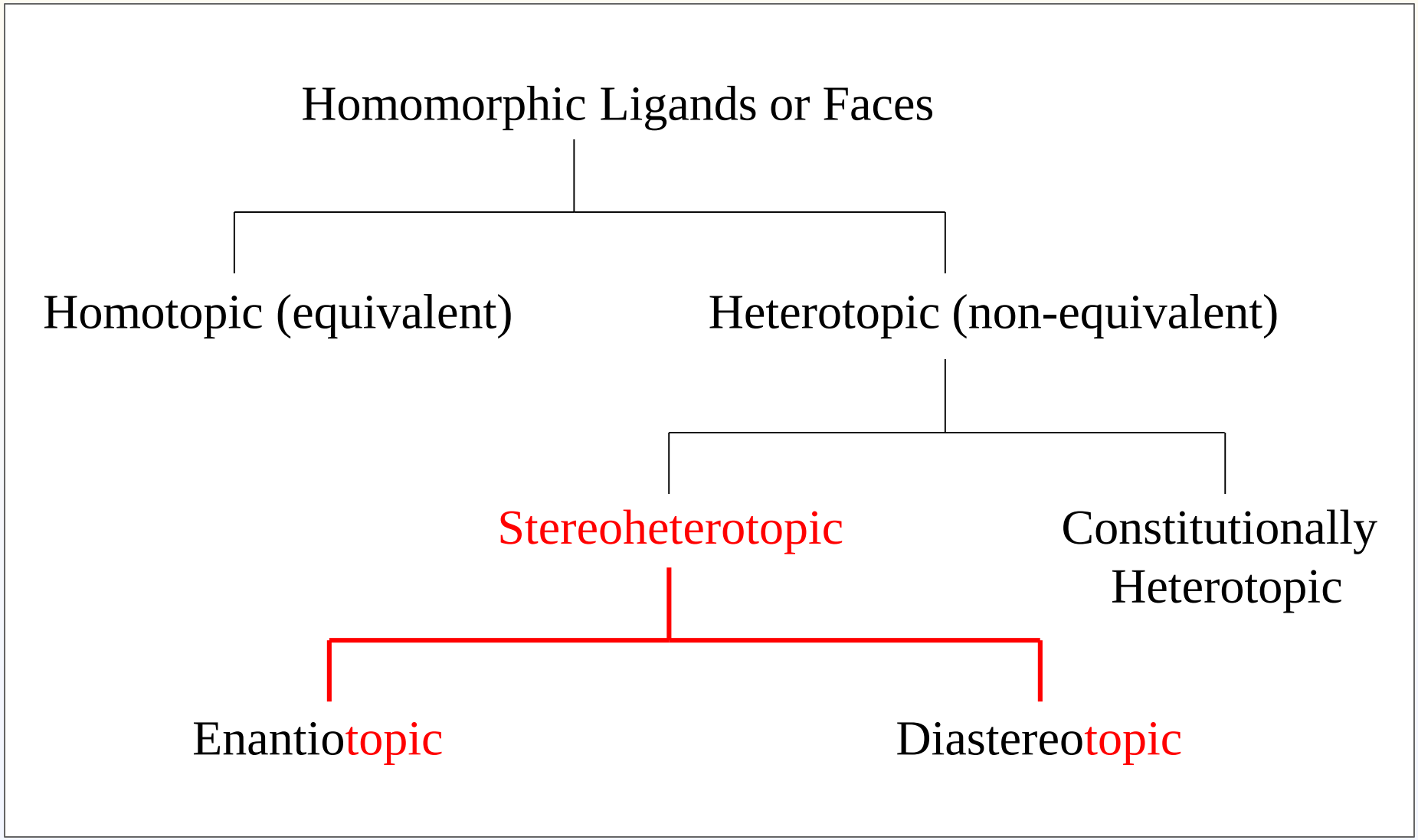
- In stereochemistry, **topicity** is the stereochemical relationship between substituents and the structure to which they are attached.
- Depending on the relationship, such groups can be **homotopic** or **heterotopic** (*enantiotopic* or *diastereotopic*).







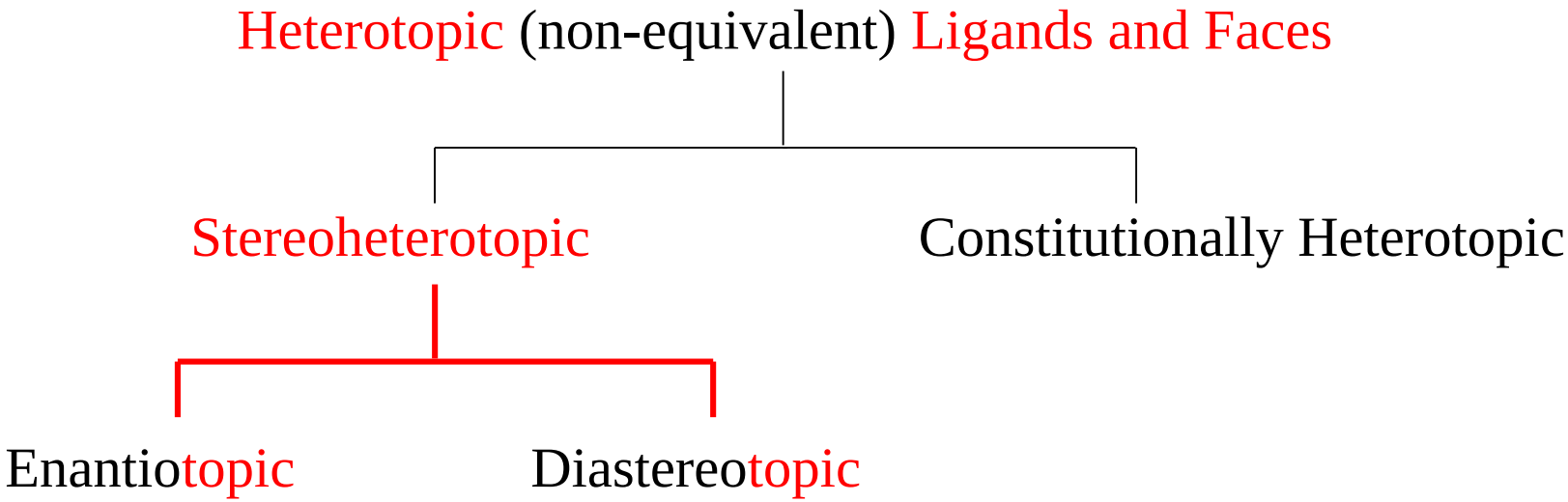
Homotopic and Heterotopic Ligands and Faces



Heterotopic Ligands and Faces

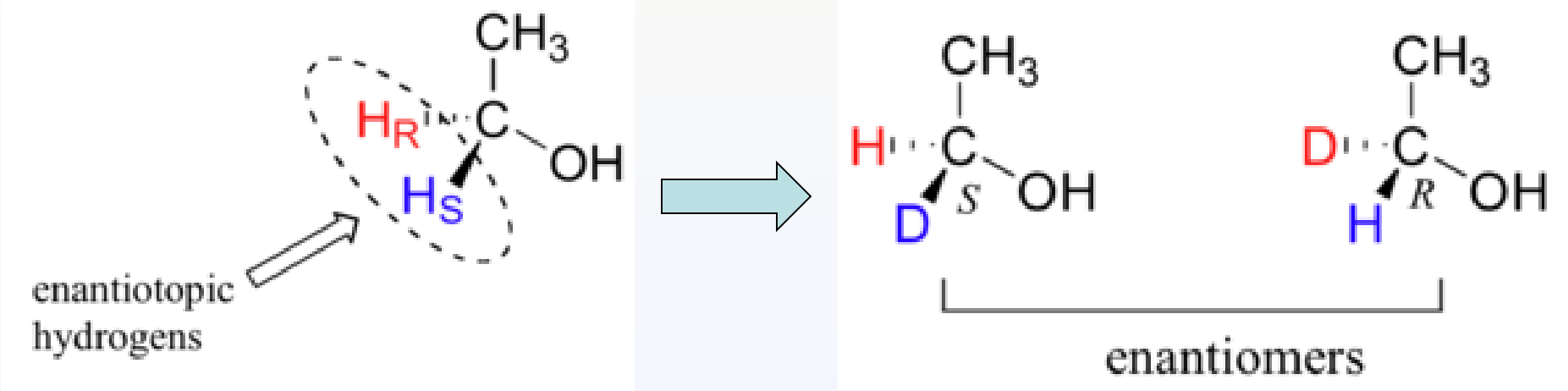
- Just as one divides stereoisomers into two sets, enantiomers (Greek *enantios* = opposite) and diastereomers, so it is convenient to divide heterotopic (non-equivalent) groups or faces into enantiotopic and diastereotopic moieties.

- Enantiotopic ligands are ligands which find themselves in mirror-image positions whereas diastereotopic ligands are in stereochemically distinct positions not related in mirror-image fashion.



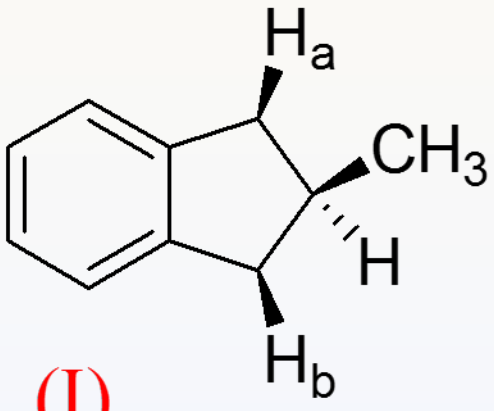
Enantiotopic Ligands

- Two ligands are enantiotopic if replacement of either one of them by a different achiral ligand, which is also different from other ligands to the prochiral element, gives rise to enantiomeric products.

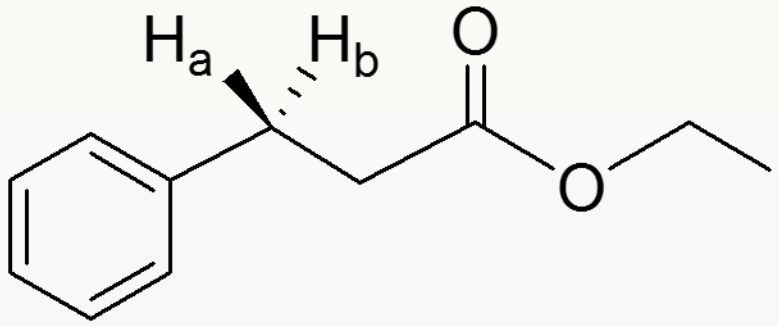


Practice Problems

Q. 1. Indicate whether the protons H_a and H_b in the following compounds are homotopic, enantiotopic or distereotopic, and also give reasons for your answer.



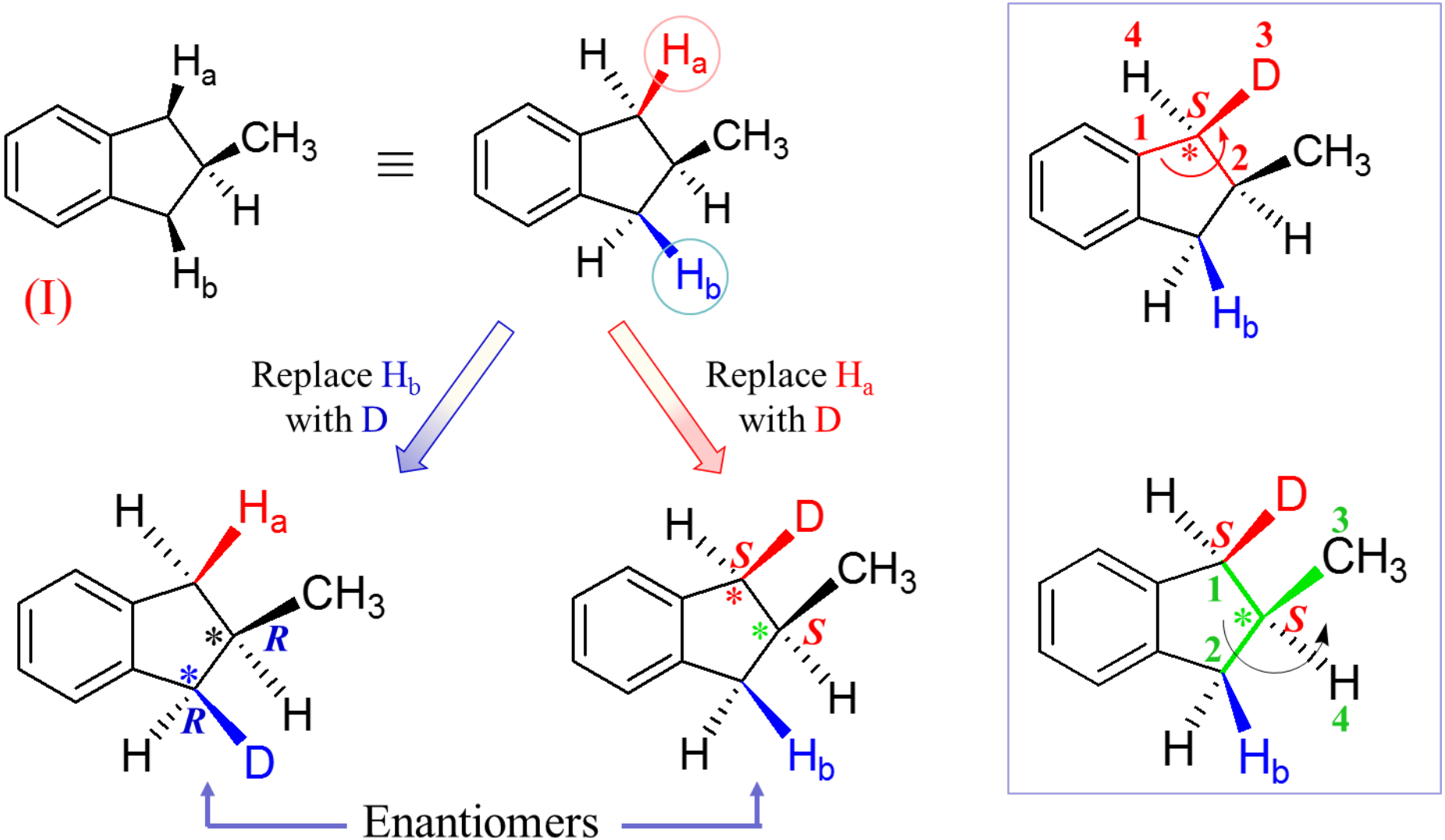
(I)



(II)

Q. 2. Label the prochiral hydrogens (H_a and H_b) in compound (II) as *pro-R* and *pro-S* and depict the top face of the carbonyl group of ester as *Re* or *Si*.

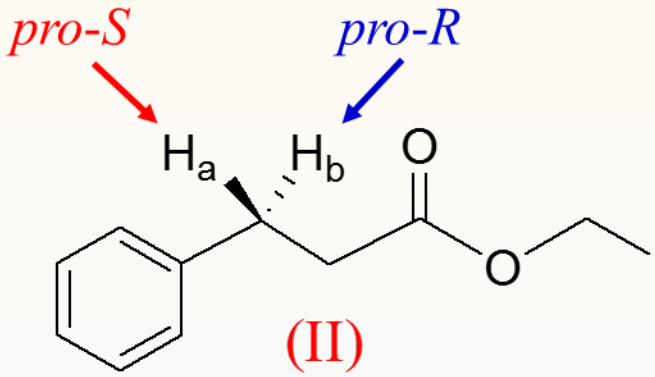
Practice Problems



So, H_a and H_b are enantiotopic hydrogens or ligands

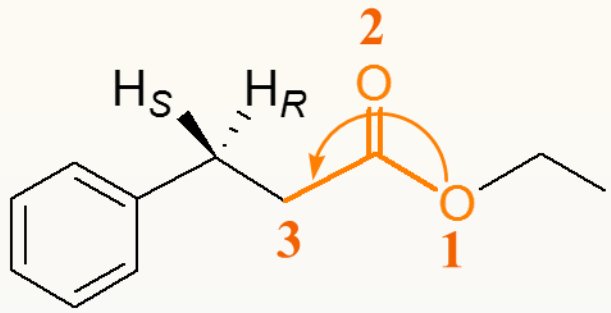
Practice Problems

Q. 2. Label the prochiral hydrogens (H_a and H_b) in compound (II) as *pro-R* and *pro-S* and depict the top face of the carbonyl group of ester as *Re* or *Si*.



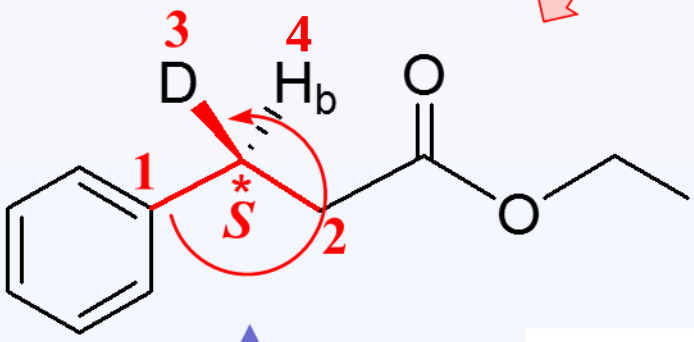
(II)

≡

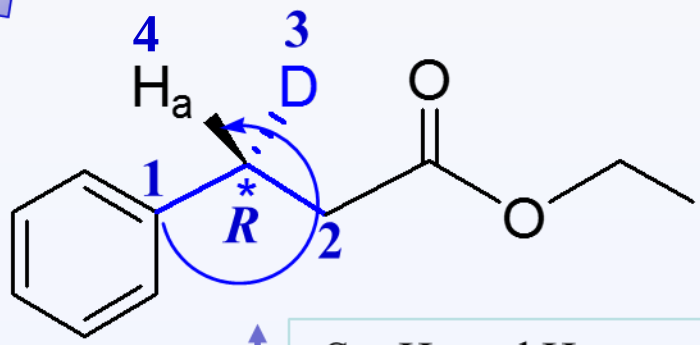


the top face is *Si* face

Replace H_a with D



Replace H_b with D

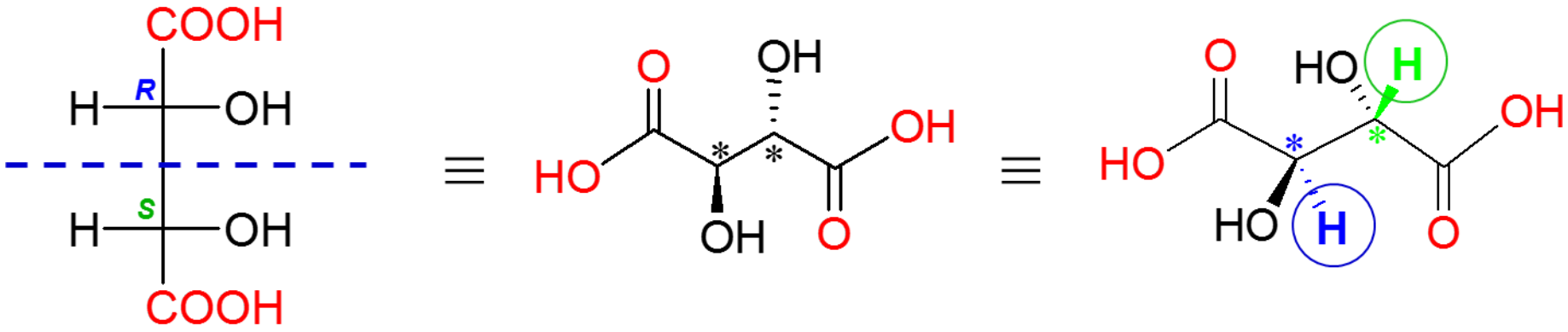


Enantiomers

So, H_a and H_b are enantiotopic hydrogens or ligands

Enantiotopic Ligands

(Substitution and Addition Criteria)



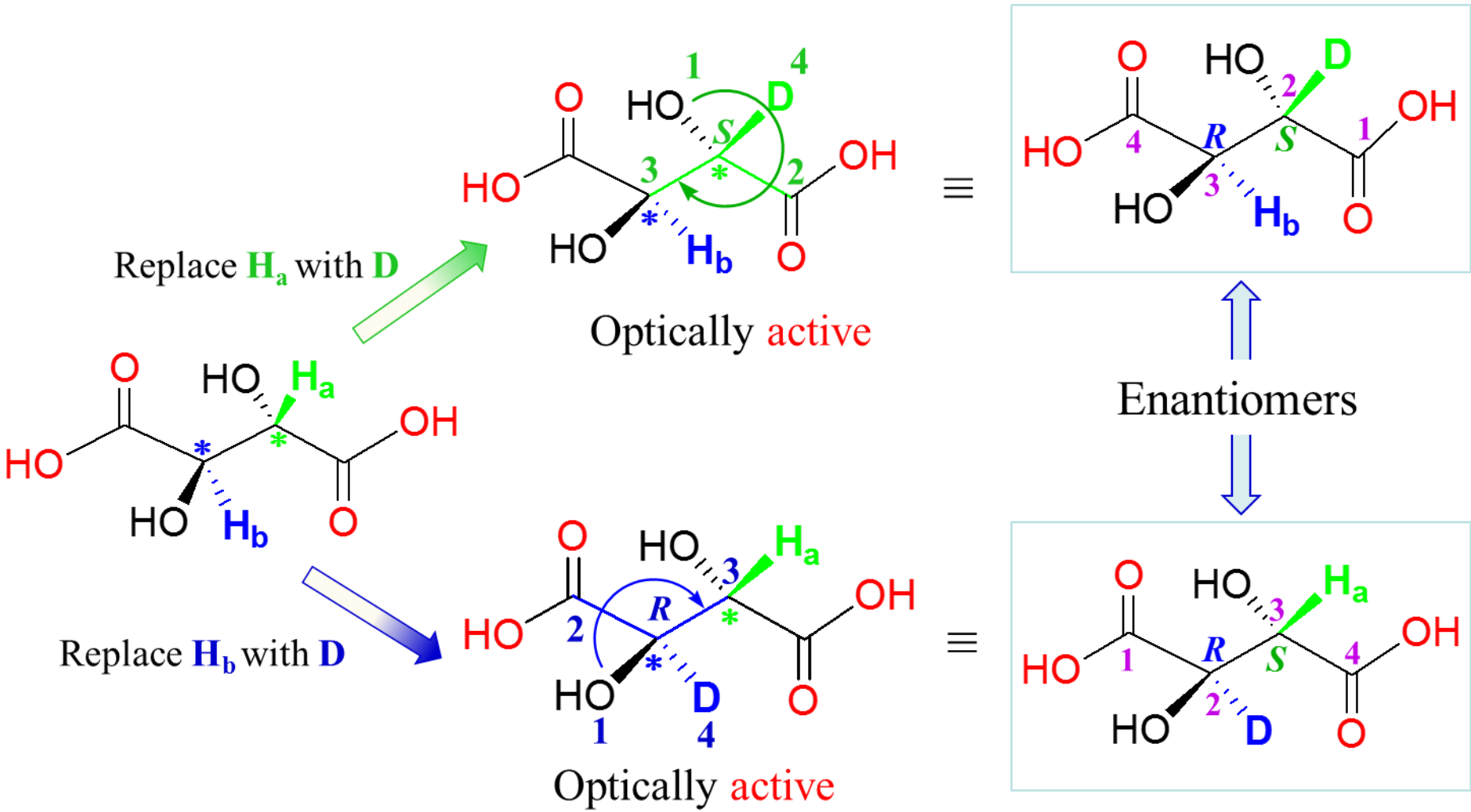
meso-Tartaric acid, or (2*R*,3*S*)-Tartaric acid

(Optically inactive)

Q. Indicate whether the hydrogens of *meso*-Tartaric acid are homotopic, enantiotopic or distereotopic, and also give reasons for your answer.

Enantiotopic Ligands

(Substitution and Addition Criteria)

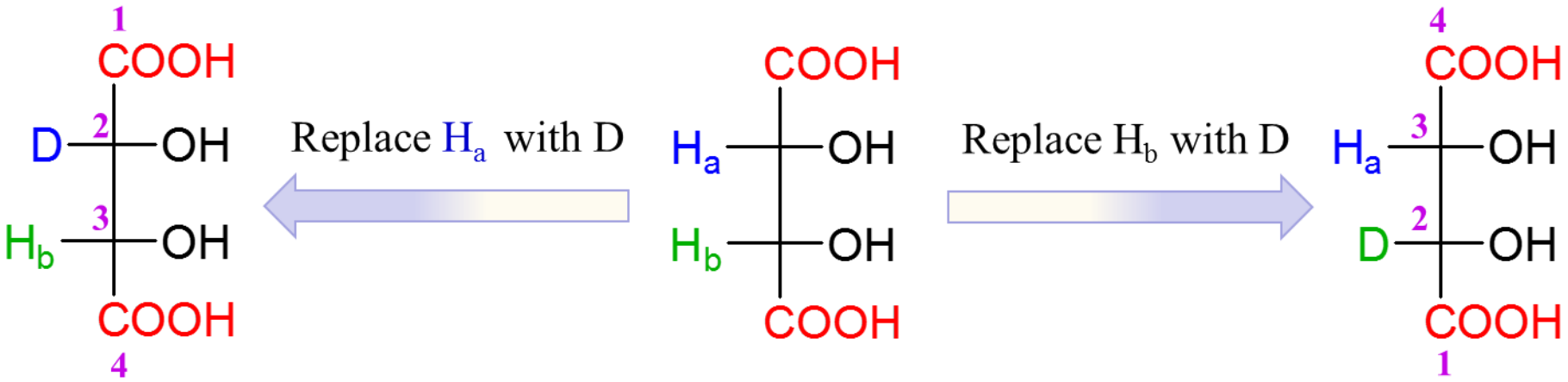


The hydrogens of *meso*-Tartaric acid are enantiotopic.

Enantiotopic Ligands

(Substitution and Addition Criteria)

- In *meso*-tartaric acid H_a and H_b hydrogens are enantiotopic.



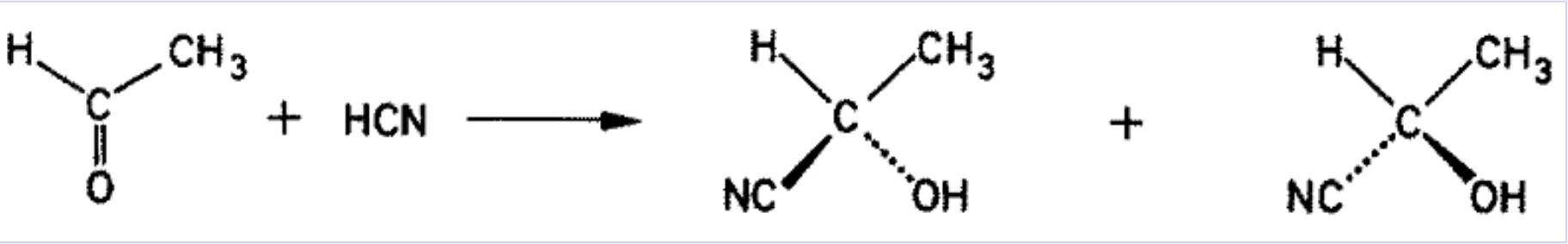
(2*R*, 3*S*)- *meso*-Tartaric acid (2*S*, 3*R*)-

Non-superimposable mirror images of each other

Enantiomers

Enantiotopic Faces

- Two corresponding faces of a molecule (usually faces of a double bond) are enantiotopic when addition of the same achiral reagent to either one or the other face will give rise to enantiomeric products.
- Thus addition of HCN to the two enantiotopic faces of acetaldehyde gives rise to the two enantiomers of lactonitrile.
- Here, also the added group must be different from any group already there.

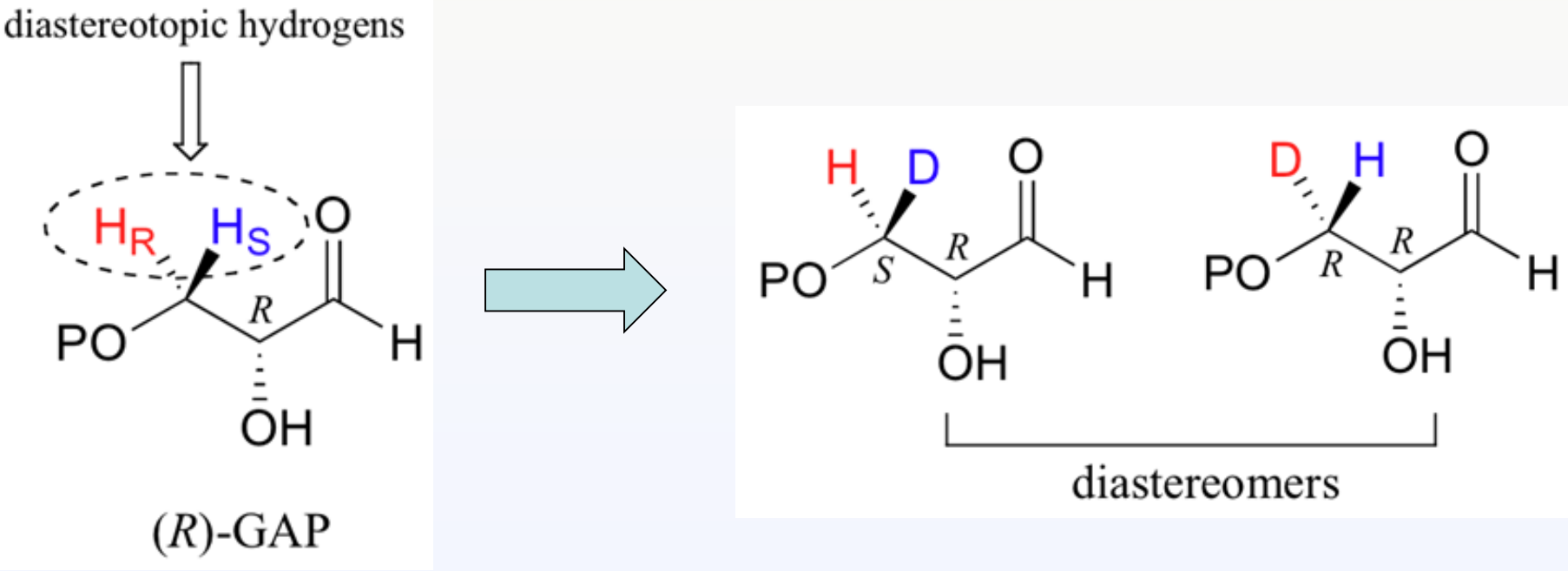


(*R*)-
Front addition

(*S*)-
Rear addition

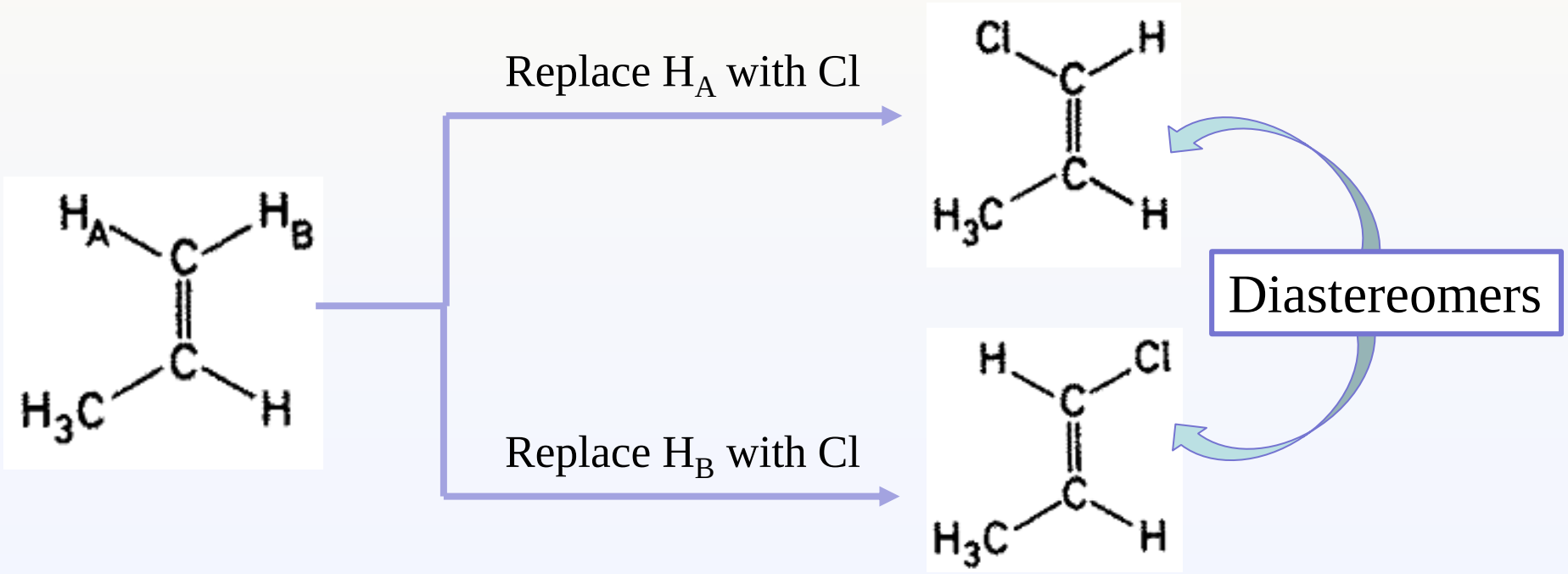
Diastereotopic Ligands

- When the replacement of either of the two (homomorphic) ligands by a different achiral test ligand gives diastereomers then such (homomorphic) ligands are termed diastereotopic.



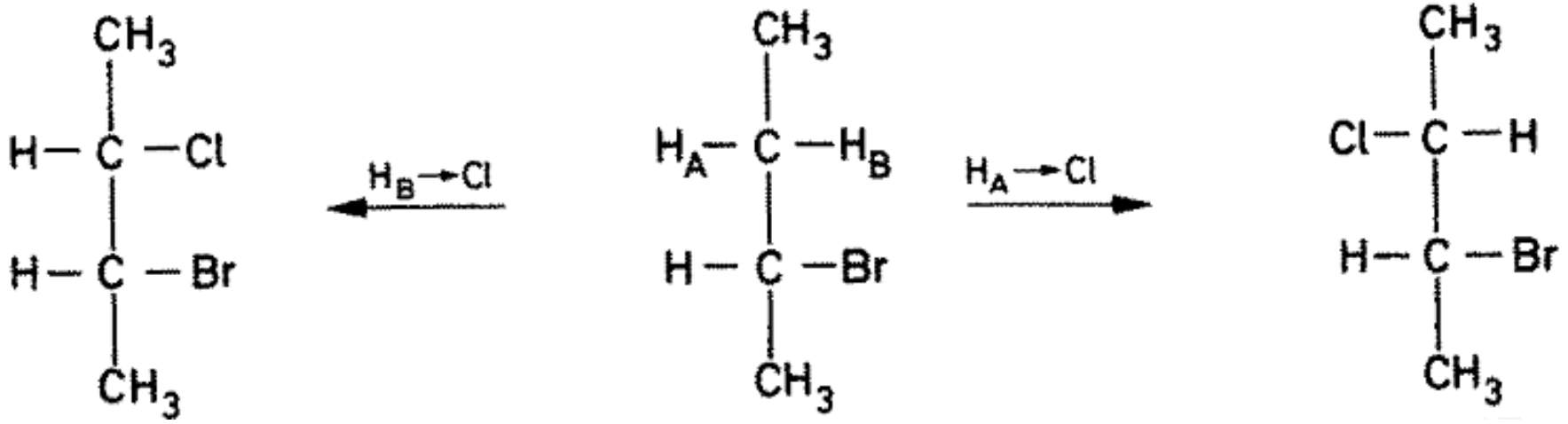
Diastereotopic Ligands

- The substitution-addition criteria may also be employed to spot diastereotopic ligands, i.e. ligands which find themselves in a stereochemically distinct but non-mirror image environment.

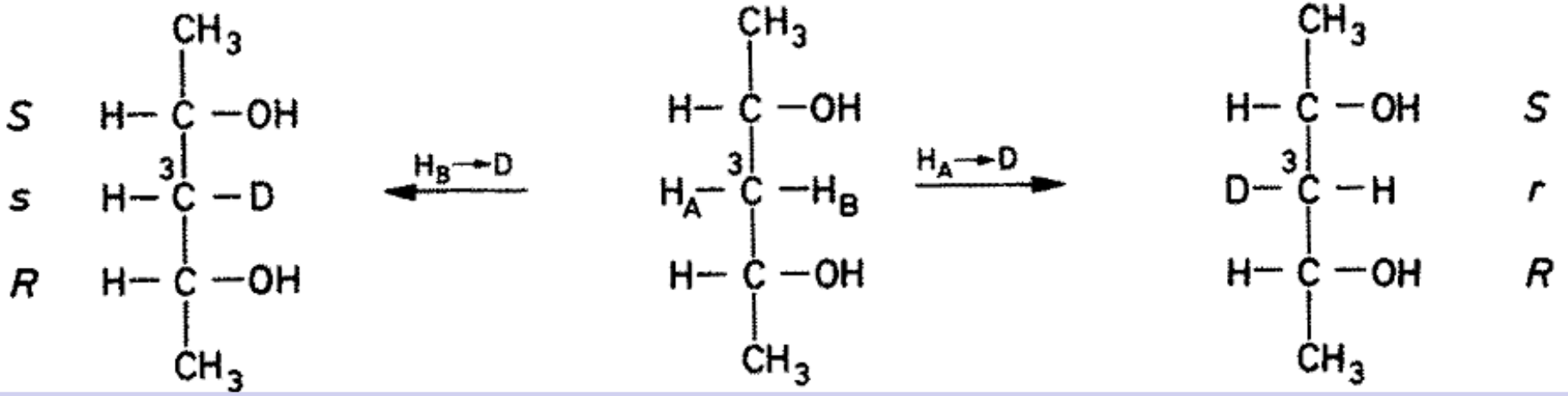


Diastereotopic Ligands

(Substitution and Addition Criteria)

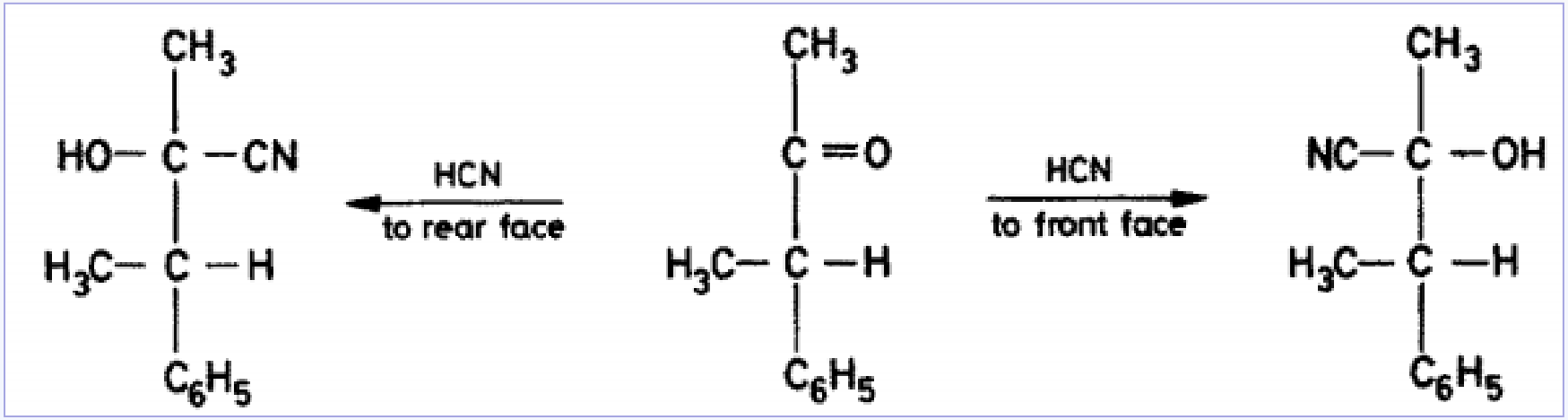


erythro - *threo* -



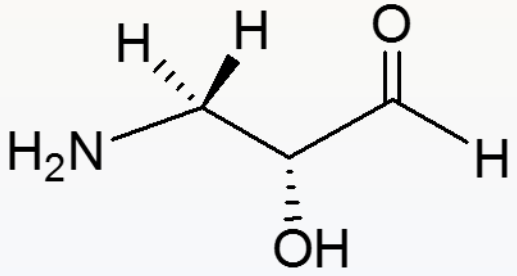
Diastereotopic Faces

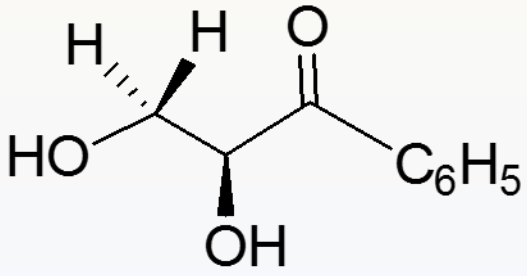
- When in aldehydes (other than formaldehyde) and unsymmetrical ketones R–CO–R′ (R ≠ R′) the groups R or R′ contain one or more stereocenters, the faces of the carbonyl group become diastereotopic. Addition to one face gives a diastereomer and addition to other face gives another diastereomer.

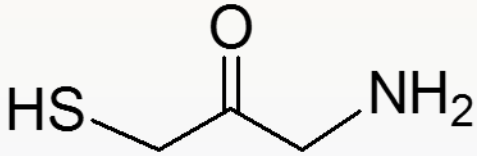


Practice Problems

Q. 1: Identify whether the hydrogens are enantiotopic, homotopic or diastereotopic in the following molecules. Also label prochiral hydrogens as H_R or H_S.



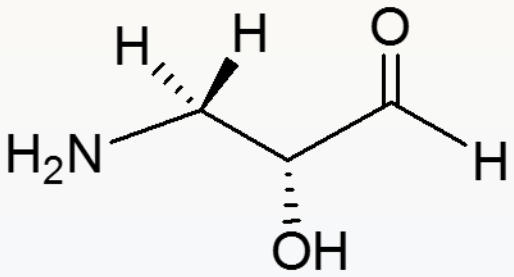


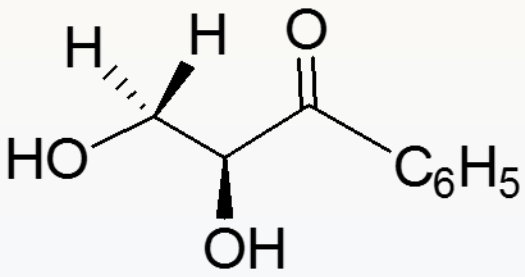


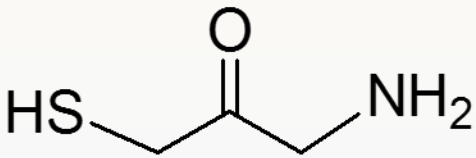
Q. 2: Using HCN addition reaction identify the faces of each of the above compounds as homotopic, enantiotopic or diastereotopic.

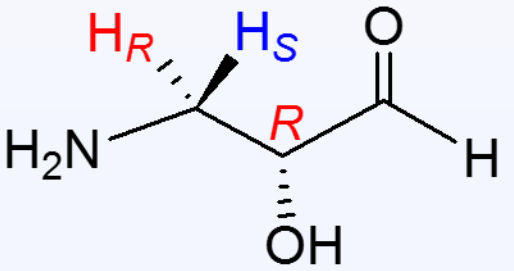
Practice Problems

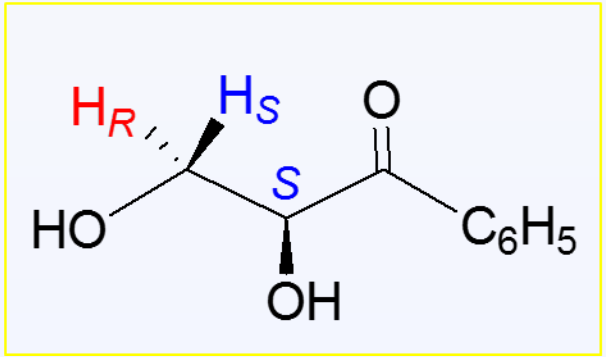
Q. 1: Identify whether the hydrogens are enantiotopic, homotopic or diastereotopic in the following molecules. Also label prochiral hydrogens as H_R or H_S.

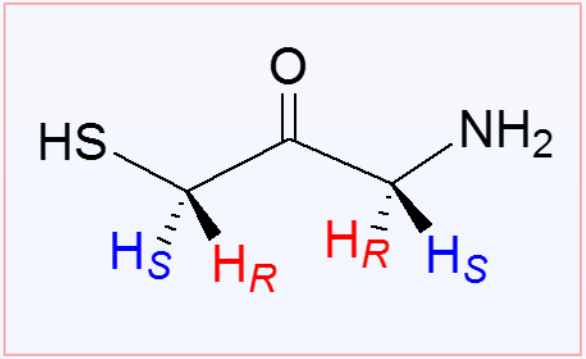








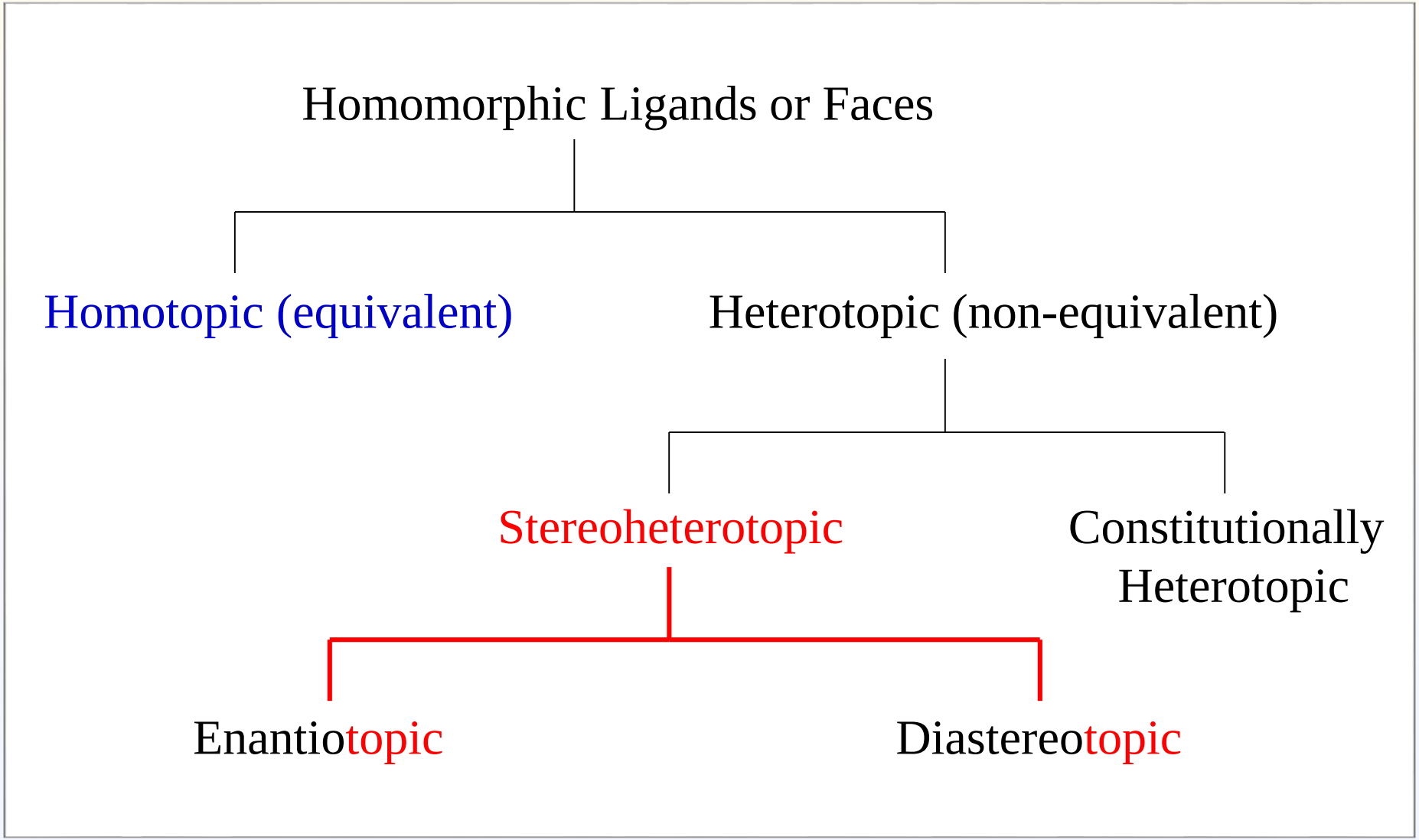




Chiral and Prochiral Molecules

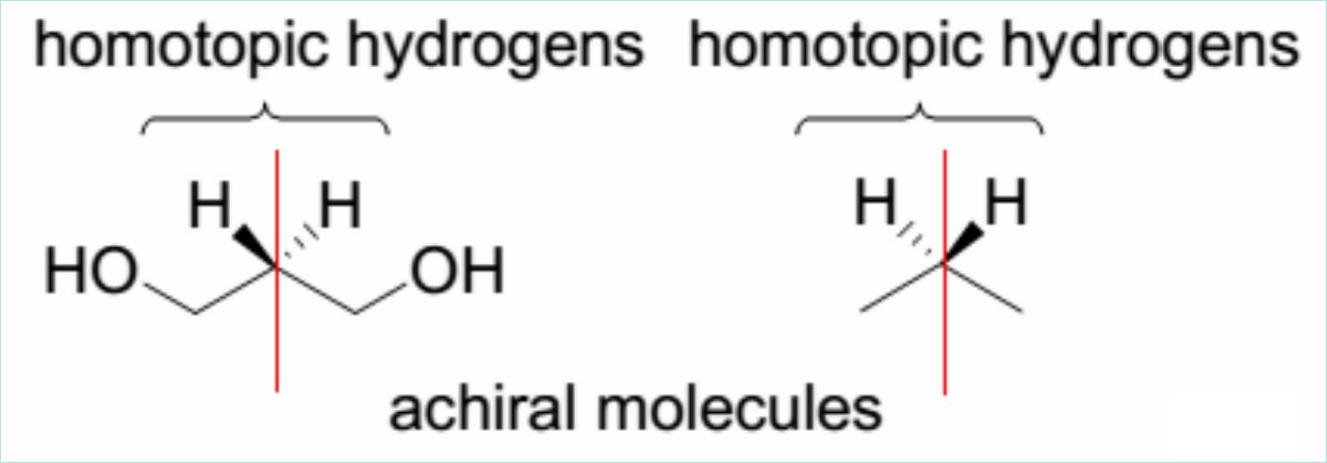
Advanced Stereochemistry

Homotopic and Heterotopic Ligands and Faces



Homotopic Ligands

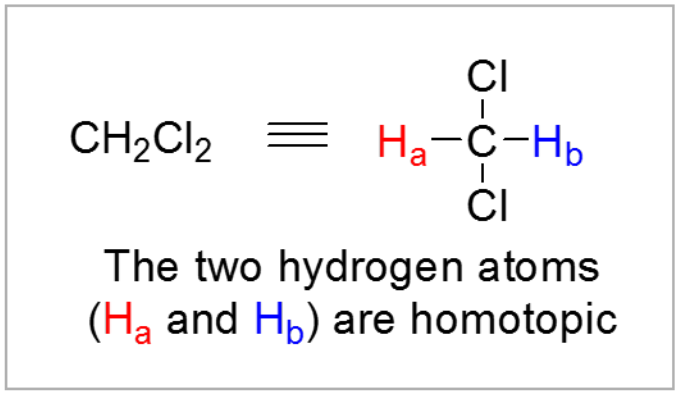
- Homotopic groups (ligands) in a chemical compound are equivalent groups.
- Homotopic atoms are always identical, in any environment.
- Homotopic NMR-active nuclei have the same chemical shift in an NMR spectrum.
- Two criteria – substitution/addition or symmetry may be employed to determine the topic relationships of homomorphic ligands.

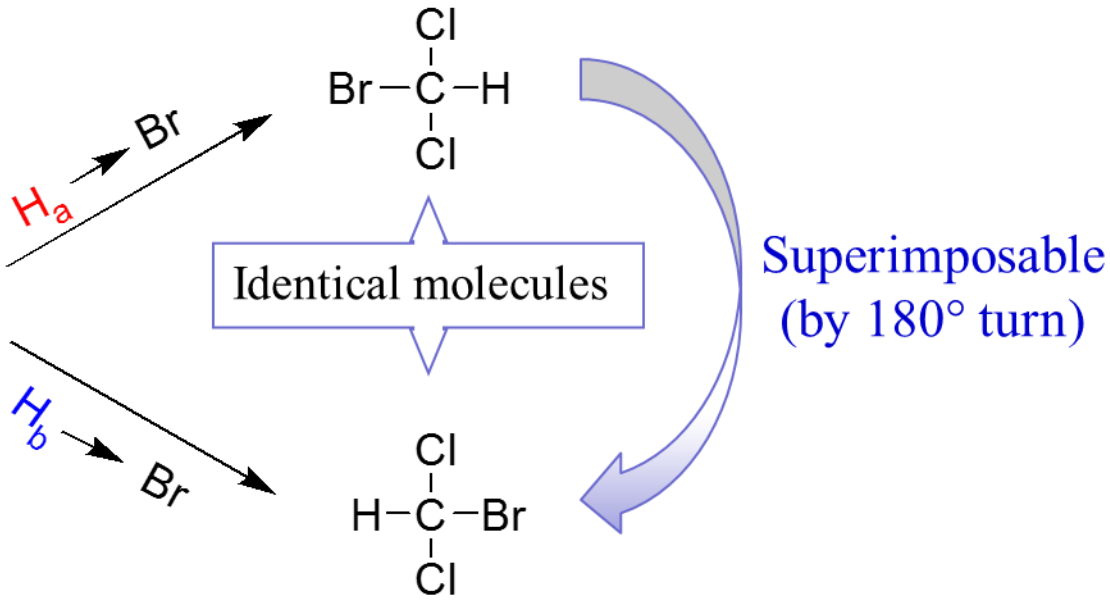


Homotopic Ligands

(Substitution And Addition Criteria)

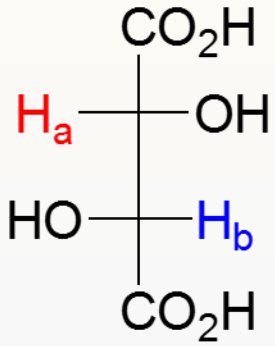
- Two homomorphic ligands are homotopic if replacement of first one and then the other by a different ligand leads to the same structure.
- The replacement ligand must be different not only from the original one but also from all other ligands attached to the same atom.





Homotopic Ligands

(Substitution And Addition Criteria)

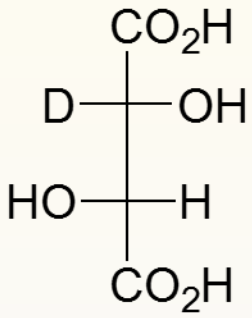


(+)-Tartaric acid is
(R, R)-Tartaric acid.

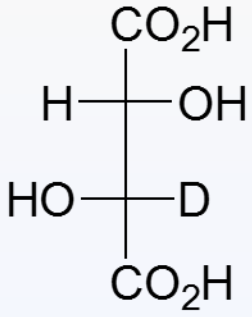
Replace **Ha** with D

When Fischer projection of one molecule is rotated 180° in the plane of paper it gives the other.

Replace **Hb** with D



Identical molecules



(2R, 3R)-Tartaric-2-d acid

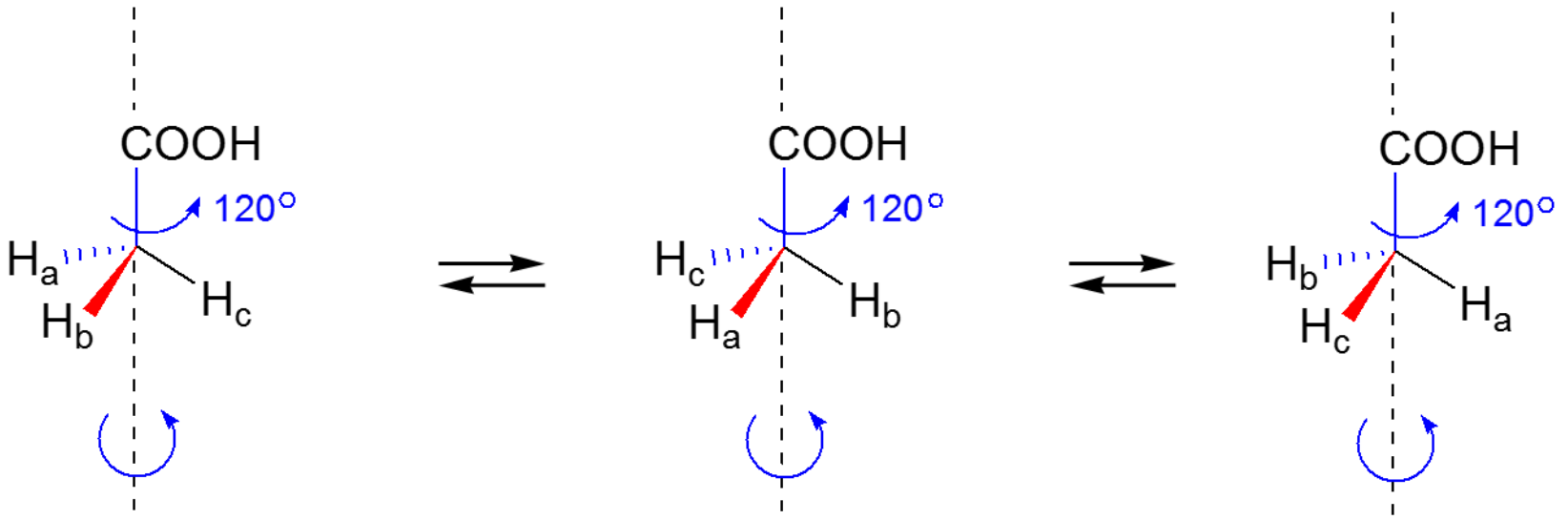
Two hydrogens, Ha and Hb are homotopic

So, Chirality Does Not Always Interfere In Having Homotopic Ligands!

Homotopic Ligands

(Symmetry Criterion)

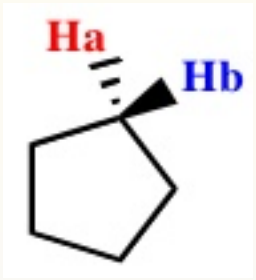
- Two ligands are homotopic if they can interchange places through operation of a C_n symmetry axis.
- In a molecule *e.g.*, acetic acid, two successive 120° rotations of a methyl group around the axis (C_3 rotational axis) allow each hydrogen to take the position of either of the other two with no structural change.



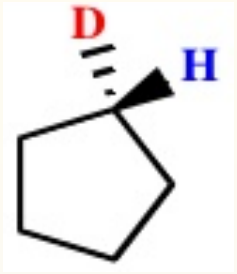
Acetic acid

All the hydrogens of methyl groups are homotopic

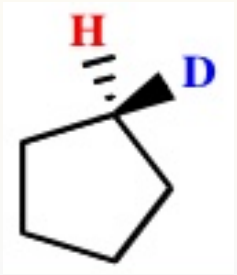
Exercise



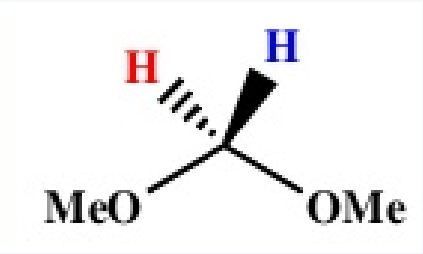
Replace **Ha** with D



Replace **Hb** with D

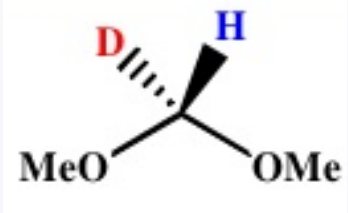


Substitution of Ha & Hb by D leads to formation of homomers and hence two hydrogens are homotopic

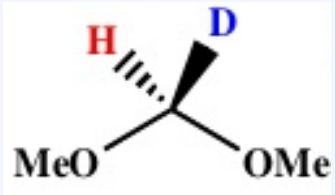


two hydrogens are homotopic

Replace **H** with D



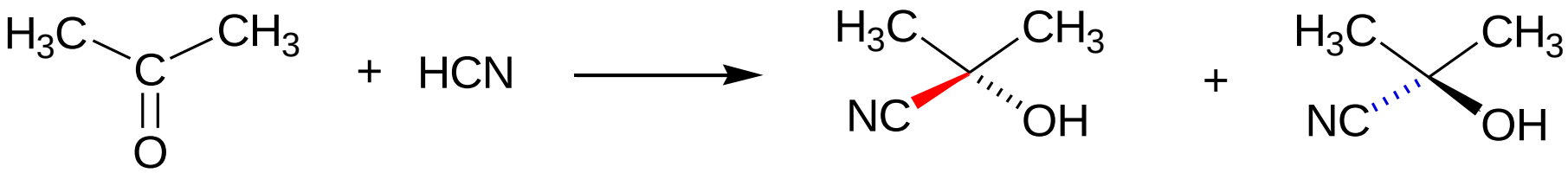
Replace **H** with D



Identical molecules

Homotopic Faces

- Two corresponding faces of a molecule (usually faces of a double bond) are homotopic when addition of the same reagent to either of the face gives the same product.
- For example, addition of HCN to acetone will give the same cyanohydrin, no matter to which face the addition occurs.



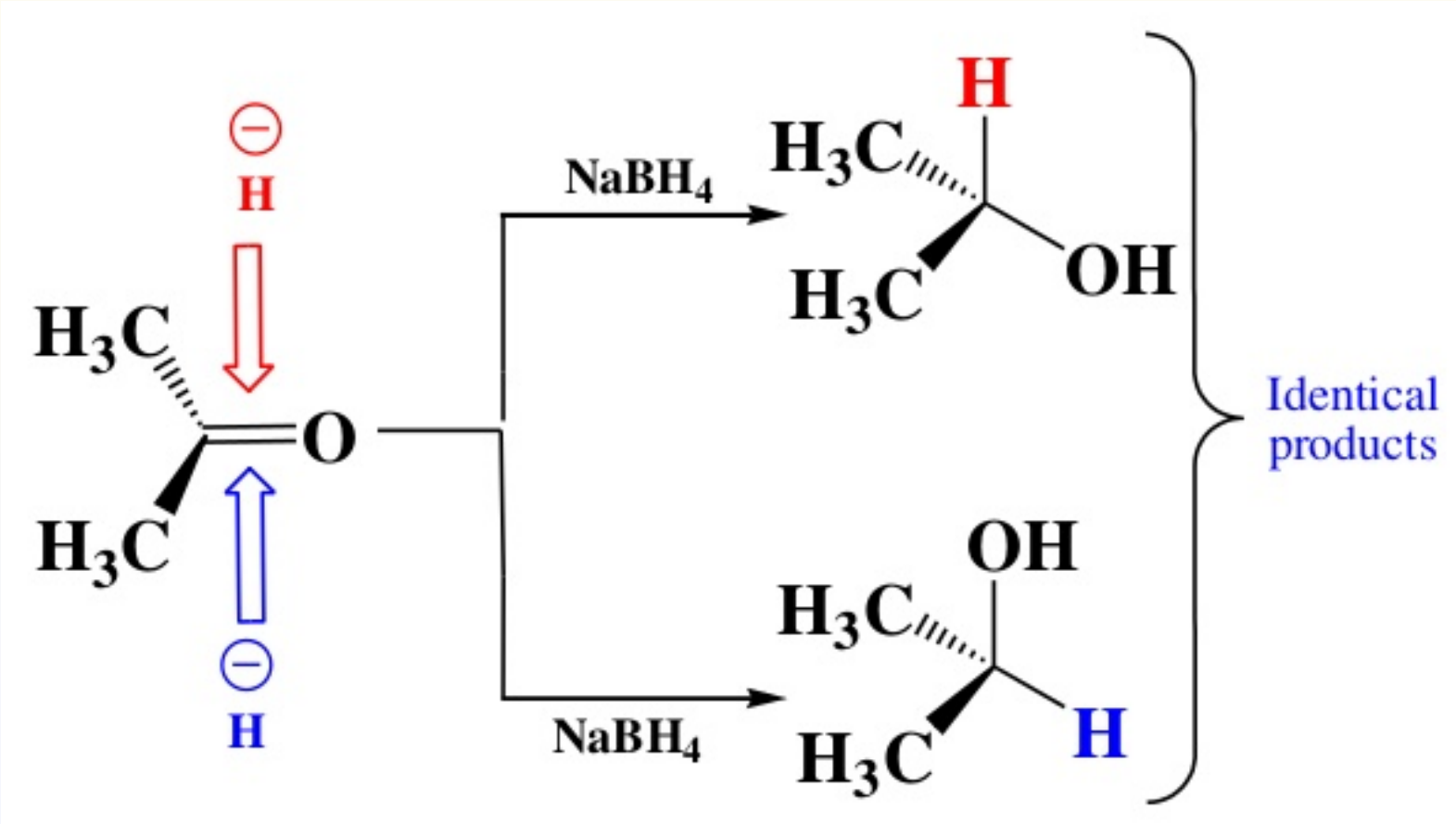
Homotopic faces

Front attack

Rear attack

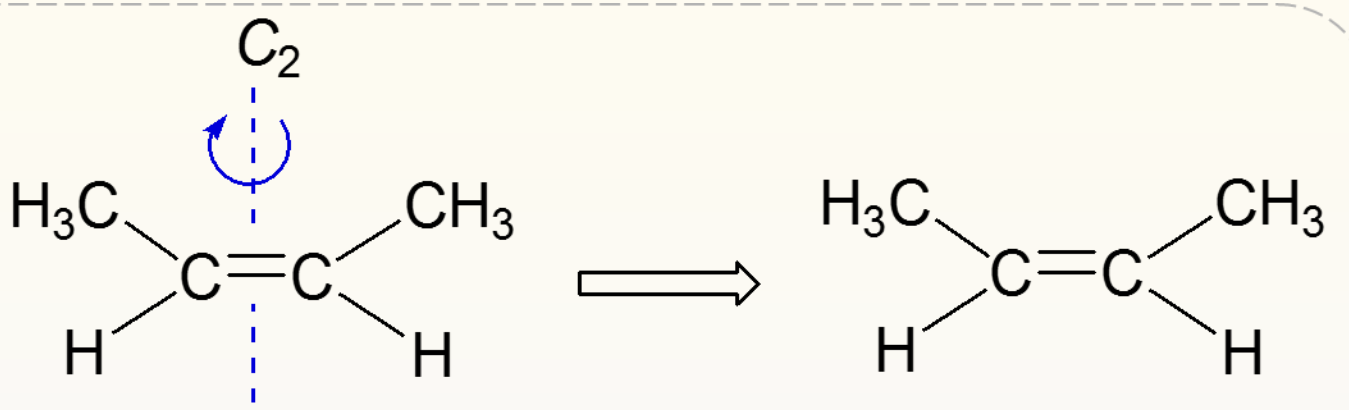
Identical molecules

Homotopic Faces

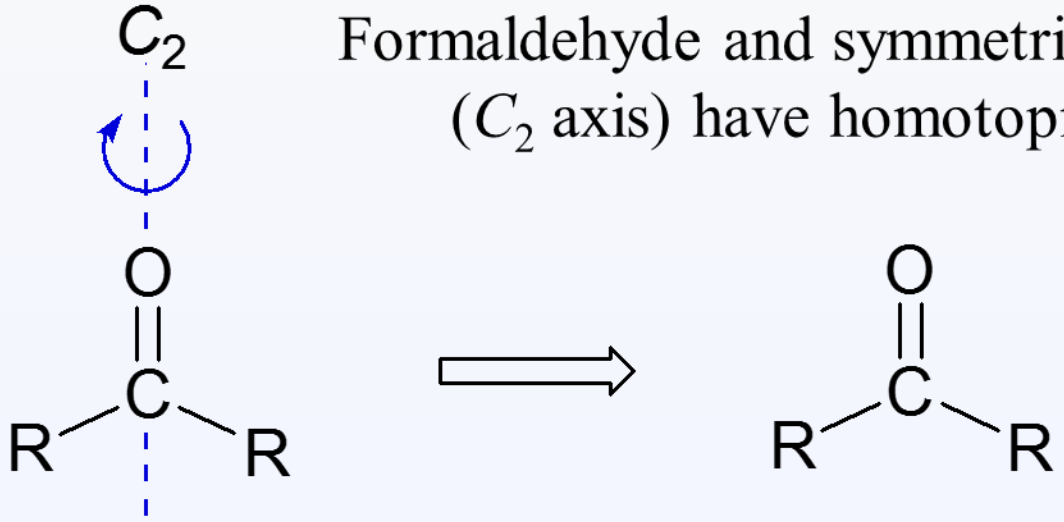


Two products are homomers. So acetone has homotopic face

Homotopic Faces



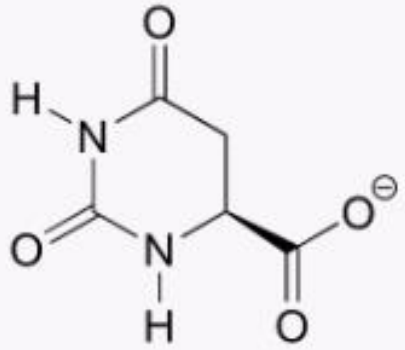
cis-2-Butene has homotopic faces (C_2 axis)



Formaldehyde and symmetrical ketones (C_2 axis) have homotopic faces

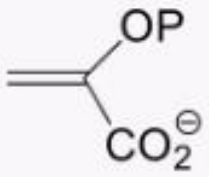
Q. Identify in the molecules below all pairs/groups of hydrogens that are homotopic, enantiotopic, or diastereotopic. When appropriate, label prochiral hydrogens as H_R or H_S.

a)



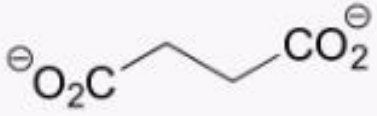
dihydroorotate
(a nucleotide biosynthesis intermediate)

b)



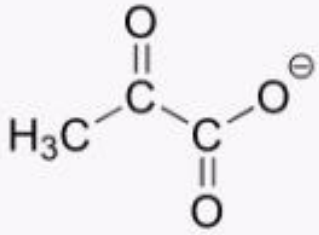
phosphoenolpyruvate
(a glycolysis intermediate)

c)



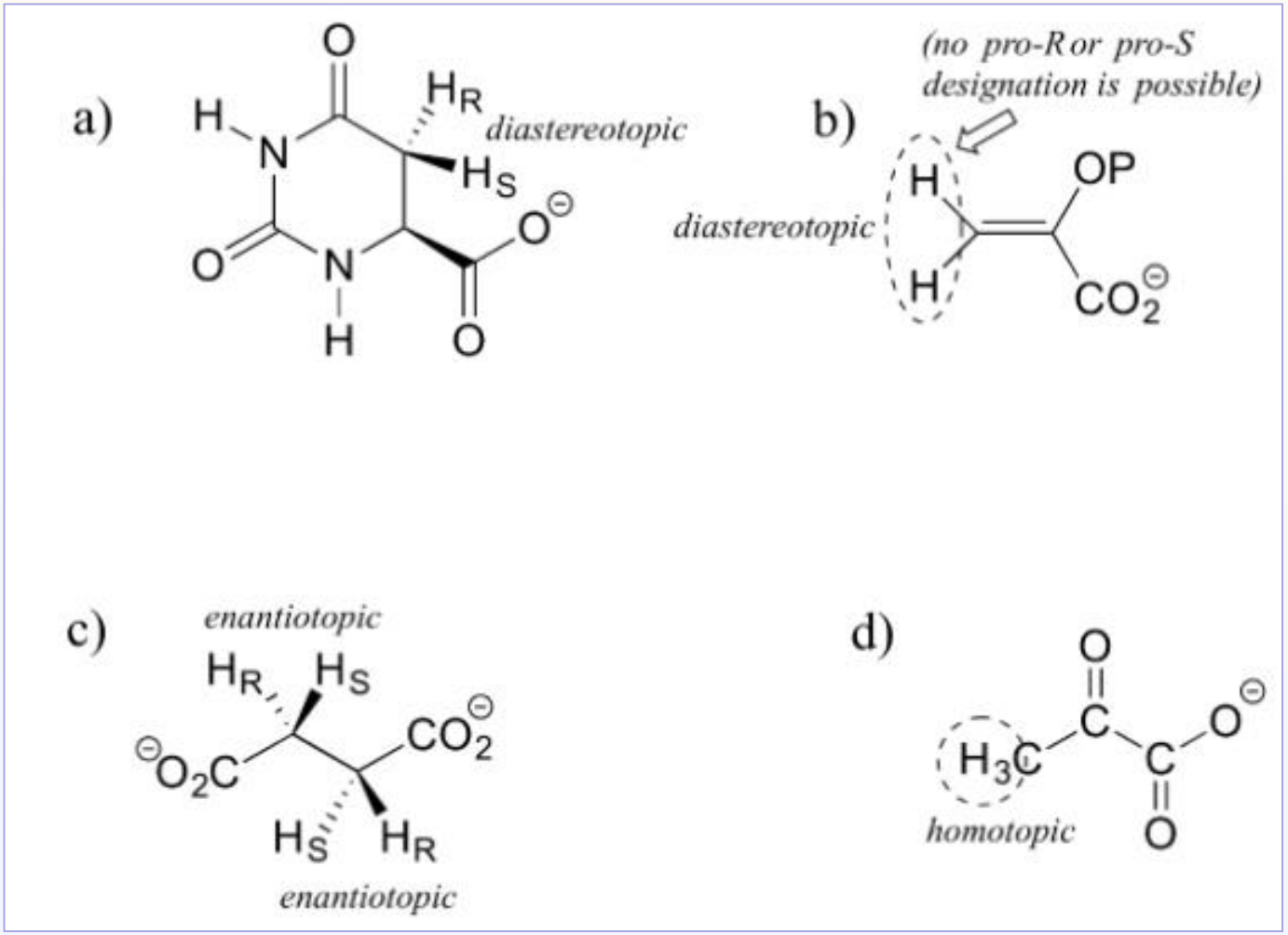
succinate
(a citric acid cycle intermediate)

d)

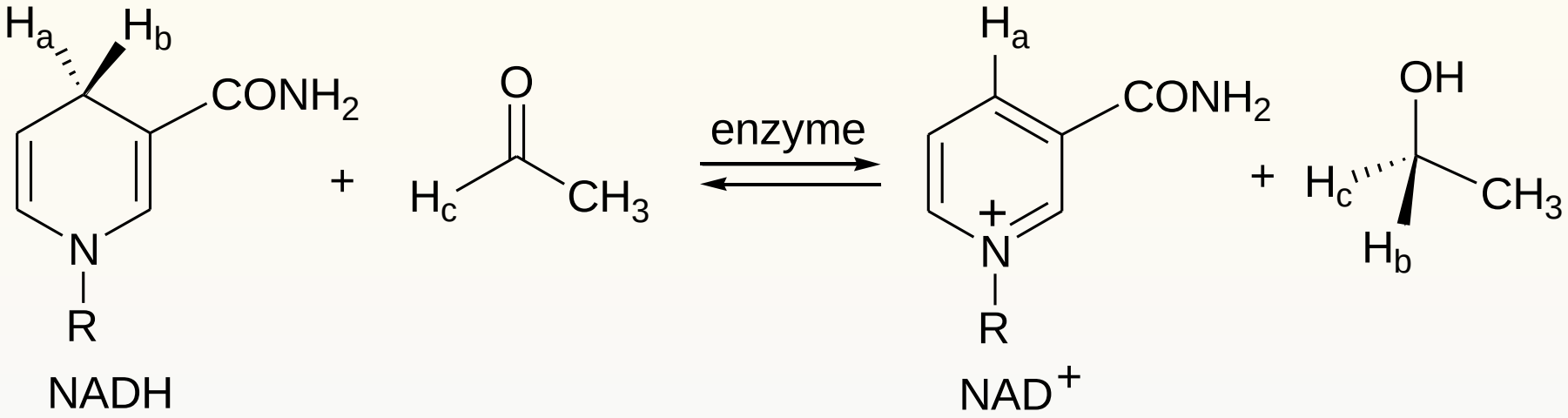


pyruvate
(endpoint of glycolysis)

Q. Identify in the molecules below all pairs/groups of hydrogens that are homotopic, enantiotopic, or diastereotopic. When appropriate, label prochiral hydrogens as H_R or H_S.



Practice Problems



Q.1 Consider the above reaction and answer the related questions:

(i) Label the prochiral hydrogens in NADH as pro-*R* and pro-*S*. 1.5

(ii) Depict the top face in acetaldehyde as *Re* or *Si* 0.75

(iii) Indicate the methylene hydrogens in ethanol as enantiotopic or diastereotopic. Give your reasoning. 1.5



Heterotopicity and Enzyme Catalyzed Reactions

What is life



Life
is full of
Activities!



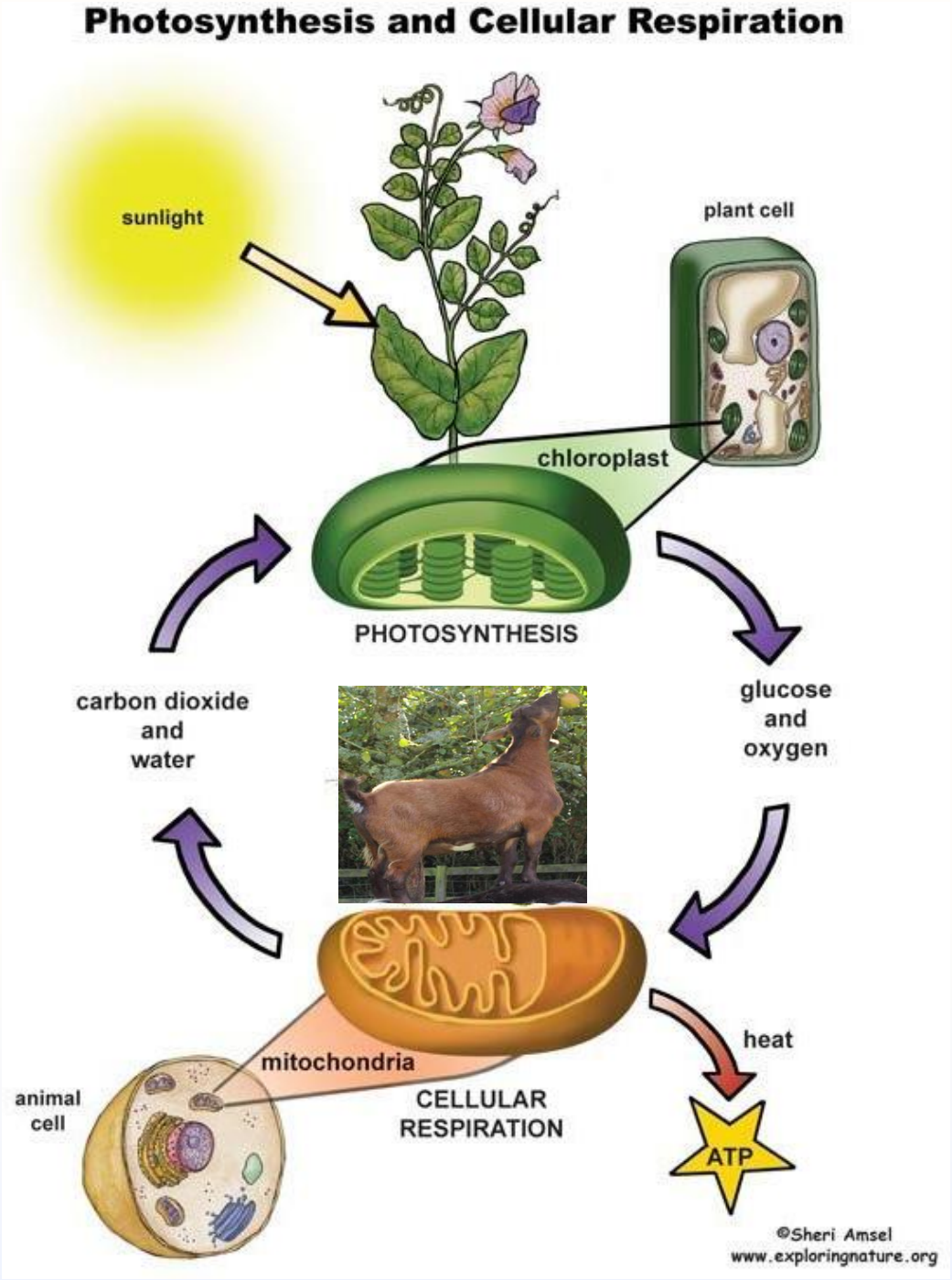
To perform work, living cells require transfusions of energy from outside sources



The Goat obtains Energy for its cells by eating plants



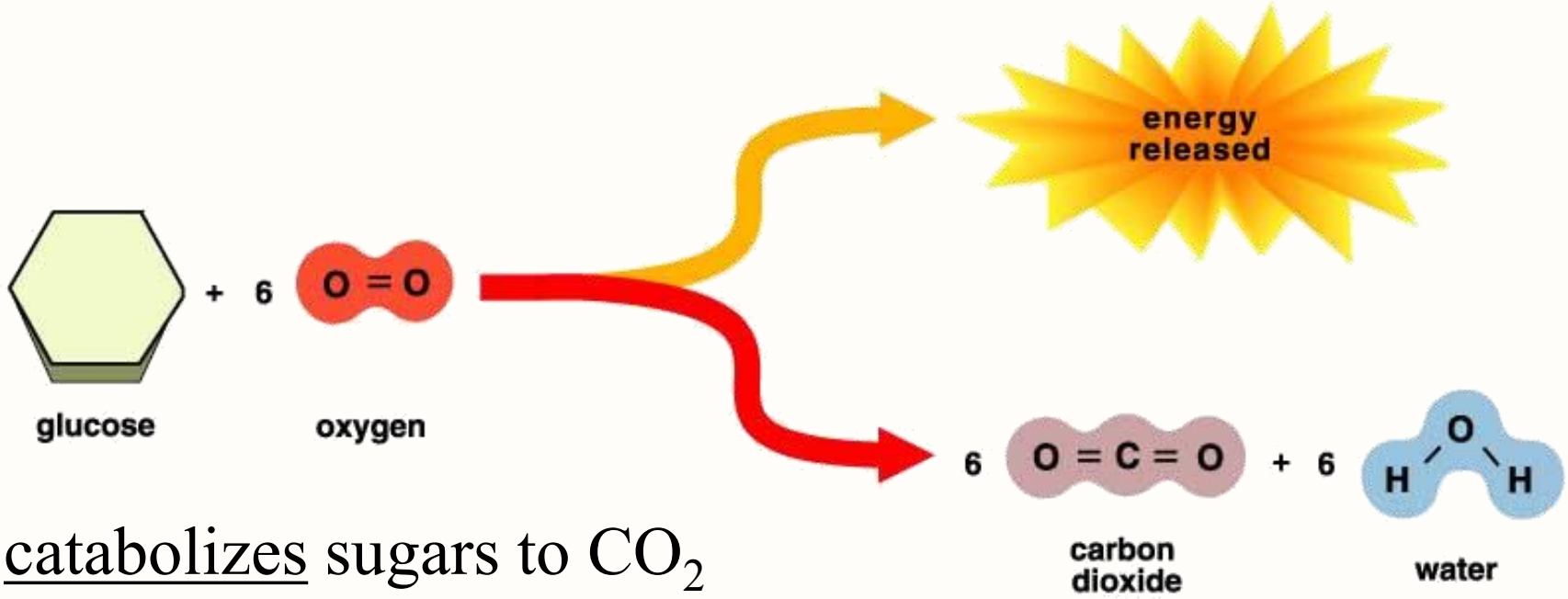
Photosynthesis and Cellular Respiration



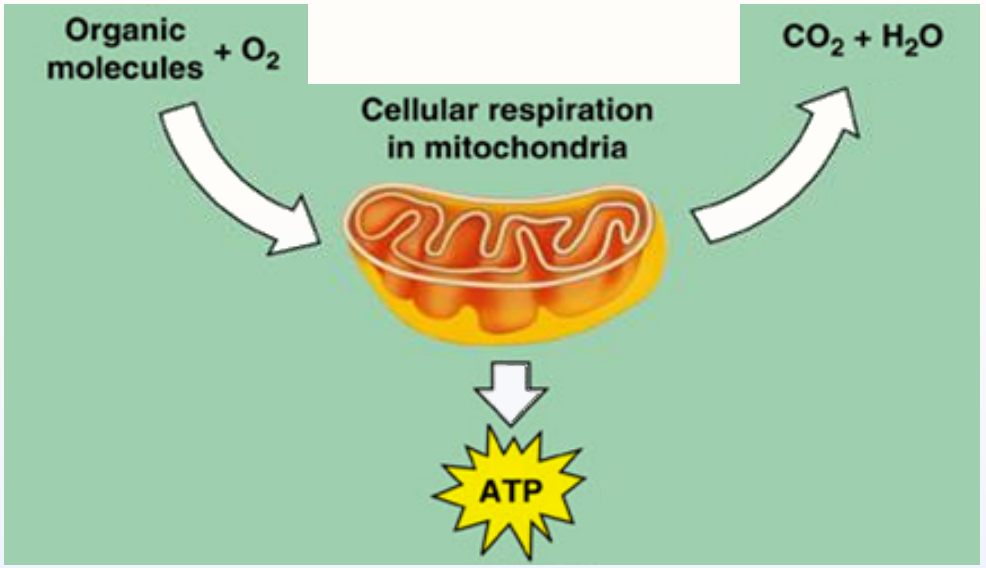
Cellular Respiration

Cellular respiration is the set of metabolic reactions and processes that take place in the cells of organisms to convert biochemical energy from nutrients into Adenosine TriPhosphate (ATP), and then release waste products.

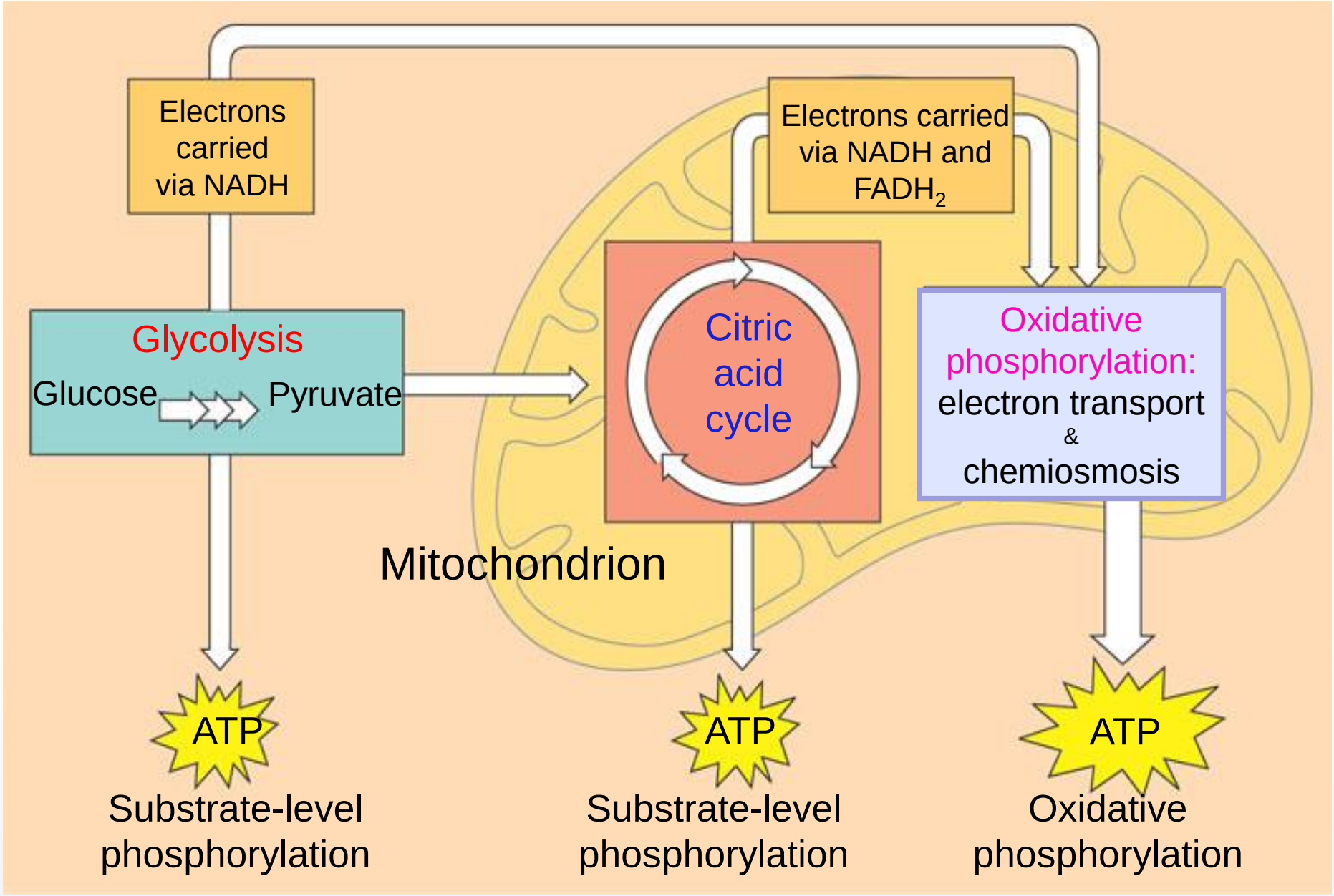
Cellular Respiration



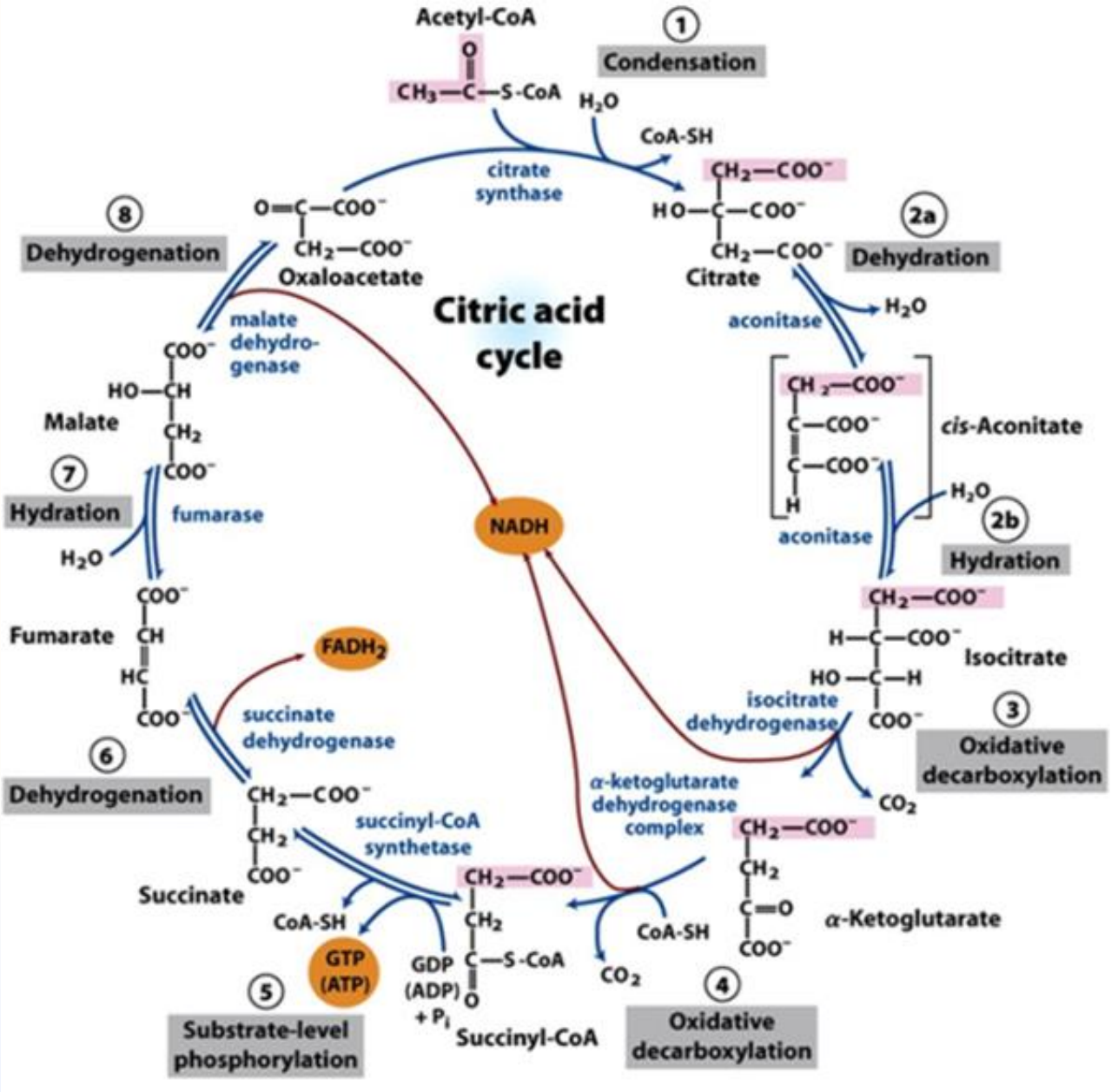
- <u>catabolizes</u> sugars to CO_2
- requires O_2
- at mitochondrion



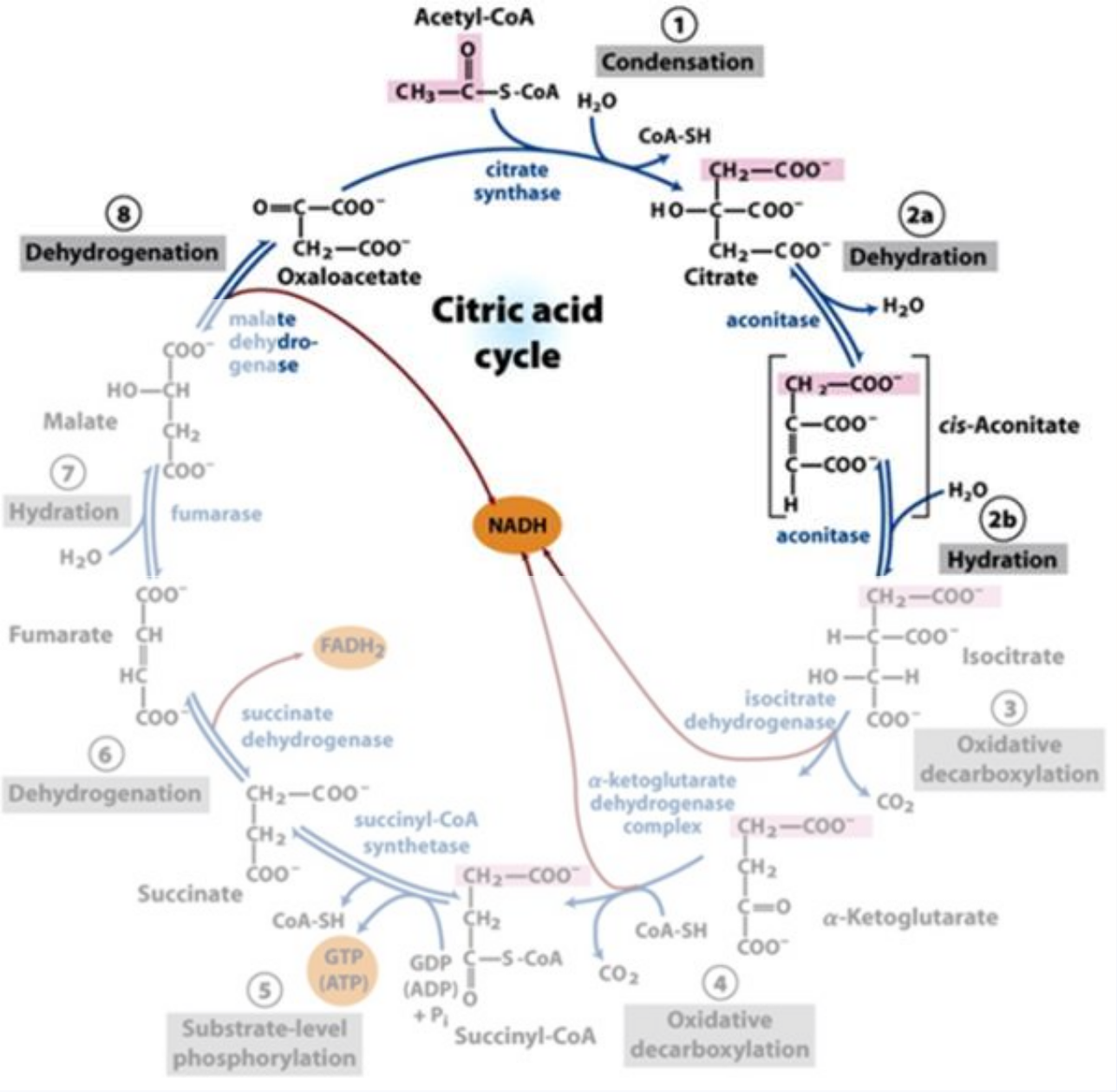
An overview of cellular respiration



Let's Have A Look At TriCarboxylic Acid Cycle

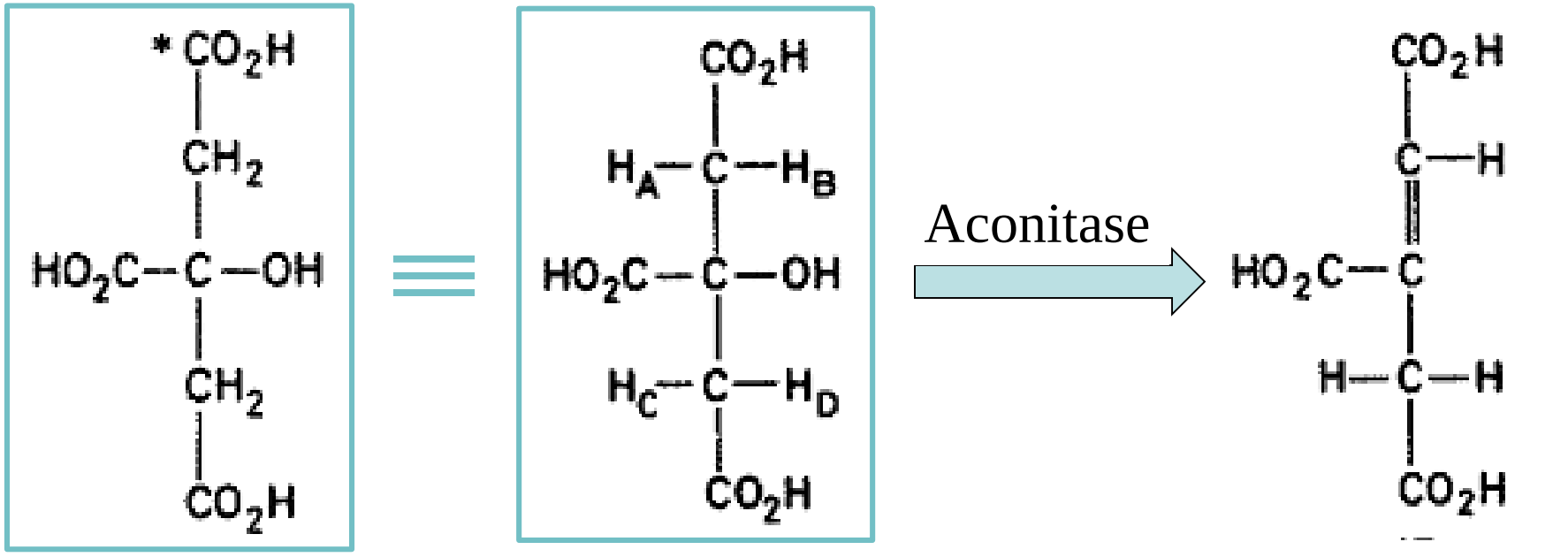


Part of TriCarboxylic Acid Cycle (TCA Cycle)



Citric Acid and *cis*-Aconitic Acid

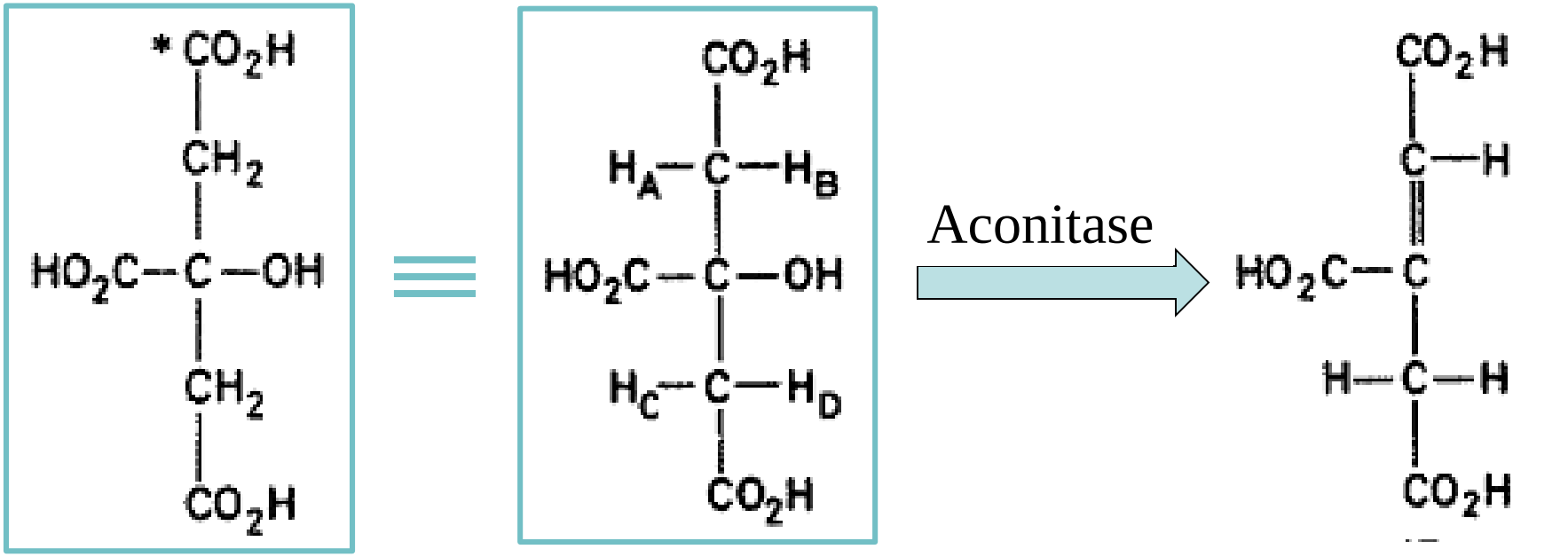
- In citric acid, can the four methylene hydrogens H_A, H_B, H_C, H_D be distinguished by virtue of their involvement in the enzymatic dehydration of citric acid to *cis-aconitic* acid?



Citric acid Citric acid *cis*-Aconitic acid

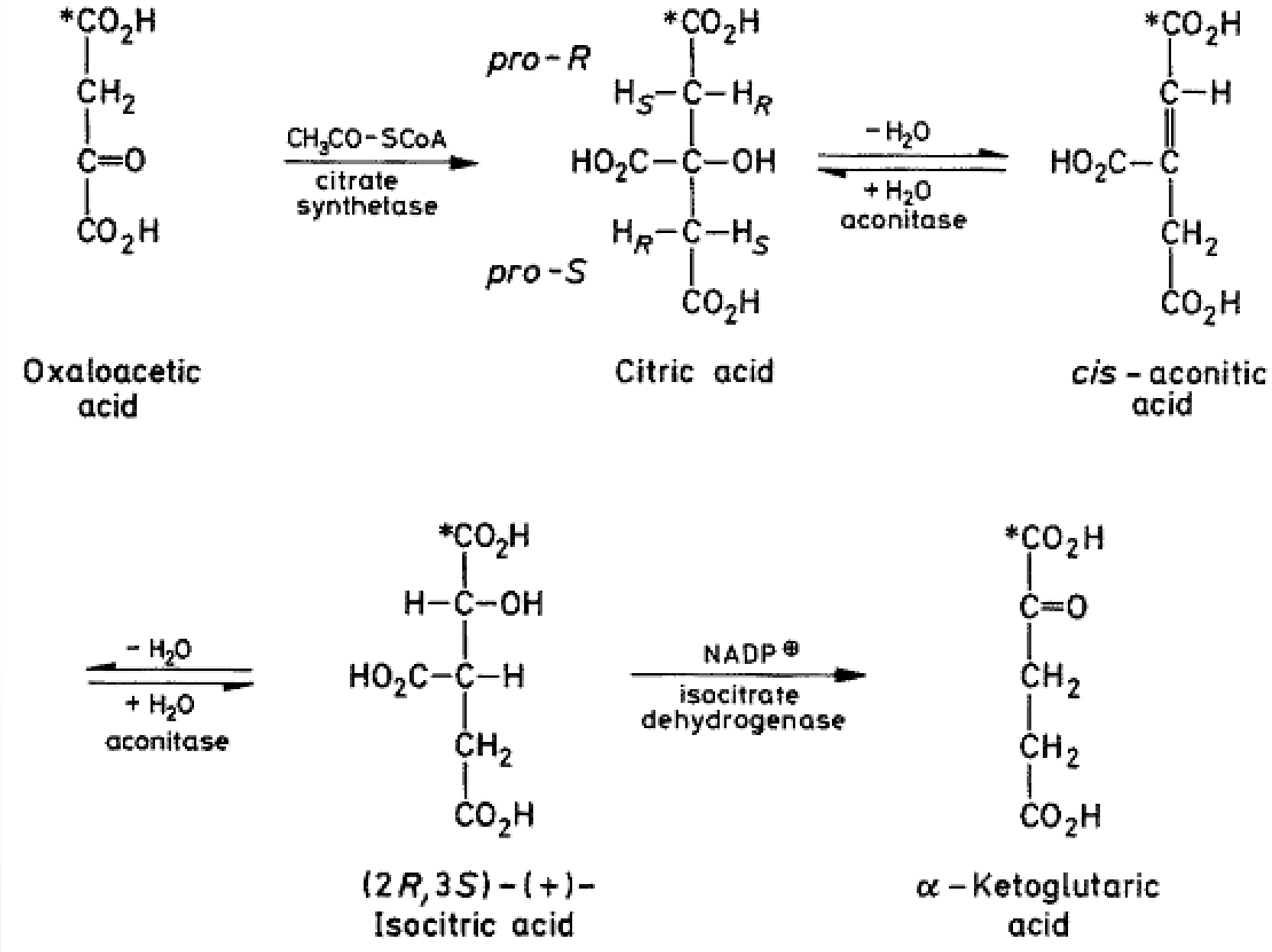
Citric Acid and *cis*-Aconitic Acid

- The methylene hydrogens H_A and H_B (or H_C and H_D) are diastereotopic. Since the enzyme is chiral, it also "sees" the enantiotopic ligands H_A and H_C (or H_B and H_D) in a diastereotopic environment and thus can distinguish between them.



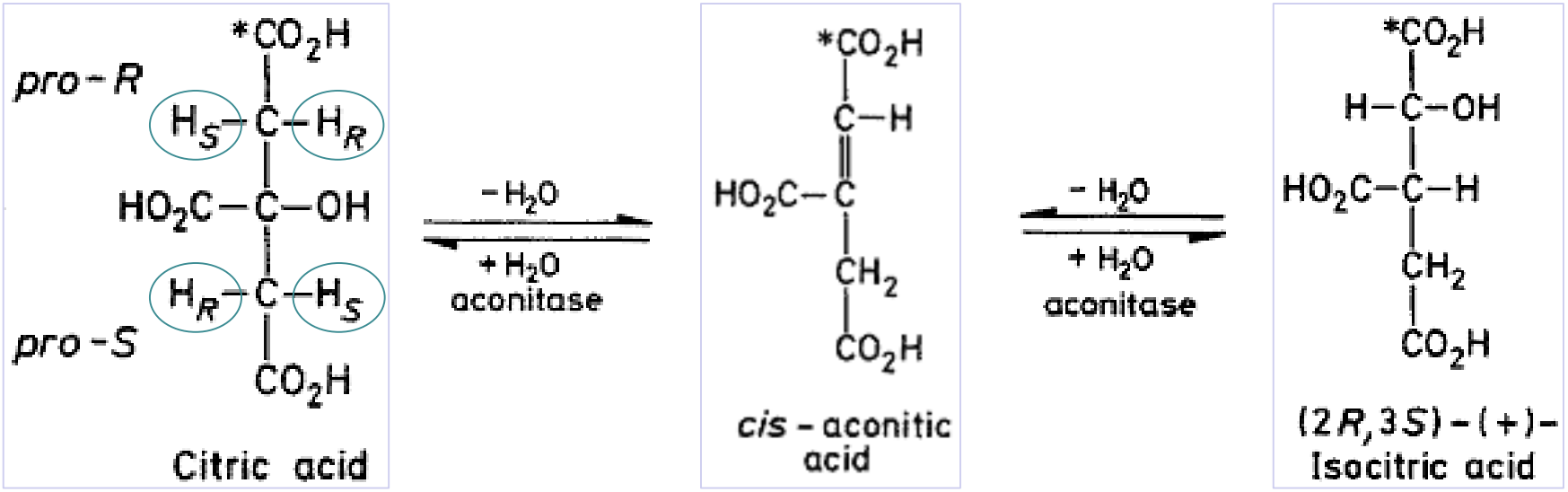
Citric acid Citric acid *cis*-Aconitic acid

Part of TriCarboxylic Acid Cycle



Part of TriCarboxylic Acid Cycle

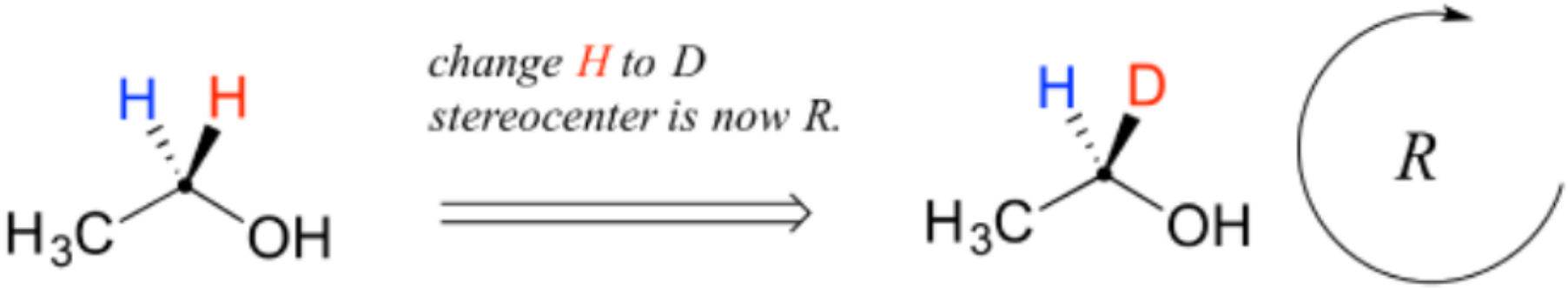
- When citric acid is equilibrated with isocitric acid via aconitic acid in the presence of aconitase but with D_2O instead of H_2O, only one of the four methylene positions acquires deuterium, hence only one of the four hydrogens can be eliminated in the dehydration step to aconitic acid and replaced in the reverse reaction.

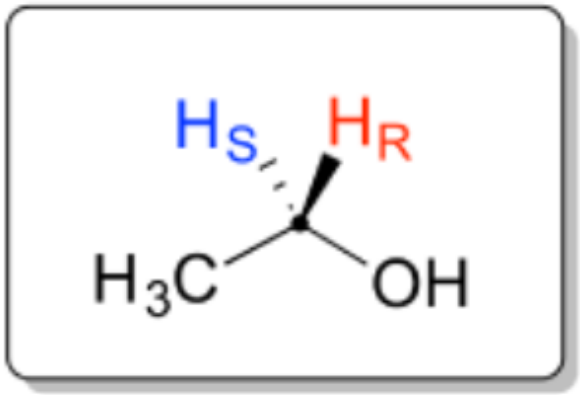


- How to identify which hydrogen atom is eliminated during the dehydration of citric acid mediated by aconitase?

Flashback

(*pro*-*R* and *pro*-*S*)



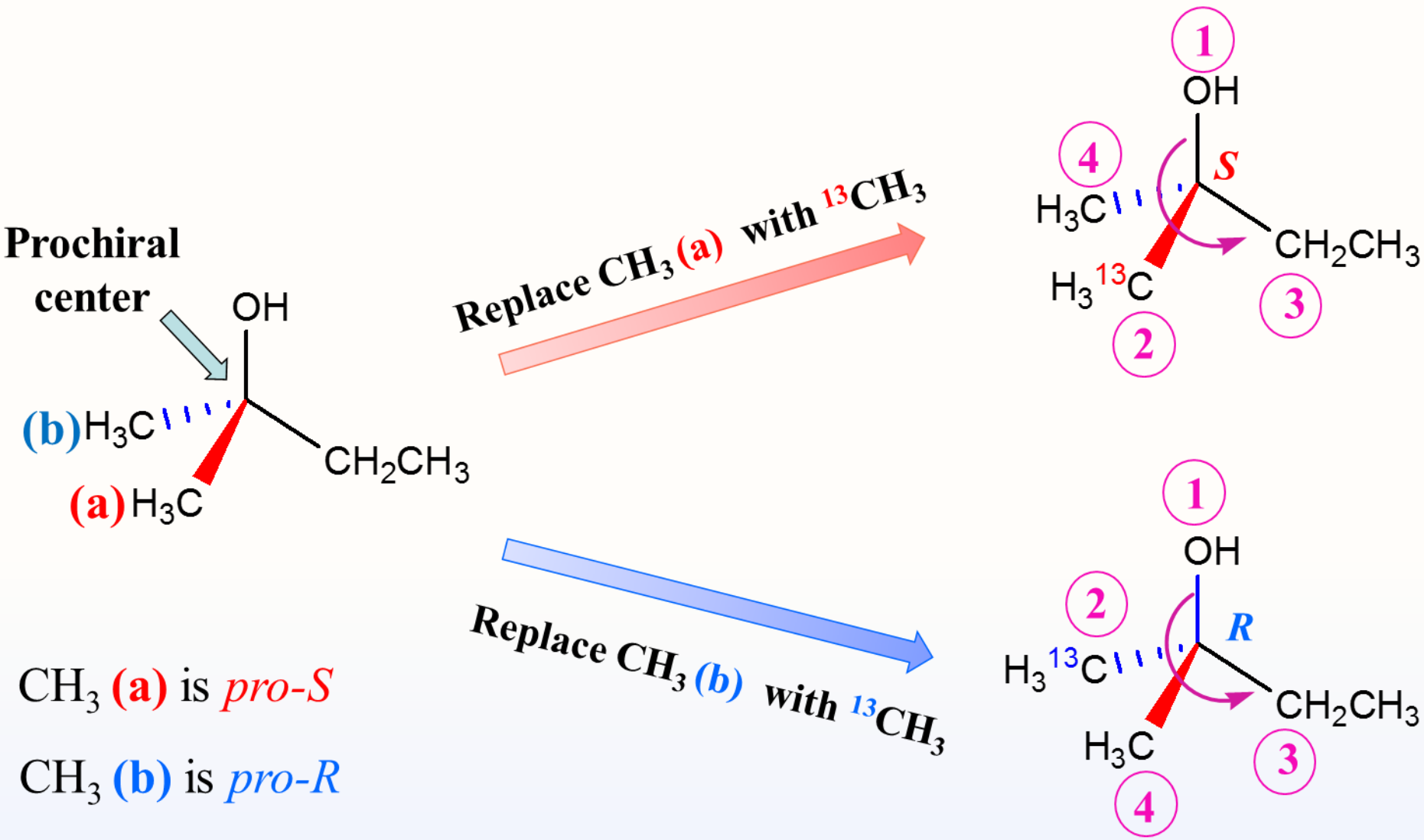


- *pro*-*R* hydrogen is labelled as H_R
- *pro*-*S* hydrogen is labelled as H_S

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(*pro*-*R* and *pro*-*S*)

- Groups other than hydrogens can be considered prochiral

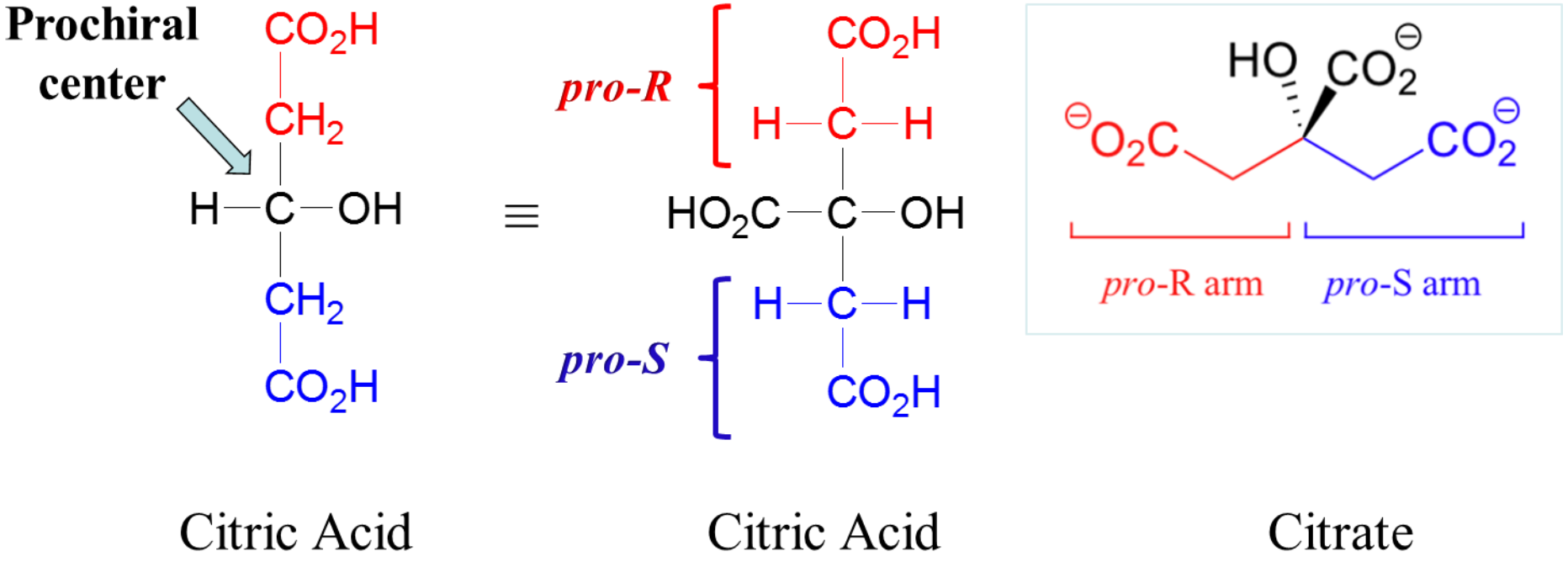


CH_3 **(a)** is *pro-S*

CH_3 **(b)** is *pro-R*

(*pro*-*R* and *pro*-*S*)

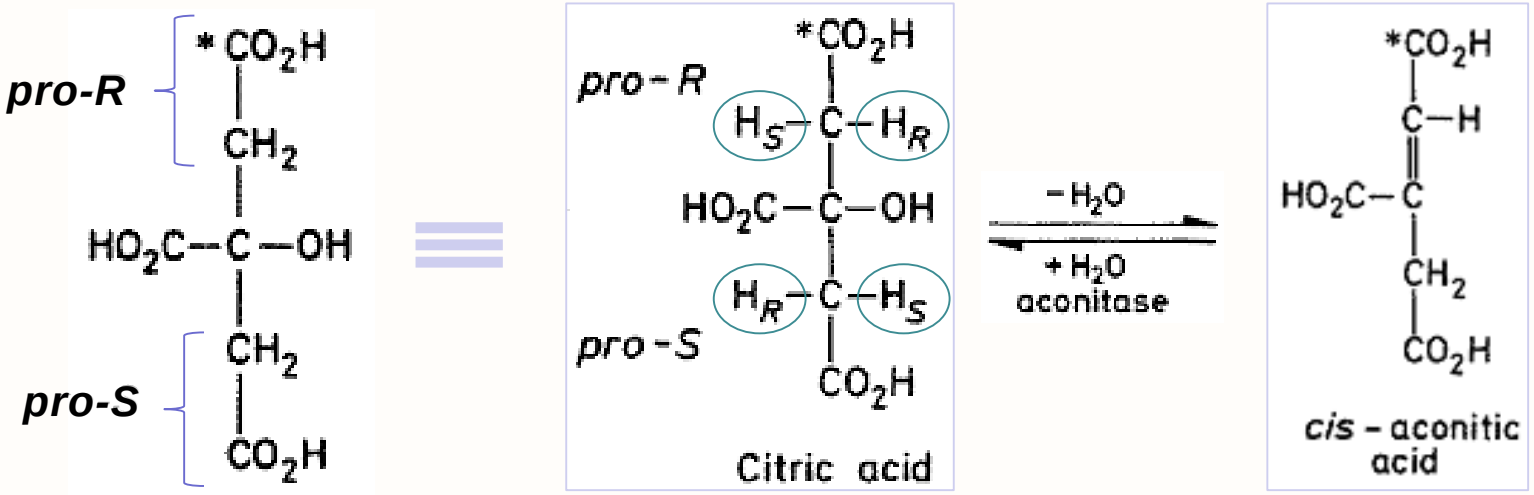
- Citrate is another example. The central carbon is a prochiral center with two 'arms' that are identical except that one can be designated *pro*-*R* and the other *pro*-*S*.



Citric Acid | Citric Acid | Citrate

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Part of TriCarboxylic Acid Cycle

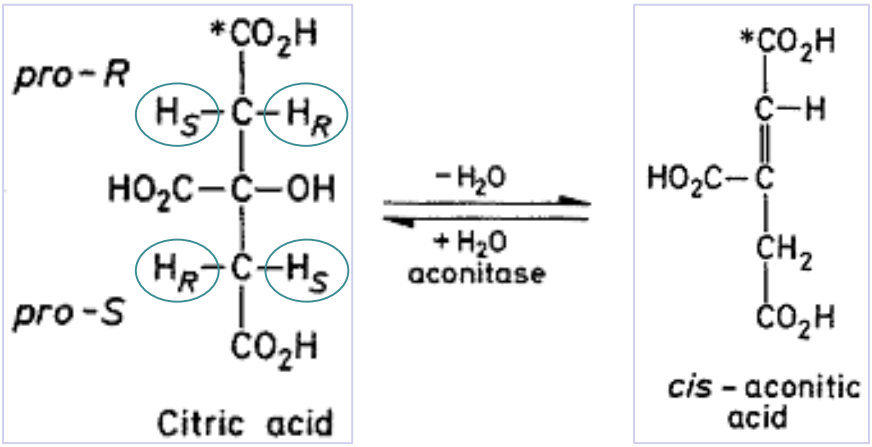


- The problem of identifying which hydrogen atom is eliminated can be factorized into two parts:

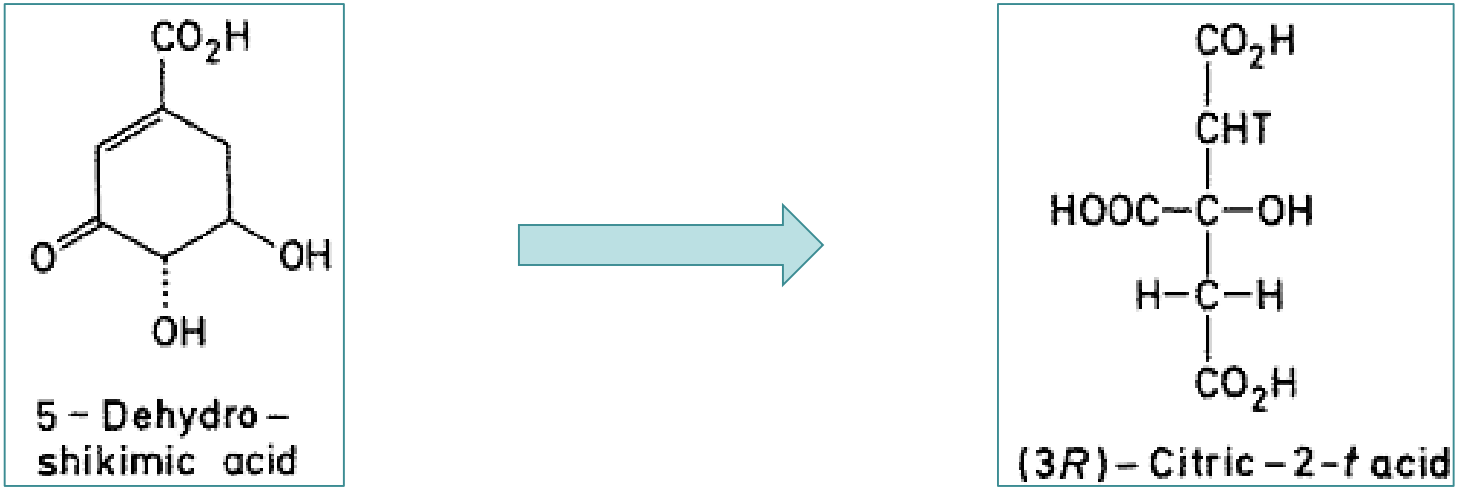
  1. Does the hydrogen come from the *pro-R* or the *pro-S* CH_2CO_2H branch of the citric acid?

  2. Within the branch specified is the hydrogen abstracted H_R or H_S?

1. Does the hydrogen come from the *pro-R* or the *pro-S* CH_2CO_2H branch of the citric acid?

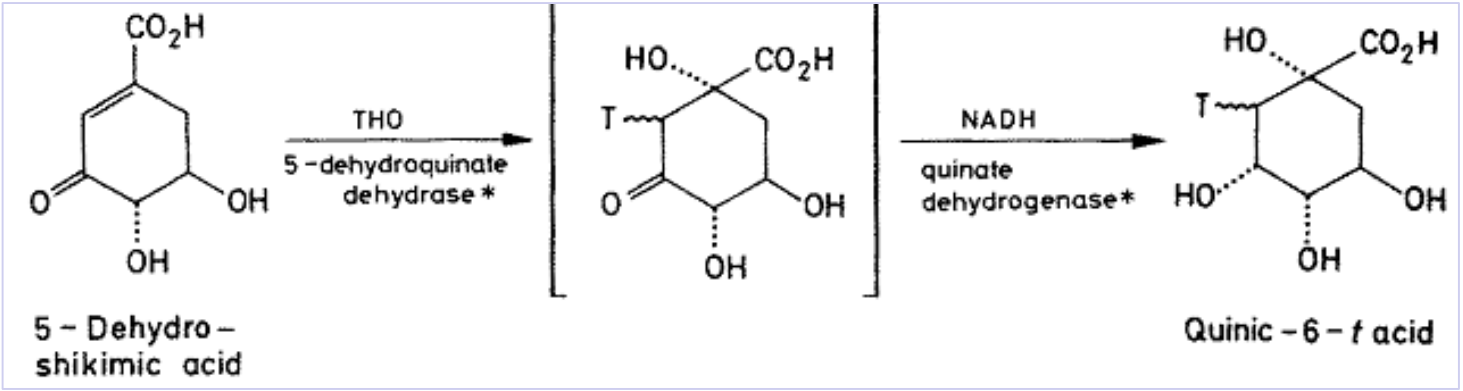


- The first problem was solved through synthesis (followed by enzymatic degradation) of tritiated citric acid [(3*R*)-Citric-2-*t* acid] stereospecifically labeled in the *pro-R* branch.
- The starting material for this synthesis is the naturally occurring 5-dehydroshikimic acid of known configuration.

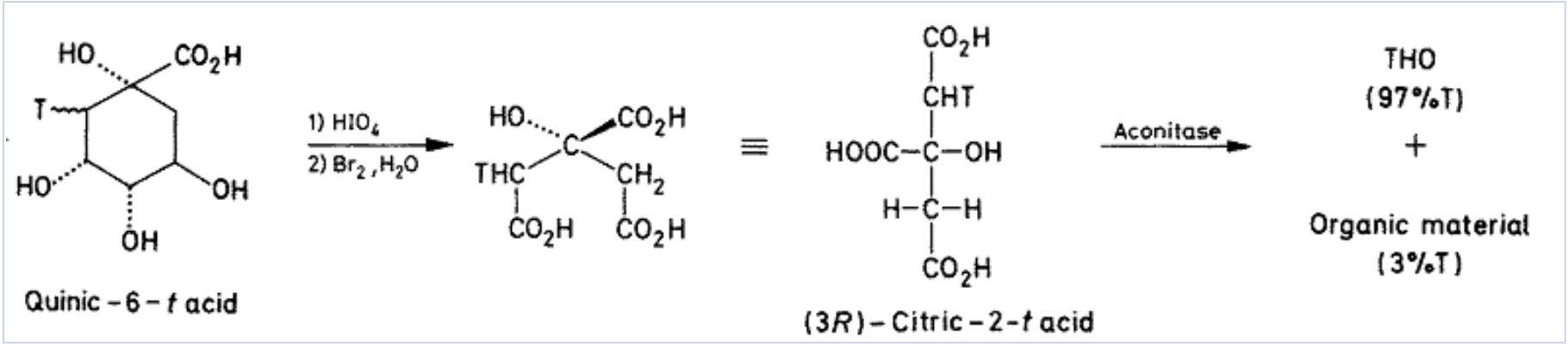


Stereospecific Synthesis of (3*R*)-Citric-2-*t* Acid

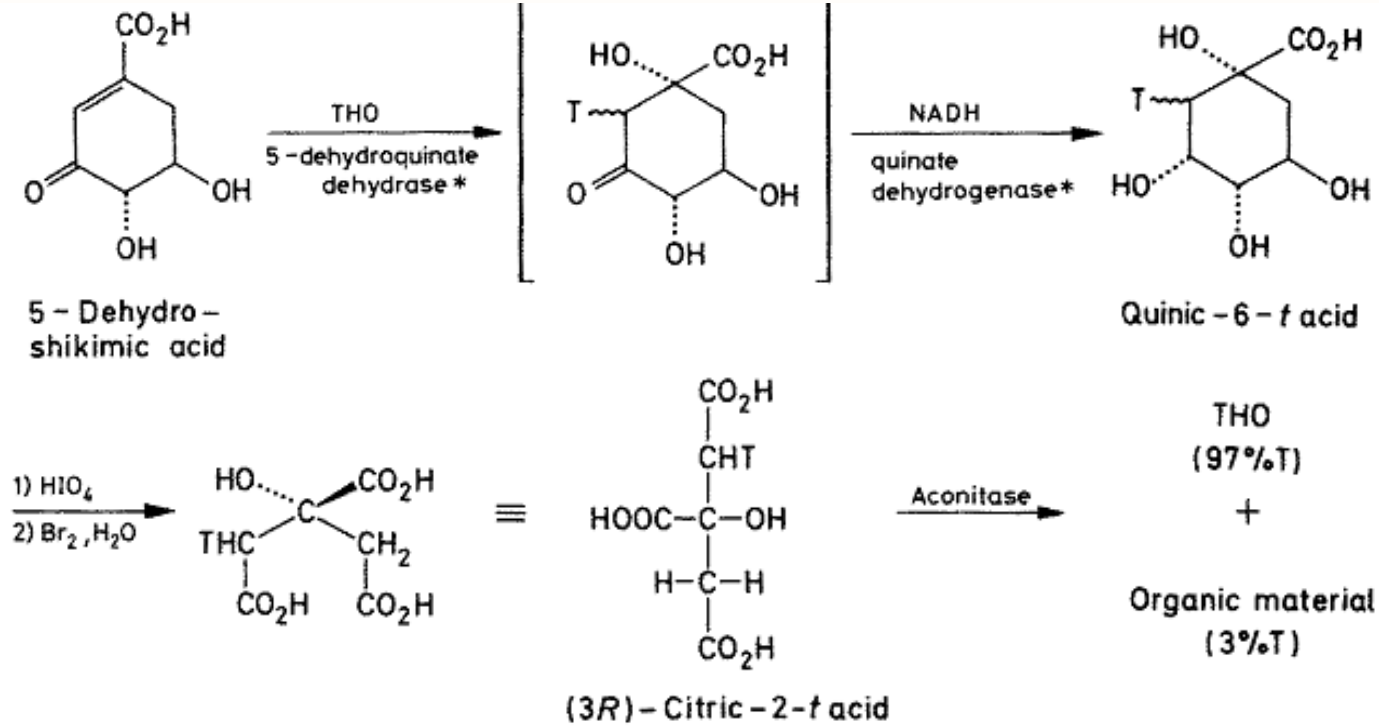
- Enzymatic hydration of the double bond in 5-dehydroshikimic acid with tritiated water gives the corresponding dihydro compound whose configuration, at C(1), is R. The hydrated material, without isolation, is reduced by NADH in the presence of a second enzyme to (1R)-quinic-6-*t* acid.



- (1R)-quinic-6-*t* acid is cleaved by periodic acid to a dialdehyde subsequently oxidized by bromine water to citric-2-*t* acid.

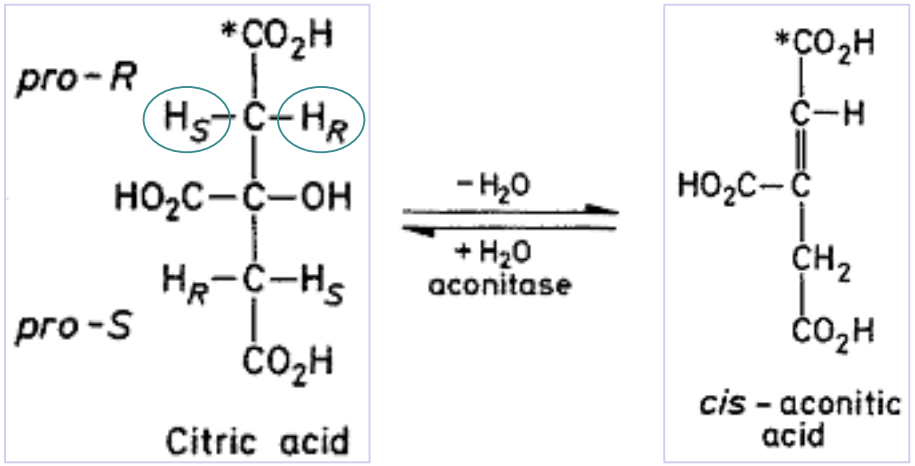


Stereospecific Synthesis of (3*R*)-Citric-2-*t* Acid

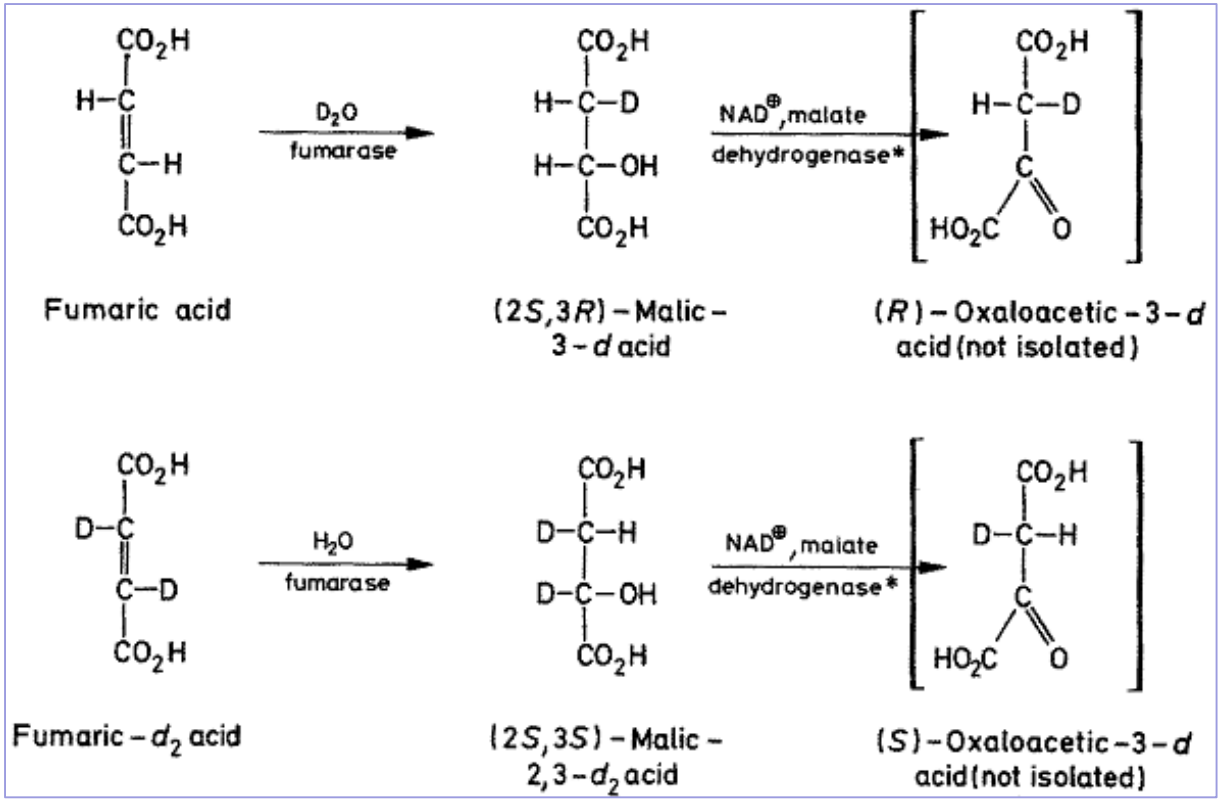


- The configuration of (3*R*)-Citric-2-*t* acid, from the method of synthesis, at C(3) must be R. The label is therefore in the *pro-R* branch.
- When the labeled citric acid is treated with aconitase, the aconitic (and/or isocitric) acid produced is nearly free of tritium whereas the water eliminated carries nearly all the label.
- Since the label was in the *pro-R* branch it follows that the *pro-R* branch is the aconitase-active one.

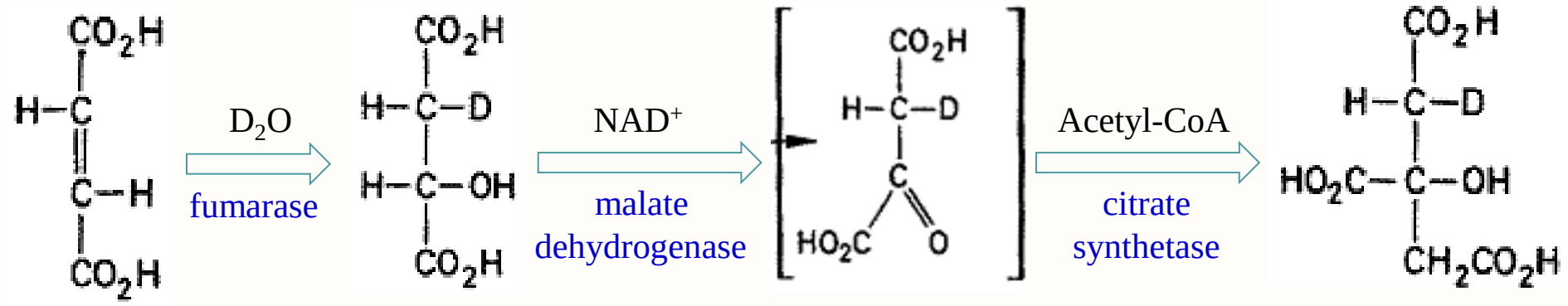
2. We turn now to the question whether the hydrogen removed within the *pro-R* branch *is pro-R* or *pro-S*.



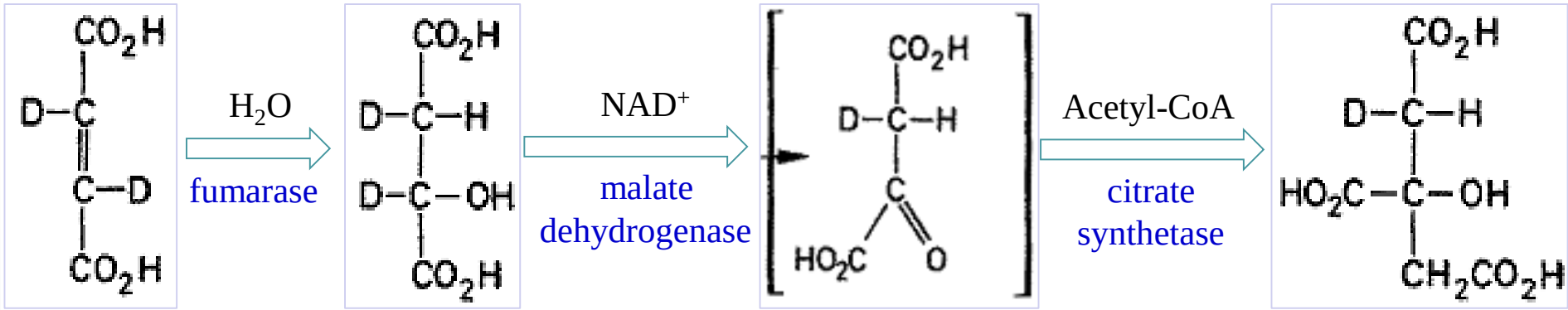
- To solve this question, (2*S*, 3*R*)- and (2*S*, 3*S*)-malic-3-*d* acids were synthesized.
- These acids were then oxidized enzymatically to oxaloacetic acid-3-d (3*R* or 3*S*, depending on the precursor).



Stereospecific Labeling of Citric Acid at C(2)



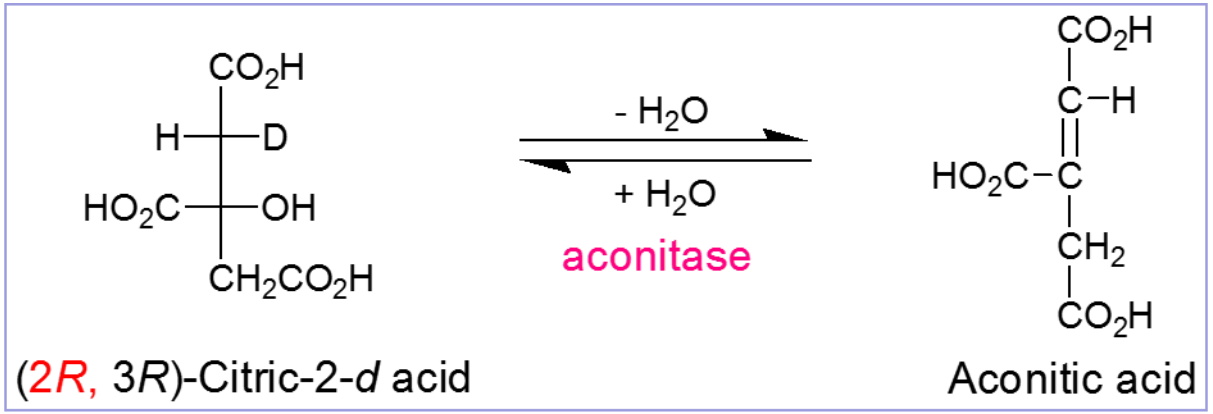
Fumaric acid | (2*S*, 3*R*)-Malic-3-*d* acid | (*R*)-Oxoloacetic-3-*d* acid (not isolated) | (2*R*, 3*R*)-Citric-2-*d* acid



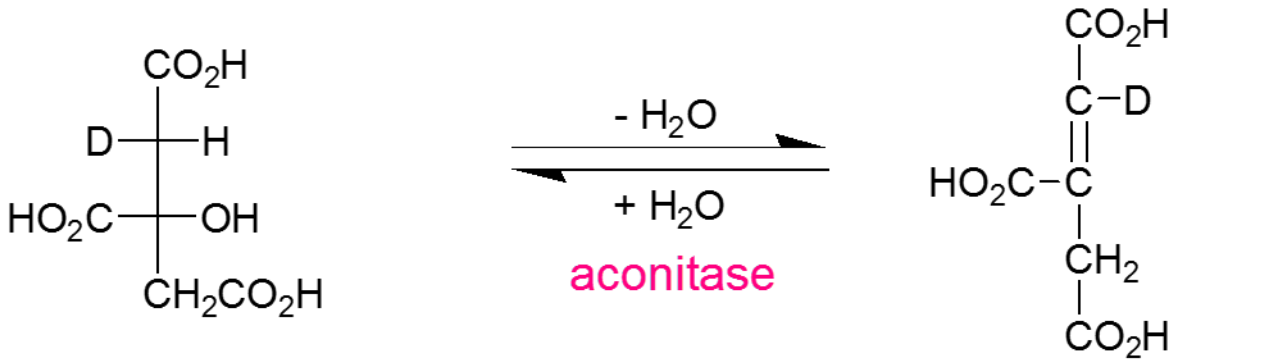
Fumaric-d_2 acid | (2*S*, 3*S*)-Malic-2,3-d_2 acid | (*S*)-Oxoloacetic-3-*d* acid (not isolated) | (2*S*, 3*R*)-Citric-2-*d* acid

These two reactions were carried out simultaneously

- Each of the two citric-2-d acids [(2*R*,3*R*)-citric-2-*d* acid and (2*S*,3*R*)-citric-2-*d* acid] was incubated with aconitase.
- The *(2R,3R)-citric-2-d* acid lost all of its deuterium in the dehydration to aconitic acid whereas the 2*S*, 3*R* isomer retained at least 80 ~o of it, i.e. the *pro-R* hydrogen at C(2) is lost.
- Thus, it may be concluded from the two experiments taken together that the *pro-R* hydrogen in the *pro-R* CH_2CO_2H branch is the one labilized by aconitase.



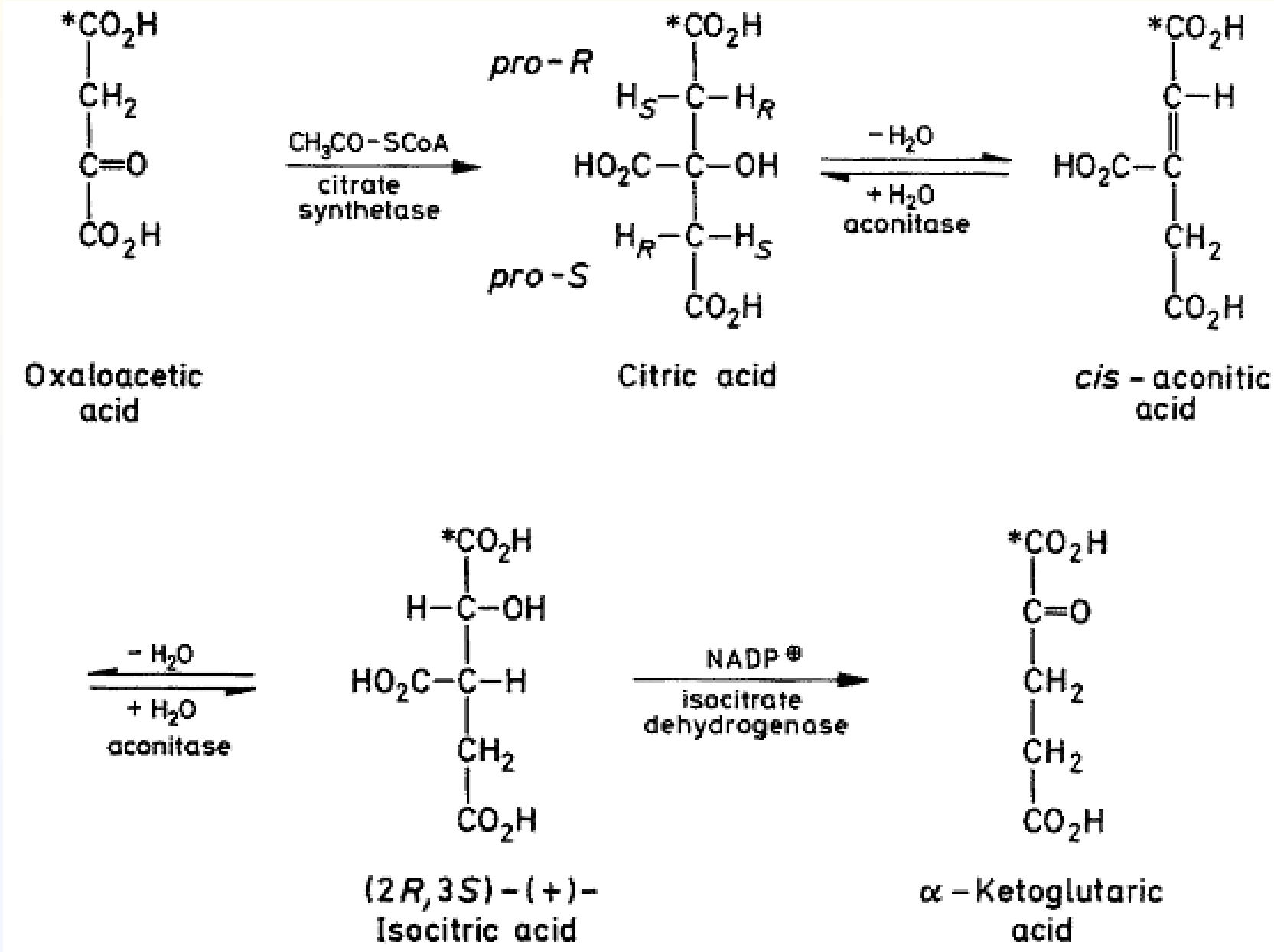
(2*R*, 3*R*)-Citric-2-*d* acid — Aconitic acid



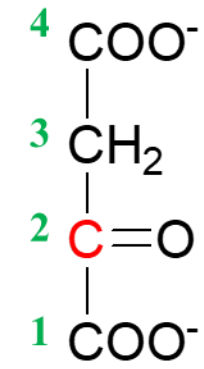
(2*S*, 3*R*)-Citric-2-*d* acid — Aconitic-*d* acid

Heterotopicity and Enzyme Catalyzed Reactions

Part of TriCarboxylic Acid Cycle

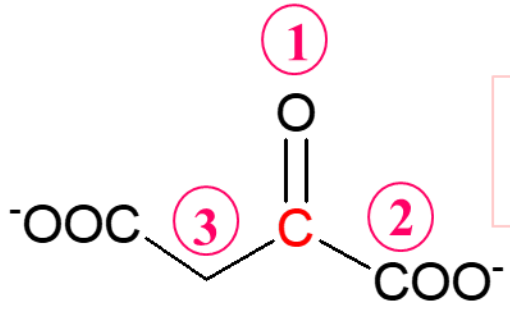


Stereochemistry of Citric Acid Cycle

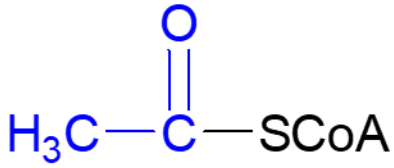


Oxaloacetate

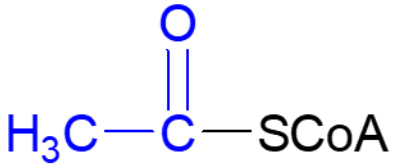
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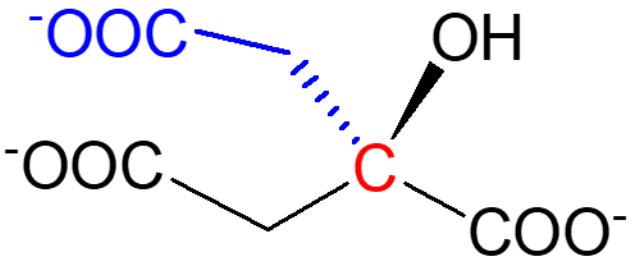
Oxaloacetate is viewed from its *Re* face



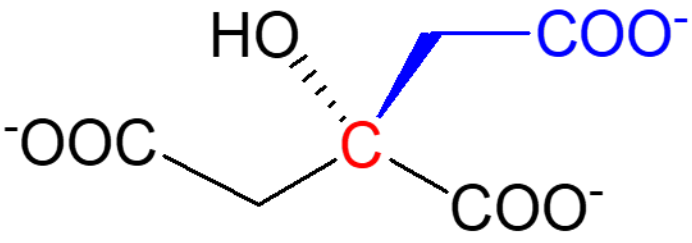
Citrate synthetase (*Si* face)



Citrate synthetase (*Re* face)

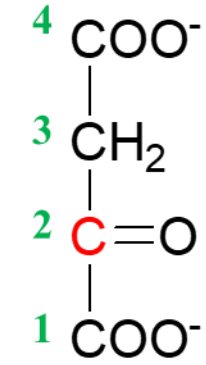


Citrate



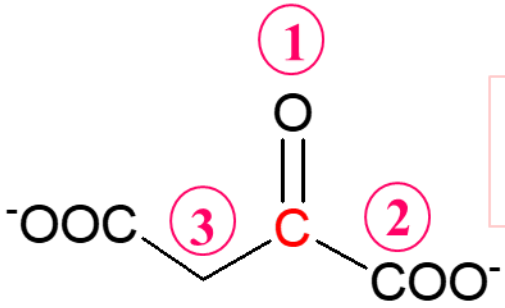
Citrate

Stereochemistry of Citric Acid Cycle

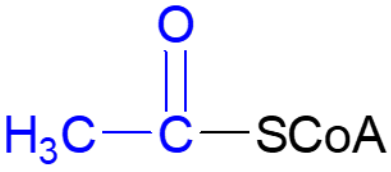


Oxaloacetate

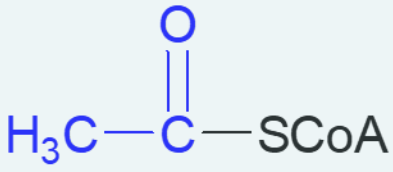
≡



Oxaloacetate is viewed from its *Re* face

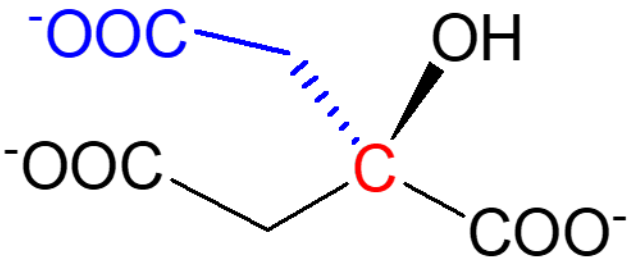


Citrate synthetase (*Si* face)

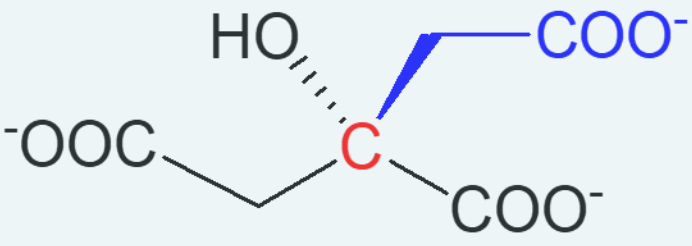


Citrate synthetase (*Re* face)

Citrate synthase (in most organisms), is *Si*-face stereospecific with respect to **C-2** of oxaloacetate.

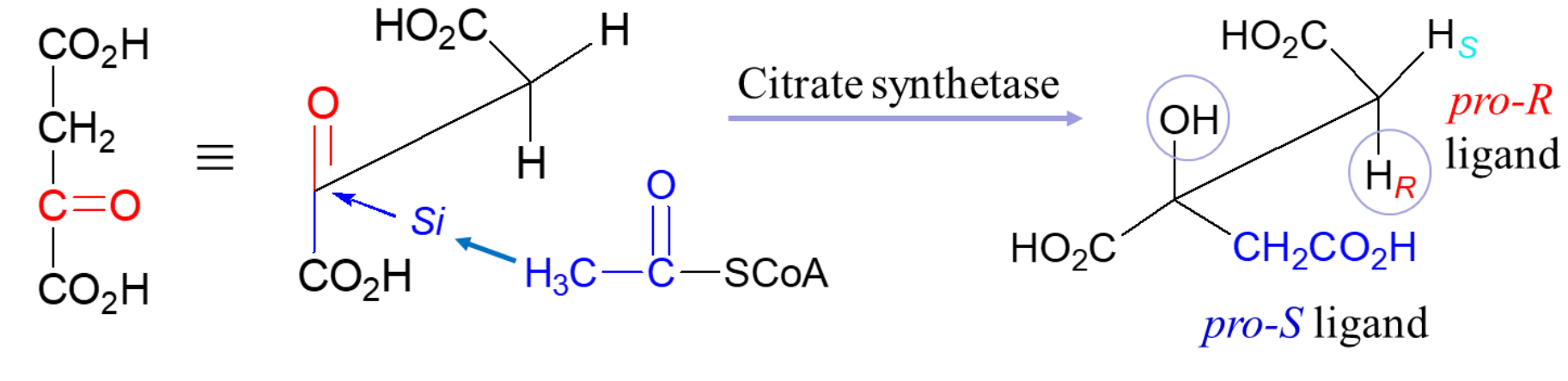


Citrate



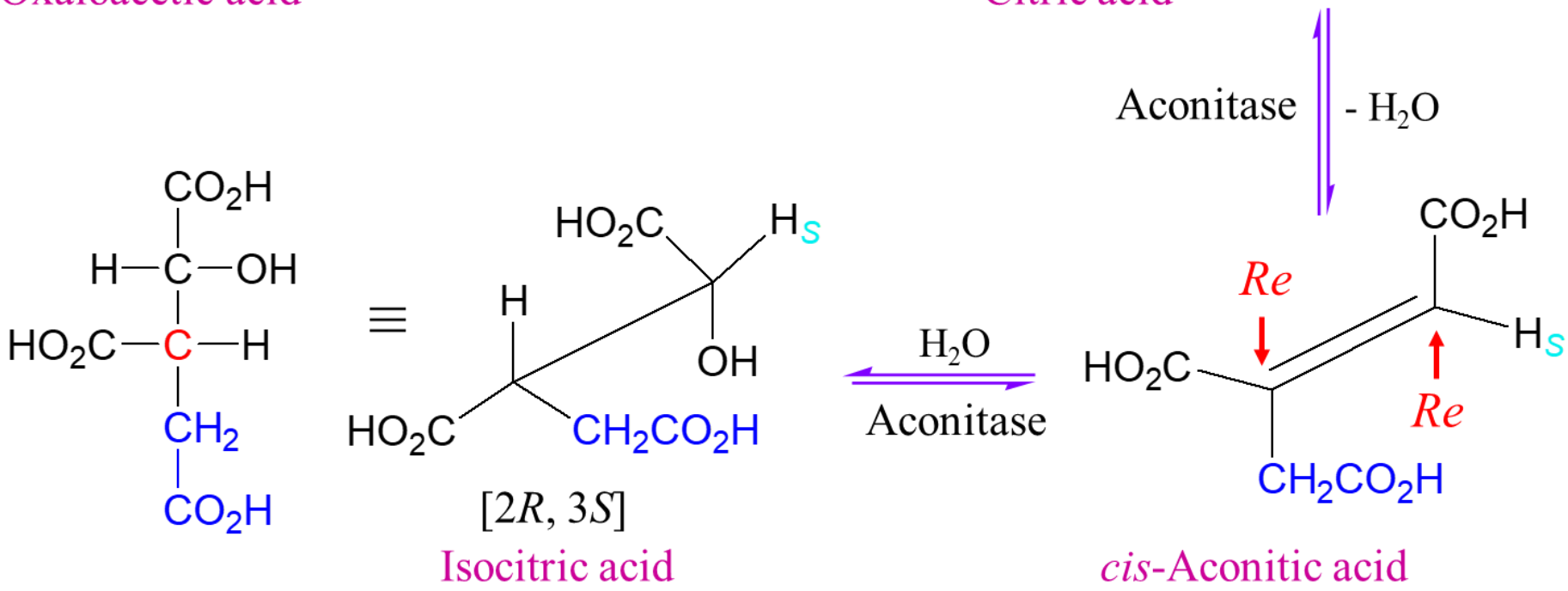
Citrate

Stereochemistry of Citric Acid Cycle



Oxaloacetic acid

Citric acid

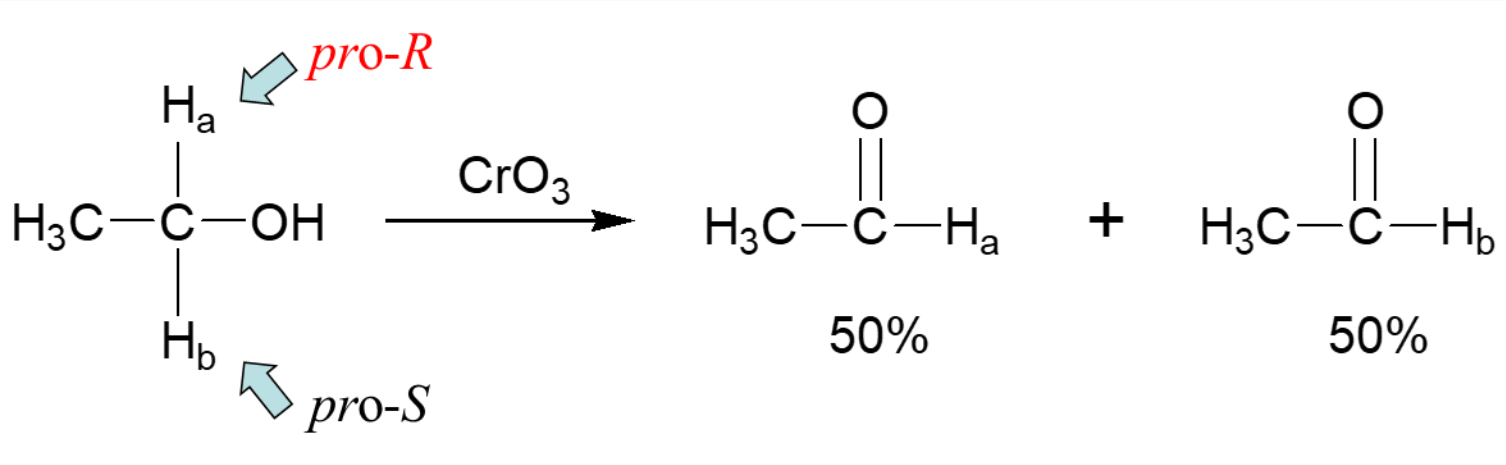


Isocitric acid

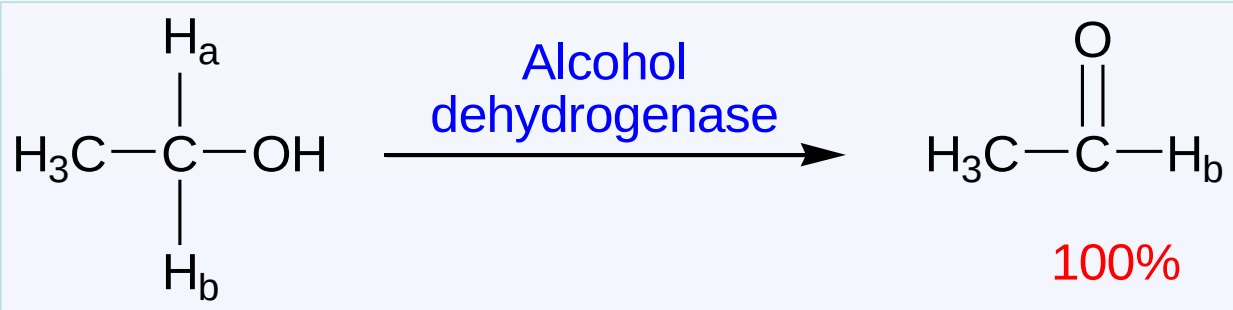
cis-Aconitic acid

Enzyme Catalyzed Reactions

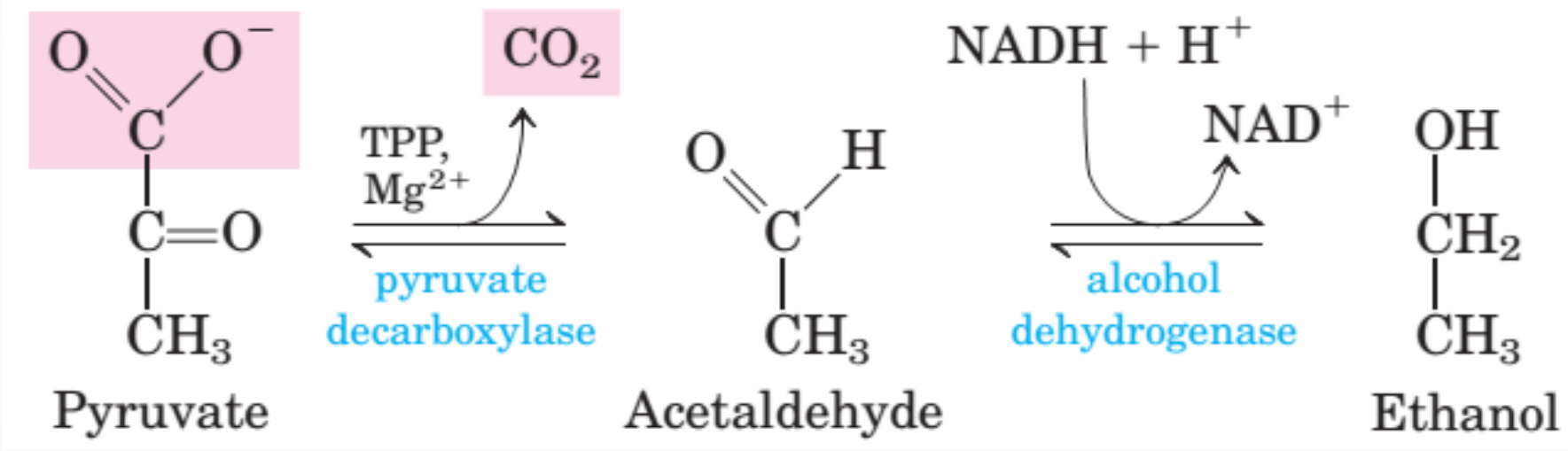
- The *pro-R* and *pro-S* can not be differentiated with achiral chemical reagents.



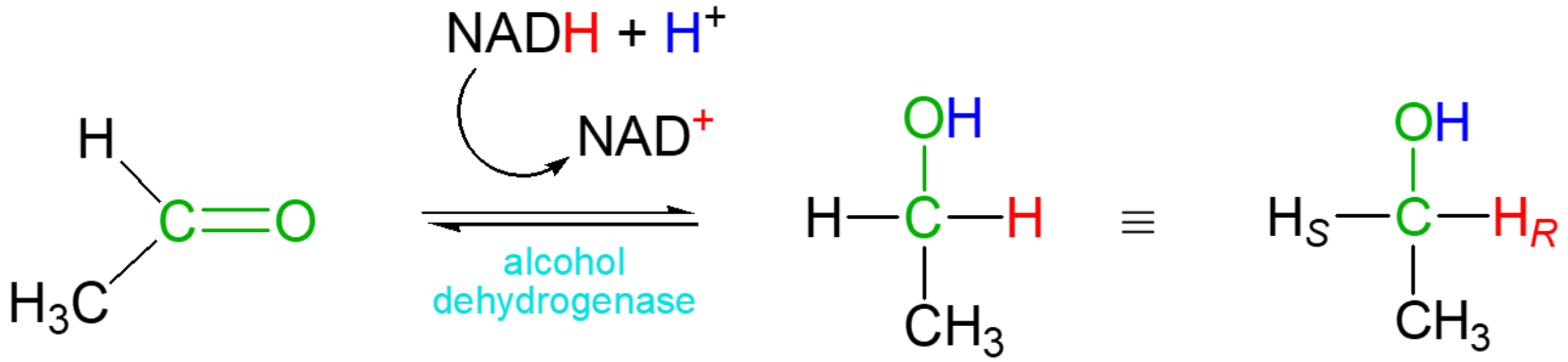
- Unlike the achiral chemical reagents, the chiral enzymes are capable of discriminating between two ligands and faces which are enantiotopic, as well as between two enantiomers.



Enzyme Catalyzed Reactions

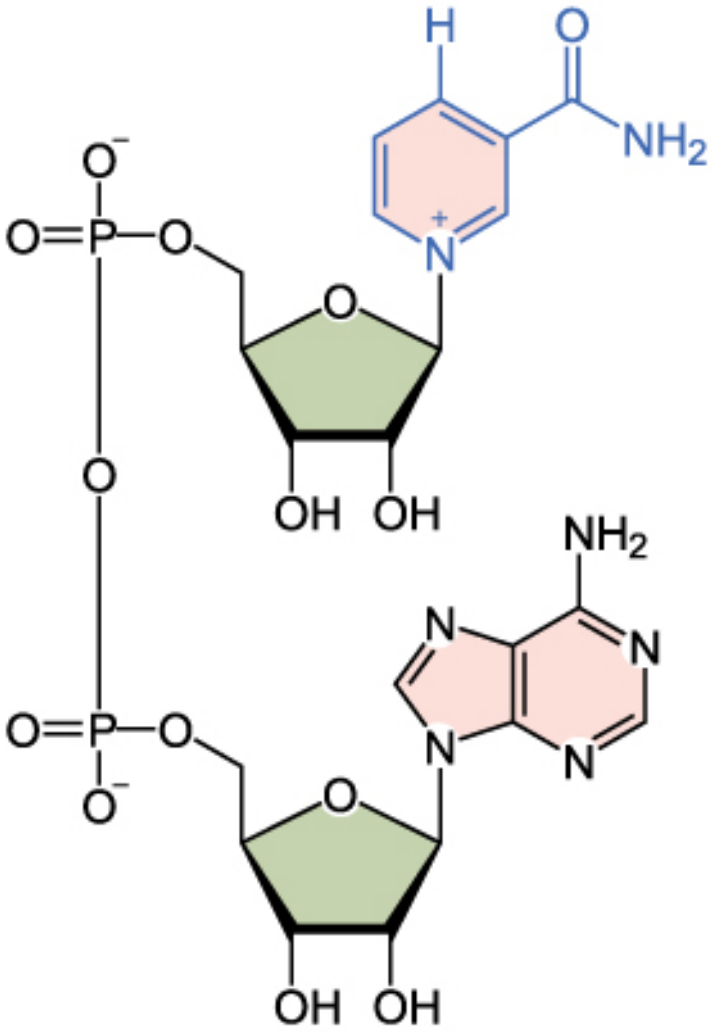
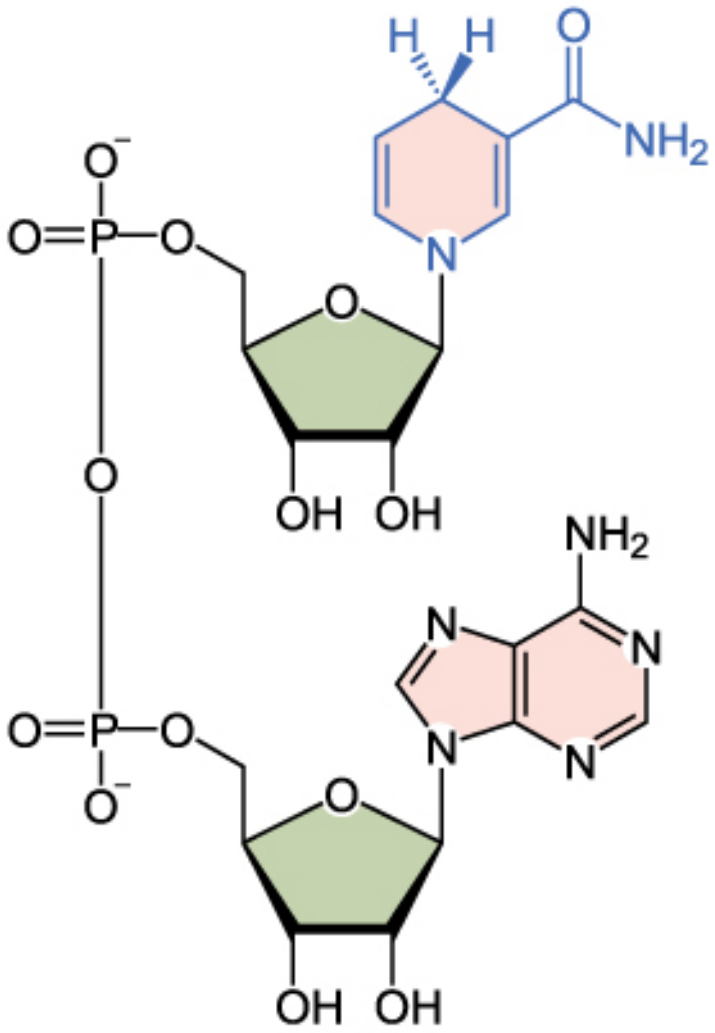


- Yeast and other microorganisms ferment glucose to ethanol and CO_2, rather than to lactate.
- Glucose is converted to pyruvate by glycolysis, and the pyruvate is converted to ethanol and CO_2 in the two-step process.



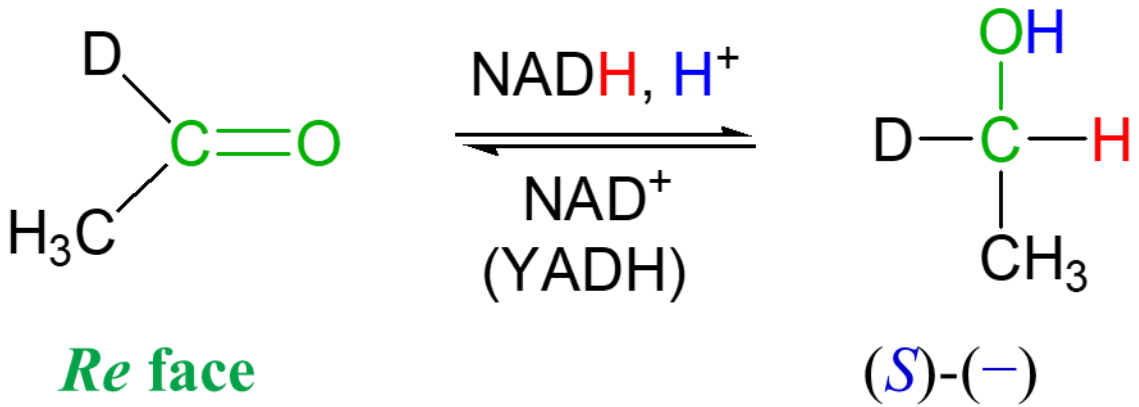
- The stereochemistry of oxidation of ethanol and reduction of acetaldehyde mediated by the enzyme yeast alcohol dehydrogenase in the presence of the oxidized (NAD^+) and reduced (NADH) forms respectively, of the coenzyme nicotinamide adenine dinucleotide is interesting.
- The stereochemically interesting feature of this reaction stems from the fact that the methylene hydrogens in CH_3CH_2OH (ethanol) and the faces of the carbonyl in CH_3CHO are enantiotopic.
- The question thus arises –

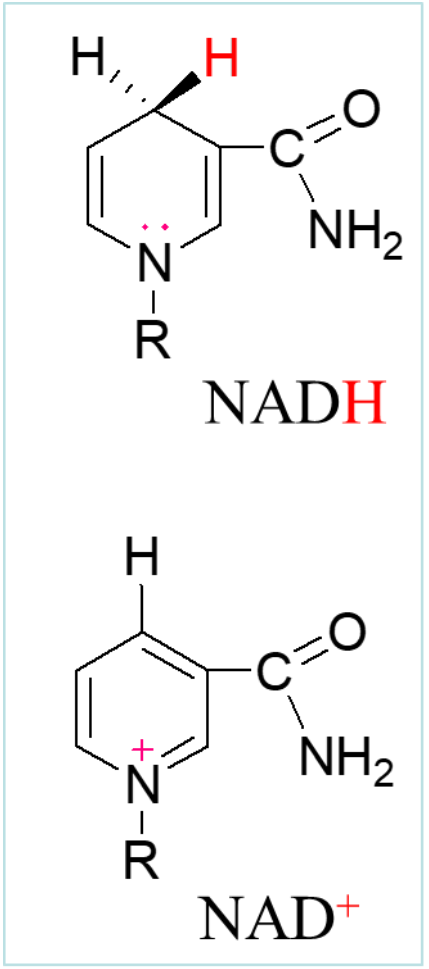
1. which of the CH_2-hydrogens is removed in the oxidation and
2. to which of the C=O faces the hydrogen attaches itself in the reduction in the presence of the enzyme and coenzyme.

NAD^+	NADH
	

Enzyme Catalyzed Oxidation-Reduction Reactions

- The reduction of ethanal-1-d with NADH in the presence of yeast alcohol dehydrogenase (YADH) gave ethanol-l-d which, upon enzymatic reoxidation by NAD^+, returned ethanal-1-d without loss of deuterium.

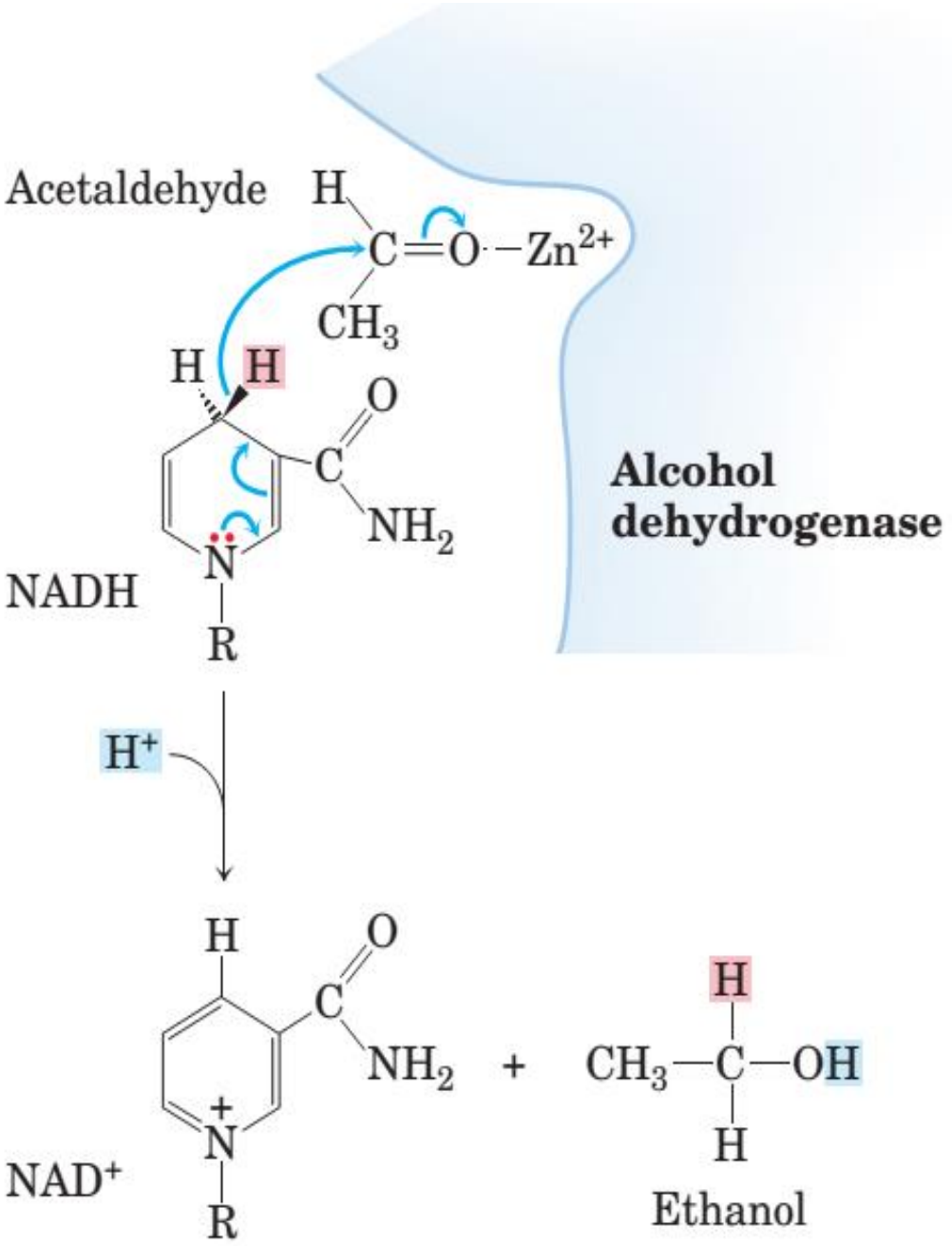




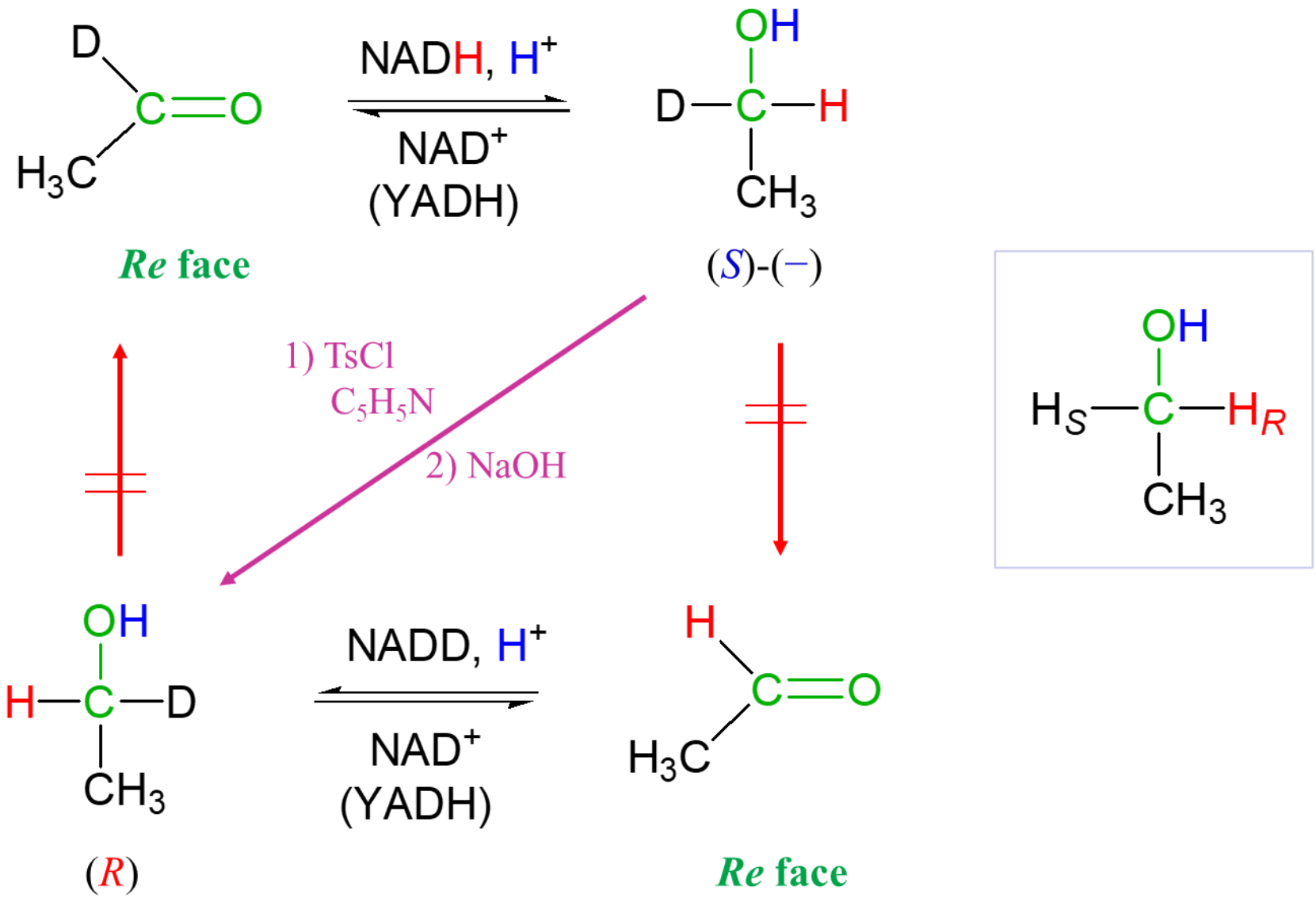
- The H and D of the CH_3CHDOH do not get scrambled but the same H which is introduced in the reduction is the one removed in the oxidation.

- Ethanol (and acetaldehyde) must fit into the active site of **yeast alcohol dehydrogenase**.

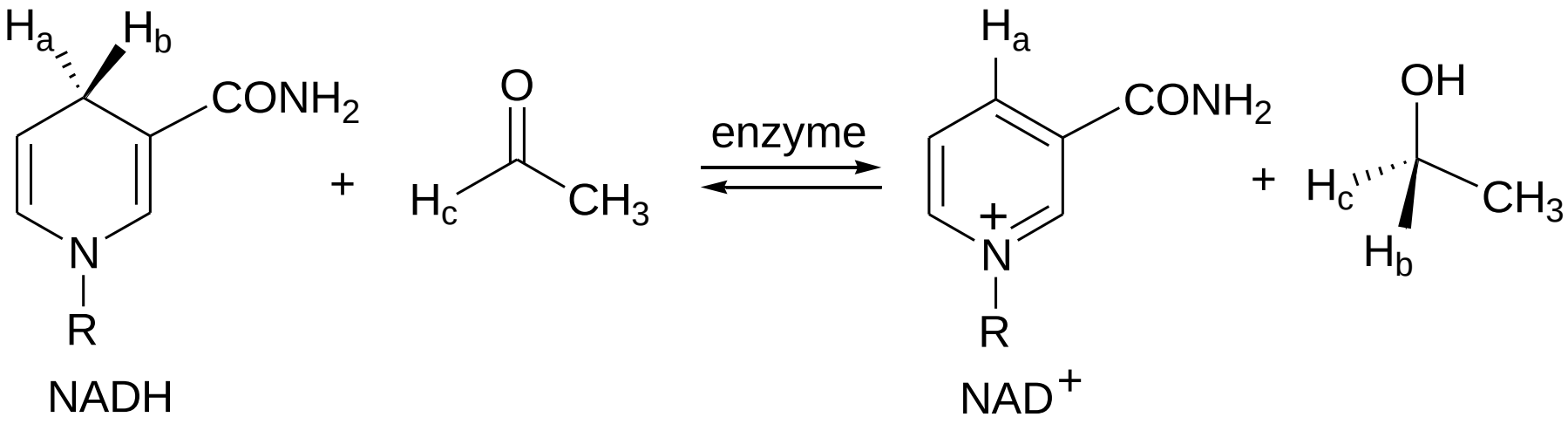
- It follows therefore that the hydrogen transferred from the NADH in the enzymatic reduction attaches itself to the ***Re* face** of the aldehyde and that this hydrogen thus becomes H_R in the ethanol; it is H_R (the *pro-R* carbinol hydrogen), in turn, which is abstracted by NAD^+ in the oxidative step.



Oxidation Of Ethanol And Reduction Of Acetaldehyde By NAD^+/NADH in the presence of YADH



Remember this Question?



Q.1 Consider the above reaction and answer the related questions:

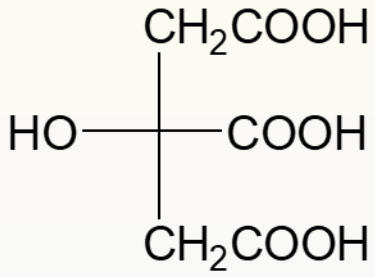
(i) Label the prochiral hydrogens in NADH as pro-*R* and pro-*S*. 1.5

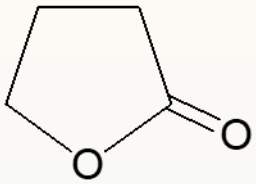
(ii) Depict the top face in acetaldehyde as *Re* or *Si* 0.75

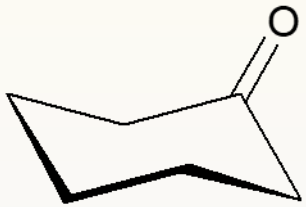
(iii) Indicate the methylene hydrogens in ethanol as enantiotopic or diastereotopic. Give your reasoning. 1.5

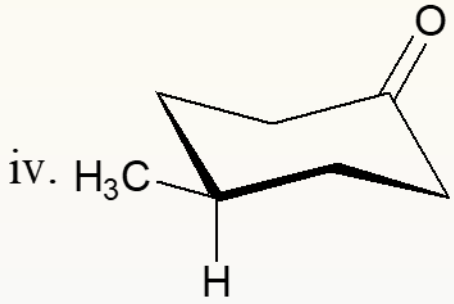
Practice Problems

Q. Locate enantiotopic and distereotopic groups or faces in the following compounds giving arguments. 2.75

i. 

ii. 

iii. 

iv. 

Q. Indicate if the protons Ha and Hb are homotopic, enantiotopic or distereotopic. 2

